USER STORIES

Test Fixture GUI Mockups

Main Window · Existing Test Found · Test Configuration · Calibration · EM Test · EM Analysis
Shear Test · Shear Analysis · Manual Test · Manual Analysis · Cyclical Test

FEATURE 1.1

Main Window

FEATURE 1.1 · MAIN WINDOW

STORY 1.1.1

MUST

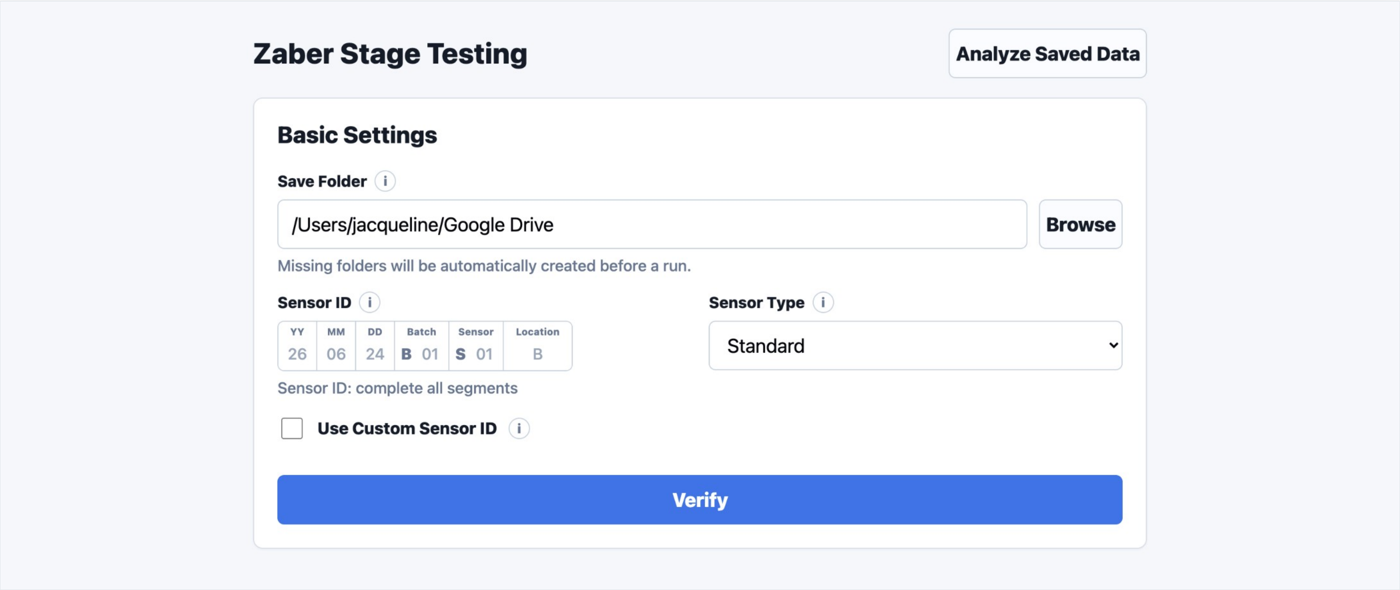
Main Window

USER STORY

As a test fixture operator, I want the Main Window to clearly display the basic setup fields so that I can prepare and run a sensor test from one central interface.

ACCEPTANCE CRITERIA

- When a user initially opens the application, they should be greeted with the main window that includes the title: "Zaber Stage Testing Setup"
- Note: Does not need to create buttons/widgets for the settings; just creates titles for each section on the page



FEATURE 1.1 · MAIN WINDOW

STORY 1.1.2

MUST

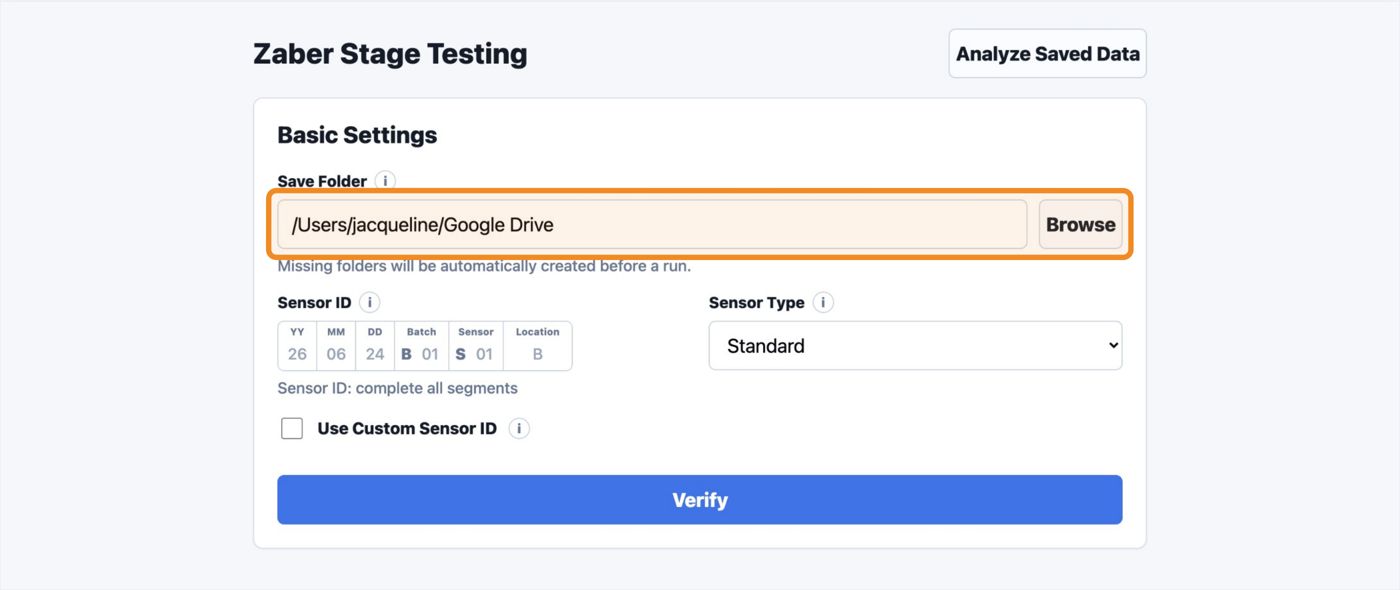
Save Folder Field

USER STORY

As a test fixture operator, I want to choose where test outputs are saved so that plots, Excel files, and analysis results are stored in the correct folder.

ACCEPTANCE CRITERIA

- The user can type a save folder path
- The user can browse and select a save folder
- If a save folder is selected, test outputs are saved there
- If no save folder is selected, the system uses the default/root directory
- If the save folder does not exist, the system creates the needed folders
- Info note below the field: "Missing folders will be automatically created before a run"



FEATURE 1.1 · MAIN WINDOW

STORY 1.1.3

MUST

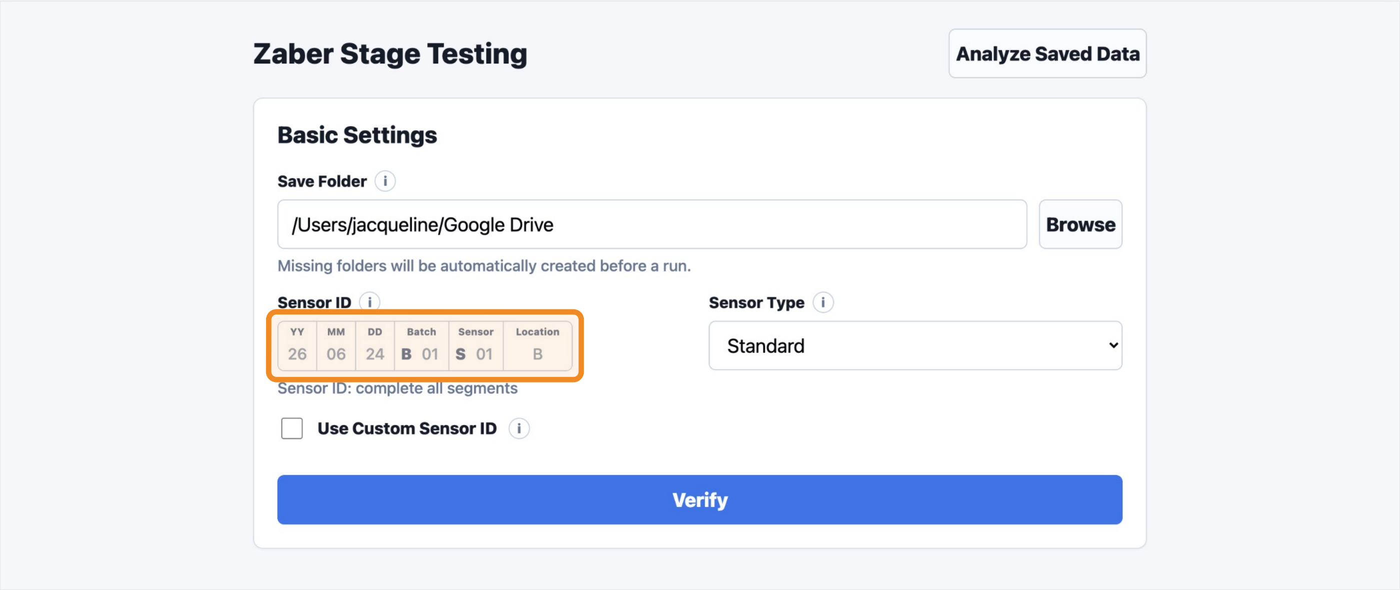
Sensor ID Field

USER STORY

As a test fixture user, I want to enter the Sensor ID through one connected segmented field so that the ID format is easier to follow and less likely to be mistyped.

ACCEPTANCE CRITERIA

- Sensor ID input is one connected row with segments: YY, MM, DD, Batch, Sensor, Location
- Batch visually includes fixed prefix "B"; user enters only the number. Same for Sensor ("S")
- Generated Sensor ID follows: YYMMDDB##S##X
    - Requirements: Year, Month, Day each 2 digits; Batch and Sensor each 2 numeric digits; Location must be A, B, or AB
    - Example valid IDs: 250506B01S01A, 250506B01S01B, 250506B01S01AB
- Completed Sensor ID is used in test completion messages, saved outputs, and analysis



FEATURE 1.1 · MAIN WINDOW

STORY 1.1.4

MUST

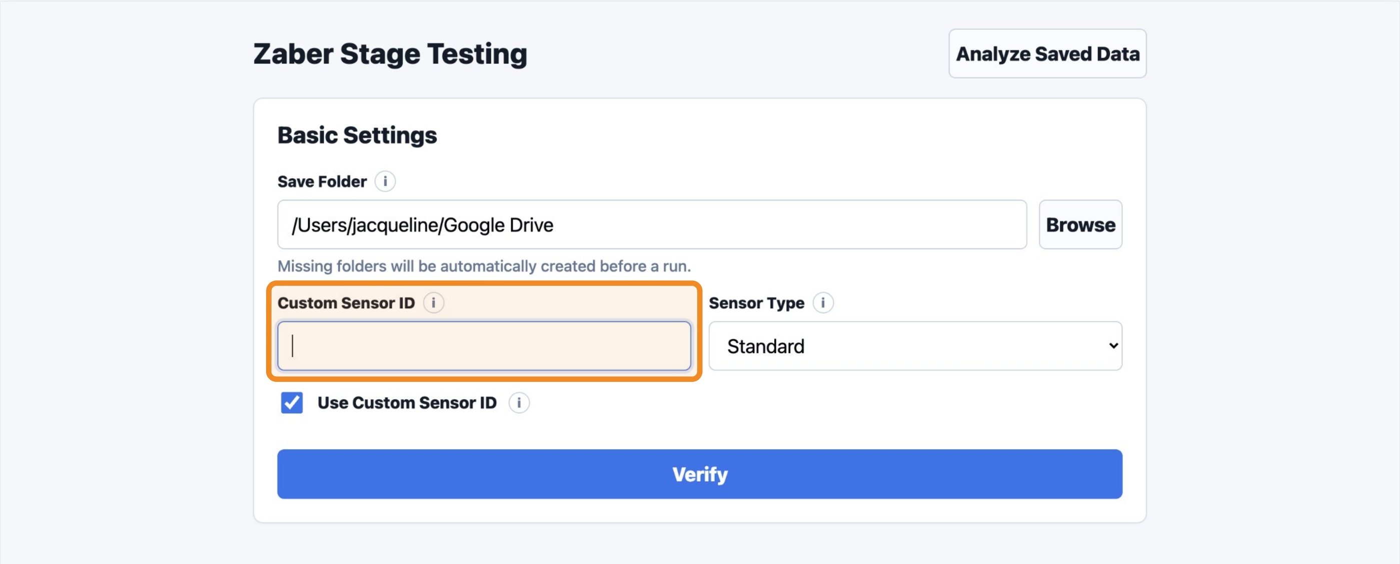
Custom Sensor ID Checkbox

USER STORY

As a test fixture user, I want the option to use a custom Sensor ID so that I can test experimental or nonstandard sensors.

ACCEPTANCE CRITERIA

- The Main Window includes a checkbox labeled "Use Custom Sensor ID"
- When unchecked, the segmented Sensor ID field is active
- When checked, the segmented Sensor ID field is disabled and a custom free-text Sensor ID input becomes active
- The custom Sensor ID is used for testing, saving, and analysis



FEATURE 1.1 · MAIN WINDOW

MUST

STORY 1.1.5

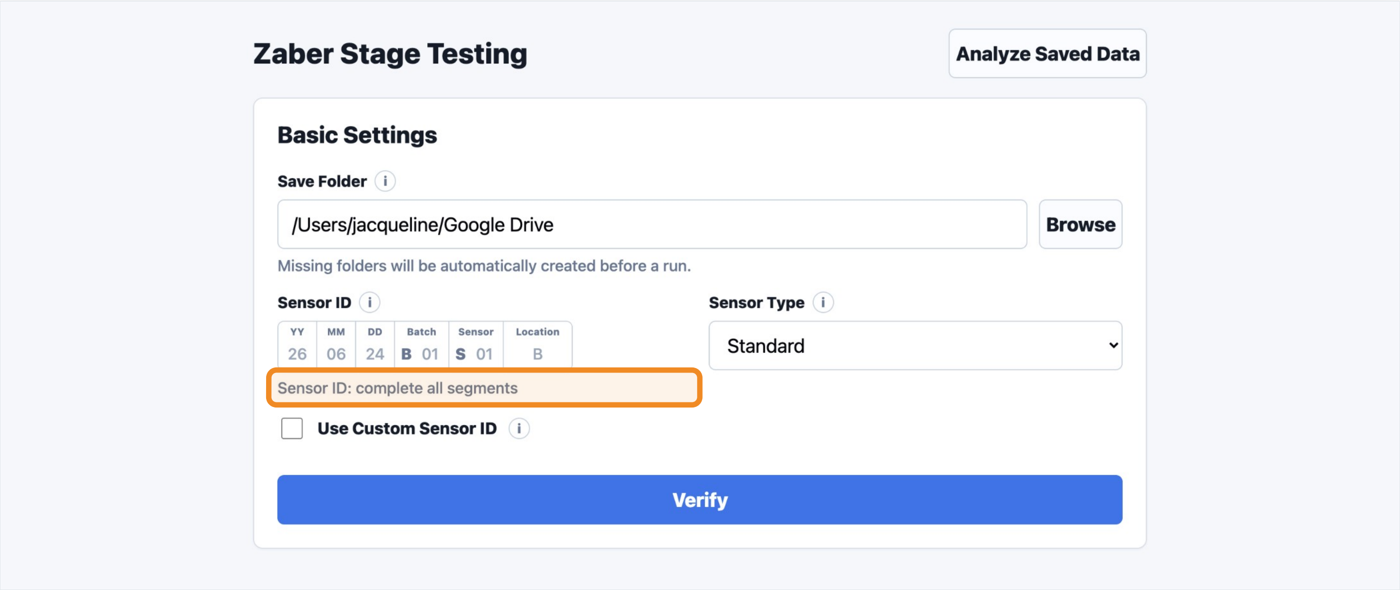
Sensor ID Validation / Preview

USER STORY

As a test fixture user, I want the GUI to validate Sensor ID formatting so that mistyped sensor IDs do not cause data to be saved under the wrong sensor.

ACCEPTANCE CRITERIA

- If all segments are complete, the GUI displays the full generated Sensor ID
- If any segment is missing, the GUI displays: "Sensor ID: complete all segments"
- The Sensor ID preview updates whenever any segment changes



FEATURE 1.1 · MAIN WINDOW

STORY 1.1.6

MUST

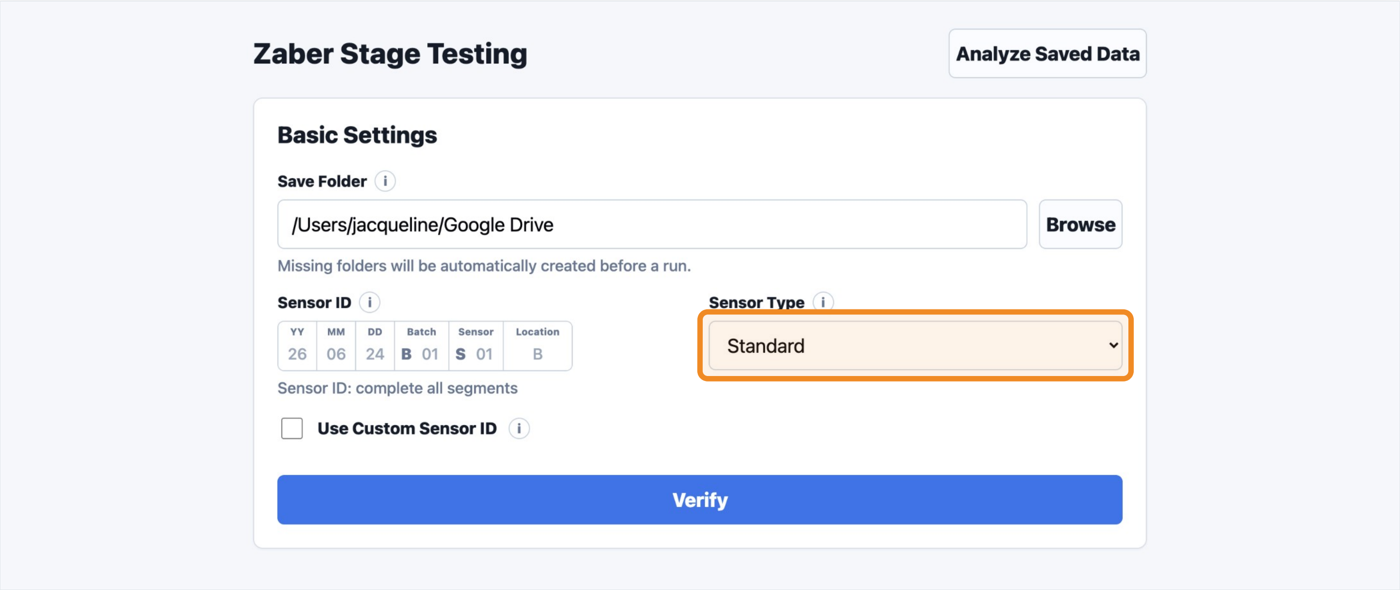
Sensor Type Dropdown

USER STORY

As a test fixture user, I want to select the sensor type so that the system can interpret the sensor data using the correct firmware/channel configuration.

ACCEPTANCE CRITERIA

- The user can select a Sensor Type from a dropdown; options: Standard, Inverted
  - Standard orders the channels [1, 2, 3, 4, 5, 6, 7, 8]
  - Inverted orders the channels [1, 2, 3, 4, 8, 7, 6, 5]
- Standard is selected by default
- The selected Sensor Type is passed into the analysis workflow
- Info note below the field explains the sensor type per firmware version / channel configuration



FEATURE 1.1 · MAIN WINDOW

MUST

STORY 1.1.7

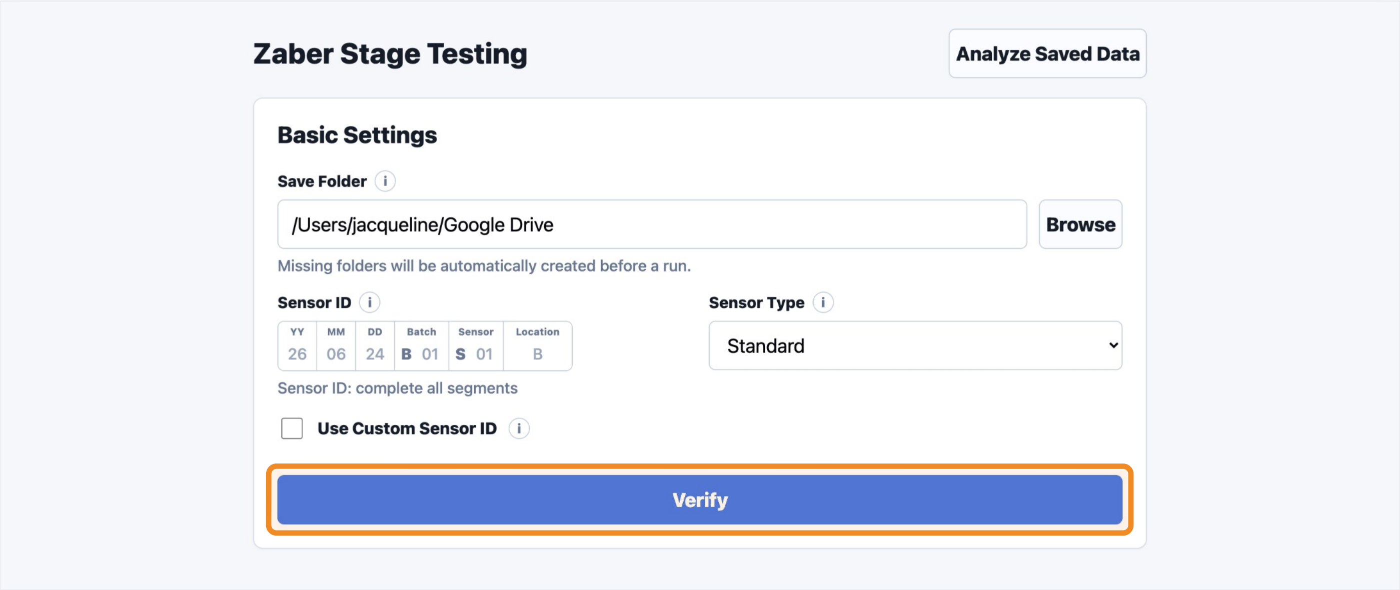
Verify Settings Button

USER STORY

As a test fixture user, I want to click Verify Settings so that I can confirm the setup parameters before the test begins.

ACCEPTANCE CRITERIA

- Clicking Verify Settings proceeds to the Test Configuration page
- If Save Folder is blank: block, warn "Fill in Save Folder before continuing"
- If Custom Sensor ID enabled and blank: block, warn "Enter a custom sensor ID"
- If Sensor Type is blank: block, warn "Fill in Sensor Type"
- If Custom Sensor ID disabled and any Sensor ID segment is incomplete: block, warn "Complete all Sensor ID segments before continuing"
- If any Sensor ID segment is invalid: block with an appropriate validation message
- If Sensor ID already has pre-existing tests from the same day, the Existing Test Found dialog opens (Feature 1.2)
- If all required fields are valid, the Test Configuration section becomes visible



STORY 1.1.8

MUST

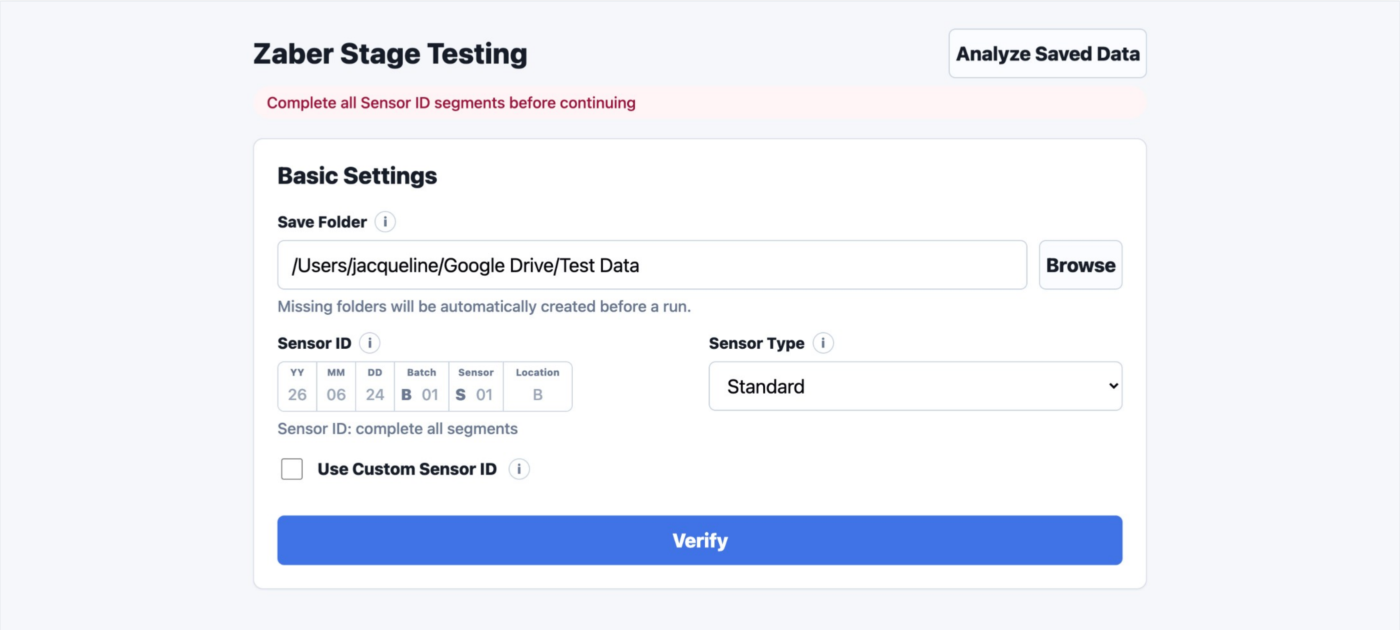
Basic Settings Reverification Behavior

USER STORY

As a test fixture user, I want the Test Configuration section to disappear when Basic Settings are changed so that tests cannot begin using outdated or unverified setup information.

ACCEPTANCE CRITERIA

- If the user edits any Basic Settings field after verification, the Test Configuration section disappears
- The GUI displays: "Basic settings changed. Please click Verify again"
- The user must click Verify again before continuing to Test Configuration
- Previously entered Test Configuration settings are not used until Basic Settings are reverified



FEATURE 1.1 · MAIN WINDOW

MUST

STORY 1.1.9

Info Icons

USER STORY

As a test fixture user, I want informational help icons throughout the window so that I can quickly understand fields, workflows, and testing terminology without needing external documentation.

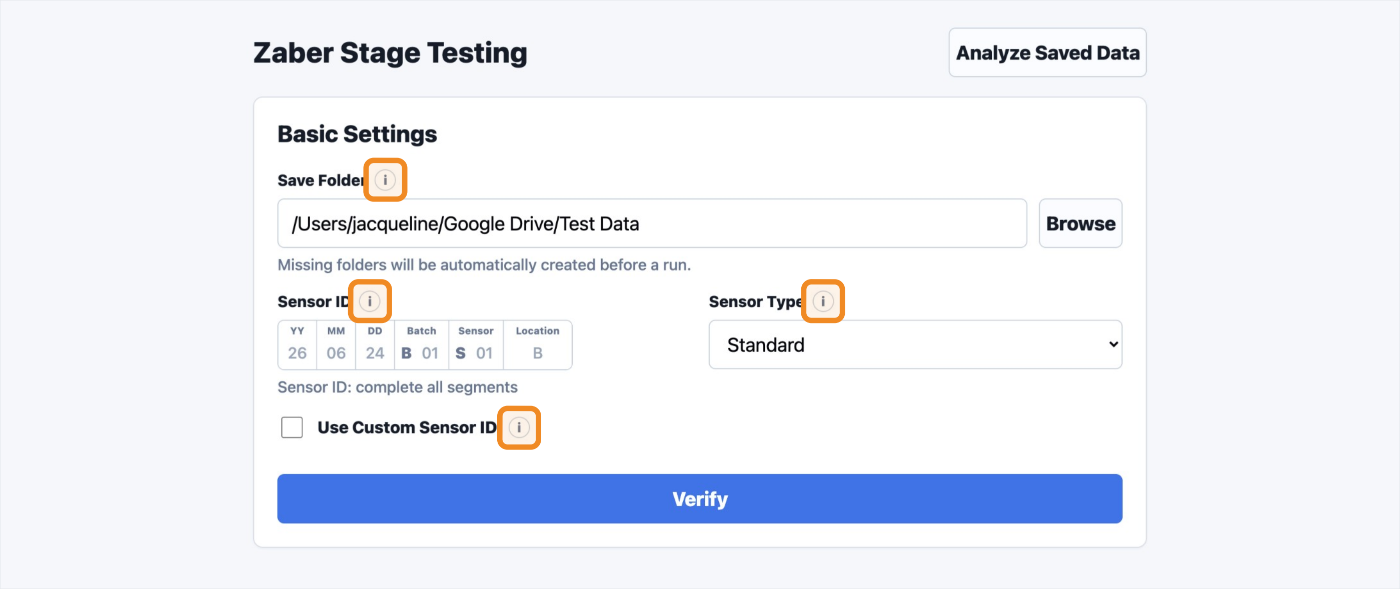
INFO ICONS ON THIS PAGE

Save Folder: Folder where run files, exported data, and analysis outputs will be saved.

Sensor ID: Format: YYMMDDB##S##X. Type two digits for date, batch, and sensor. Location can be A, B, AB, or BA.

Use Custom Sensor ID: When enabled, the Sensor ID builder is replaced with a free-text custom Sensor ID field.

Sensor Type: Standard uses normal channel ordering. Inverted reverses the channel order.



FEATURE 1.2

Existing Test Found Page

FEATURE 1.2 · EXISTING TEST FOUND

MUST

STORY 1.2.1

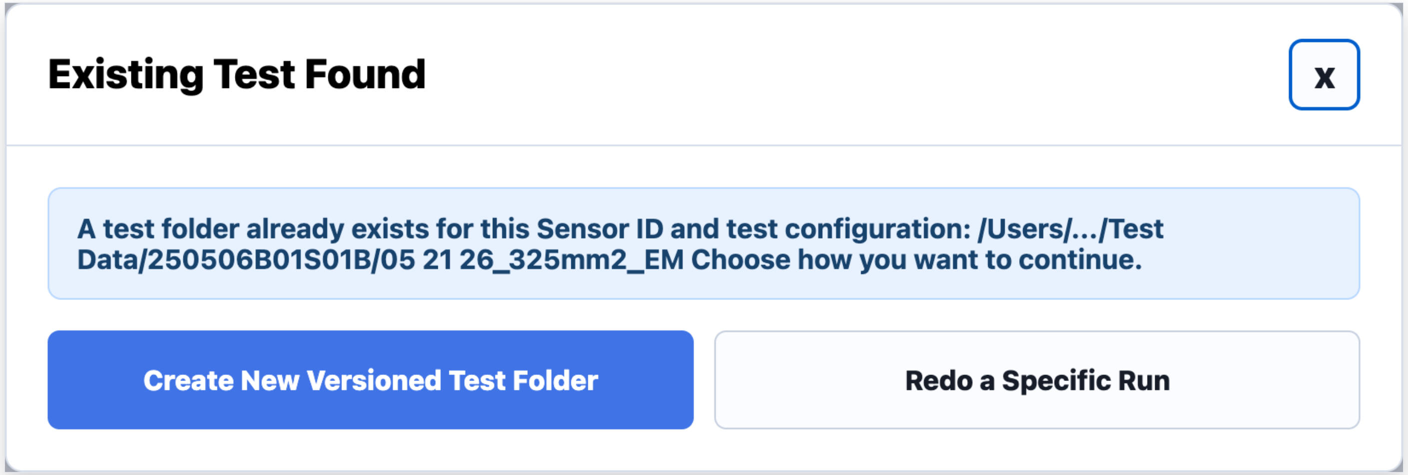
Existing Test Found Popup

USER STORY

As a test fixture user, I want the software to warn me when a matching test folder already exists for a specified sensor so that I can decide how to continue without accidentally overwriting previous data.

ACCEPTANCE CRITERIA

- If an existing test-session folder already exists, the GUI displays an "Existing Test Found" popup window
- The popup displays the full existing test path (e.g., /Users/.../Test Data/12345/05 21 26_325mm2_EM)
- The popup displays: "A test folder already exists for this Sensor ID and test configuration. Choose how you want to continue."



FEATURE 1.2 · EXISTING TEST FOUND

MUST

STORY 1.2.2

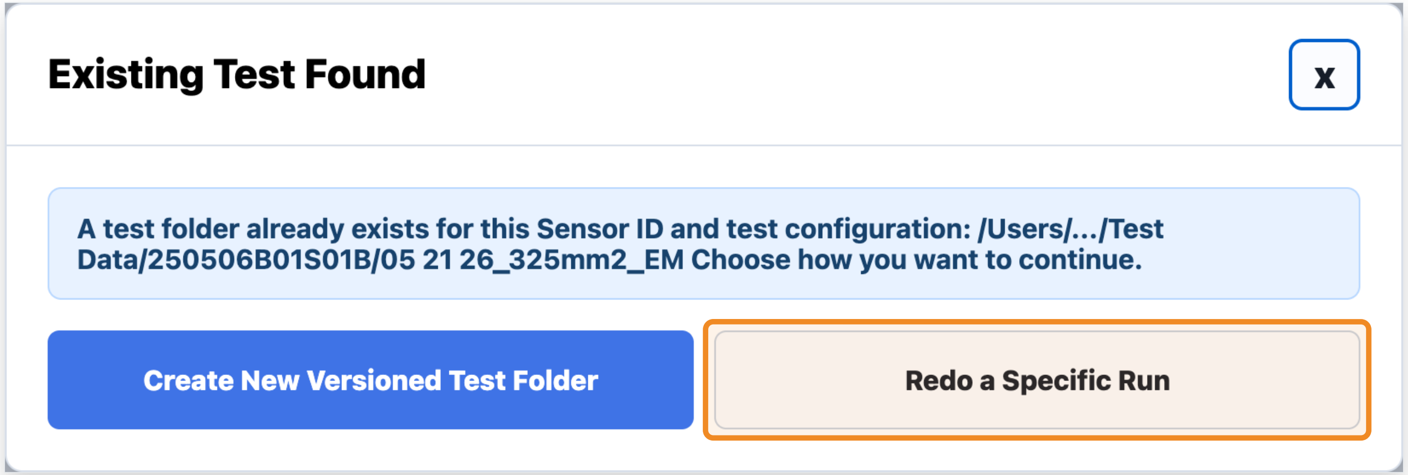
Redo a Specific Run Button

USER STORY

As a test fixture user, I want to redo a specific run from an existing test so that I can repeat failed or invalid runs without restarting the entire test session.

ACCEPTANCE CRITERIA

- The Existing Test Found popup includes a button: "Redo a Specific Run"
- Redo a Specific Run functionality only applies to EM Testing workflows (not supported for Shear, Manual, or Cyclical testing).
    - If the existing test was created from a Shear, Manual, or Cyclical test, the "Redo a Specific Run" button is disabled.
- Clicking the button opens the Test Configuration Window in Redo Run mode
- In Redo Run mode, EM Test is automatically selected and cannot be modified
- In Redo Run mode, Number of Runs is automatically set to 1 and cannot be modified
- The "Run to Redo" field becomes enabled; user specifies which run to repeat
- The system only repeats the selected run instead of restarting the entire test sequence



FEATURE 1.2 · EXISTING TEST FOUND

MUST

STORY 1.2.3

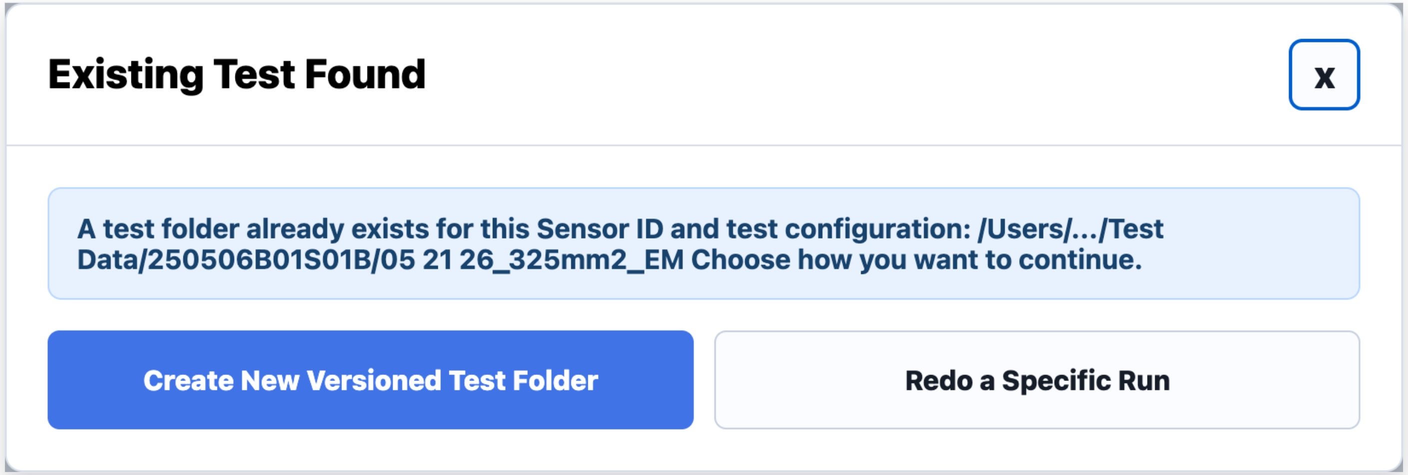
Overwrite Existing Test Button

USER STORY

As a test fixture user, I want the option to overwrite an existing test folder so that I can intentionally replace previous test data when needed.

ACCEPTANCE CRITERIA

- The Existing Test Found popup includes a button: "Overwrite Existing Test"
- Clicking the button opens the Test Configuration Window and allows the workflow to overwrite the existing test-session folder and outputs
- The system displays a confirmation warning before overwriting existing data



FEATURE 1.2 · EXISTING TEST FOUND

MUST

STORY 1.2.4

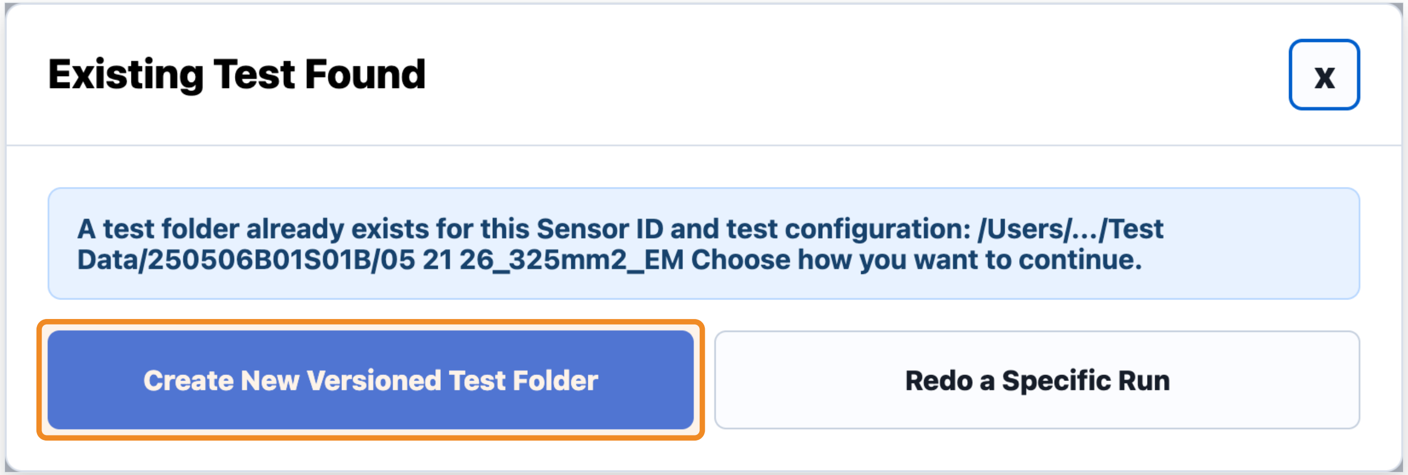
Create New Versioned Test Folder Button

USER STORY

As a test fixture user, I want to create a new versioned test folder so that I can preserve previous test data while starting a new test session.

ACCEPTANCE CRITERIA

- The Existing Test Found popup includes a button: "Create New Versioned Test Folder"
- Clicking the button opens the Test Configuration Window and creates a new versioned test-session folder instead of overwriting
- The new versioned folder is used for all new FUT, CAP, and analysis outputs



FEATURE 1.2 · EXISTING TEST FOUND

STORY 1.2.5

MUST

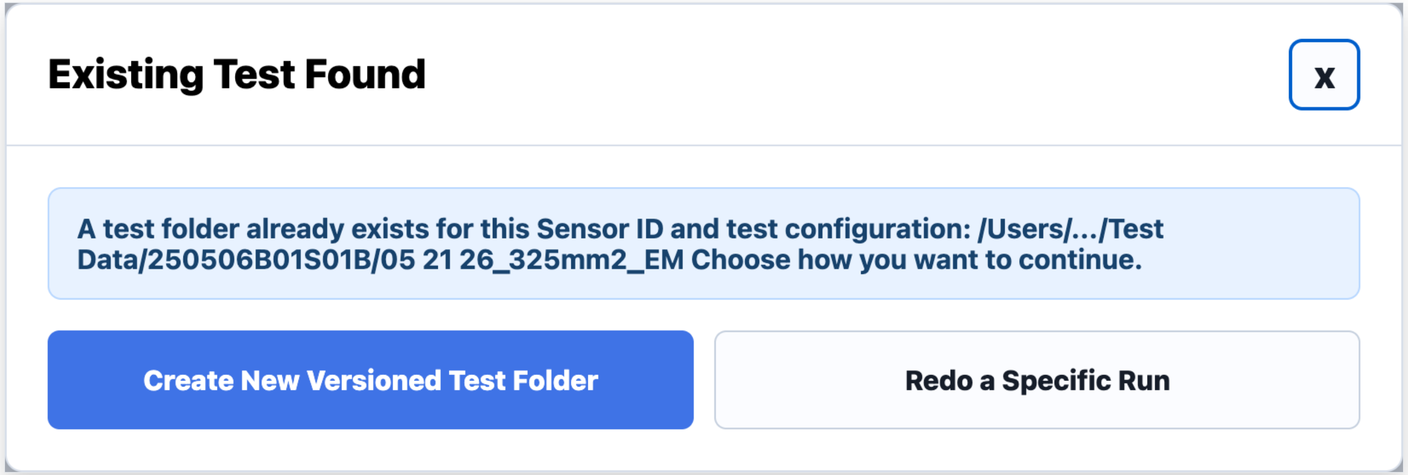
Cancel Button

USER STORY

As a test fixture user, I want to cancel the Existing Test Found workflow so that I can safely stop the current action without modifying any files or starting a test.

ACCEPTANCE CRITERIA

- The Existing Test Found popup includes a button: "Cancel"
- Clicking the button closes the popup window
- No test begins
- No files or folders are modified



FEATURE 1.3

Test Configuration Page

FEATURE 1.3 · TEST CONFIGURATION

MUST

STORY 1.3.1

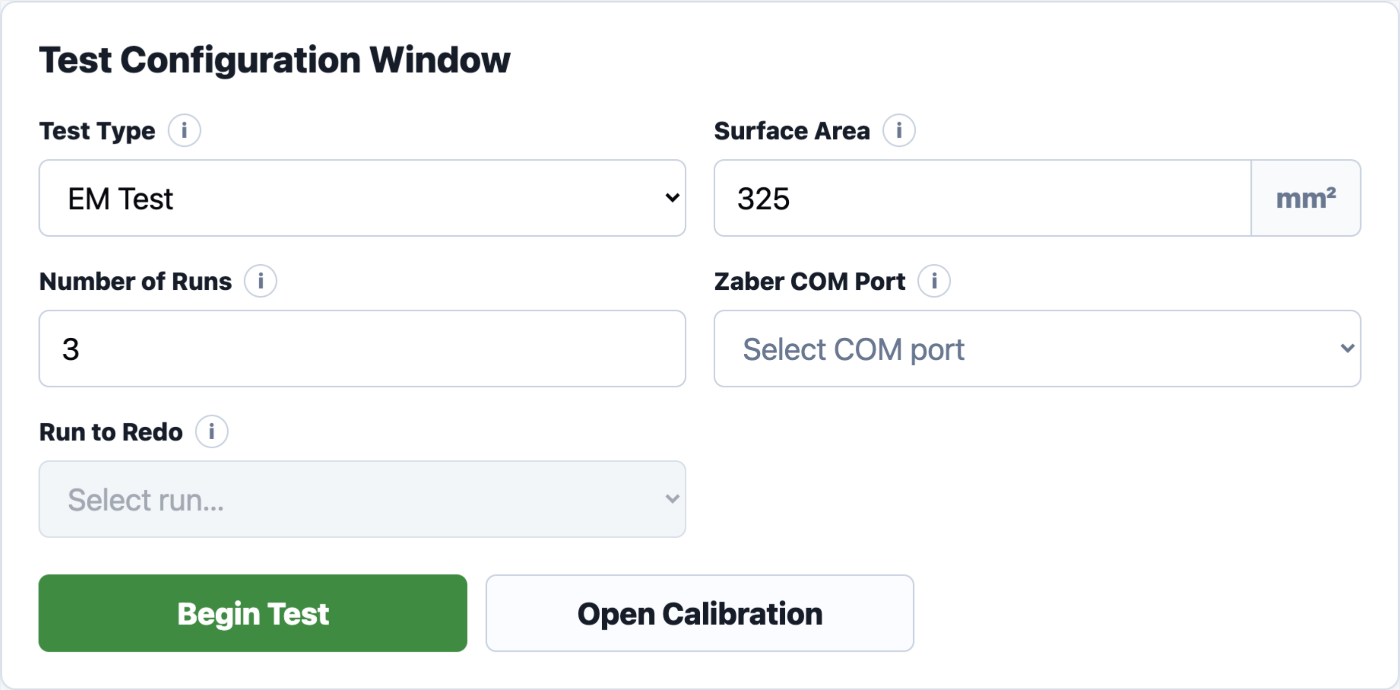
Test Configuration Window

USER STORY

As a test fixture user, I want the Test Configuration window to clearly display the test setup fields so that I can prepare and run a sensor test.

ACCEPTANCE CRITERIA

- Clicking the "Verify Settings" button opens the Test Configuration window with the title: "Test Configuration Window"



FEATURE 1.3 · TEST CONFIGURATION

MUST

STORY 1.3.2

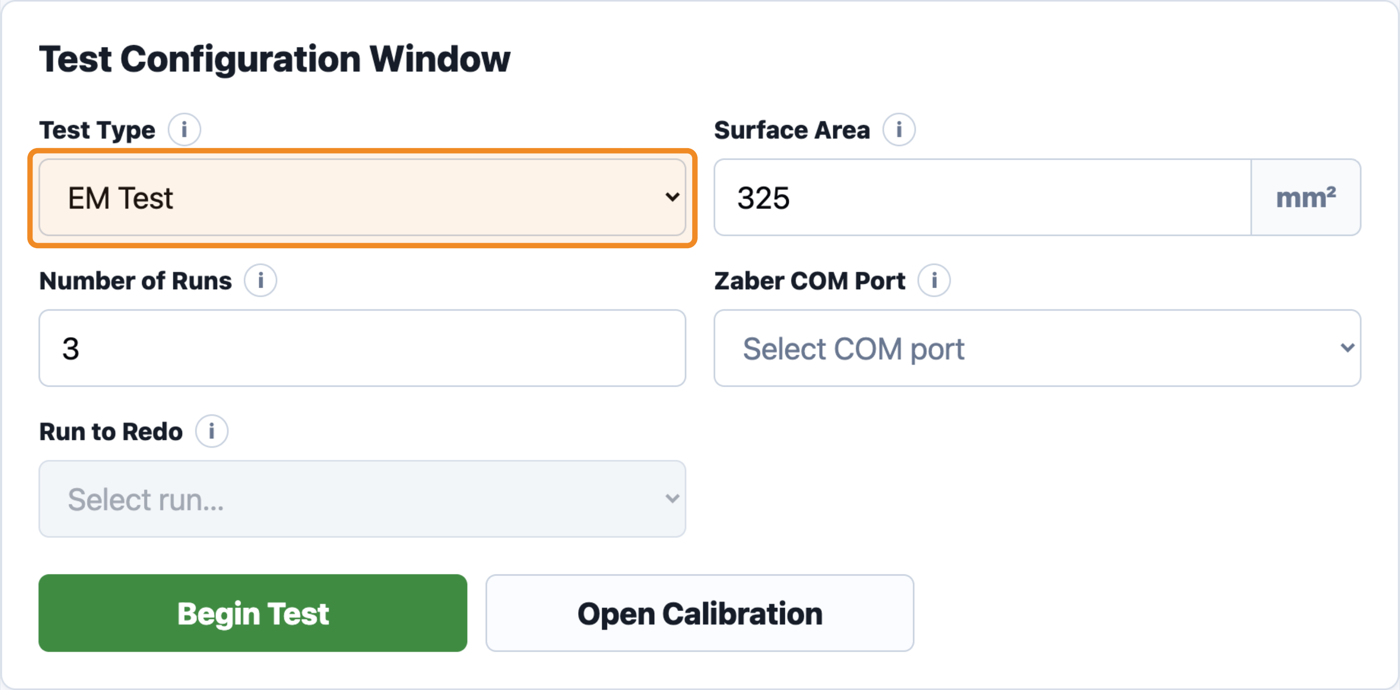
Test Type Dropdown

USER STORY

As a test fixture user, I want to select the Test Type so that the correct testing workflow and analysis pipeline are used.

ACCEPTANCE CRITERIA

- User can select Test Type from a dropdown; options: EM Test, Shear Test, Manual Test, Cyclical Test
- The selected Test Type is passed into the testing and analysis workflow
- EM Test: enables Number of Runs, Zaber COM Port, and Pause Between Runs
- Shear Test: disables Number of Runs, Zaber COM Port, and Pause Between Runs
- Manual Test: disables Number of Runs and Run to Redo; Zaber COM Port stays enabled
- Cyclical Test: disables Number of Runs and Run to Redo; Zaber COM Port stays enabled; cyclical waveform controls become visible



STORY 1.3.3

MUST

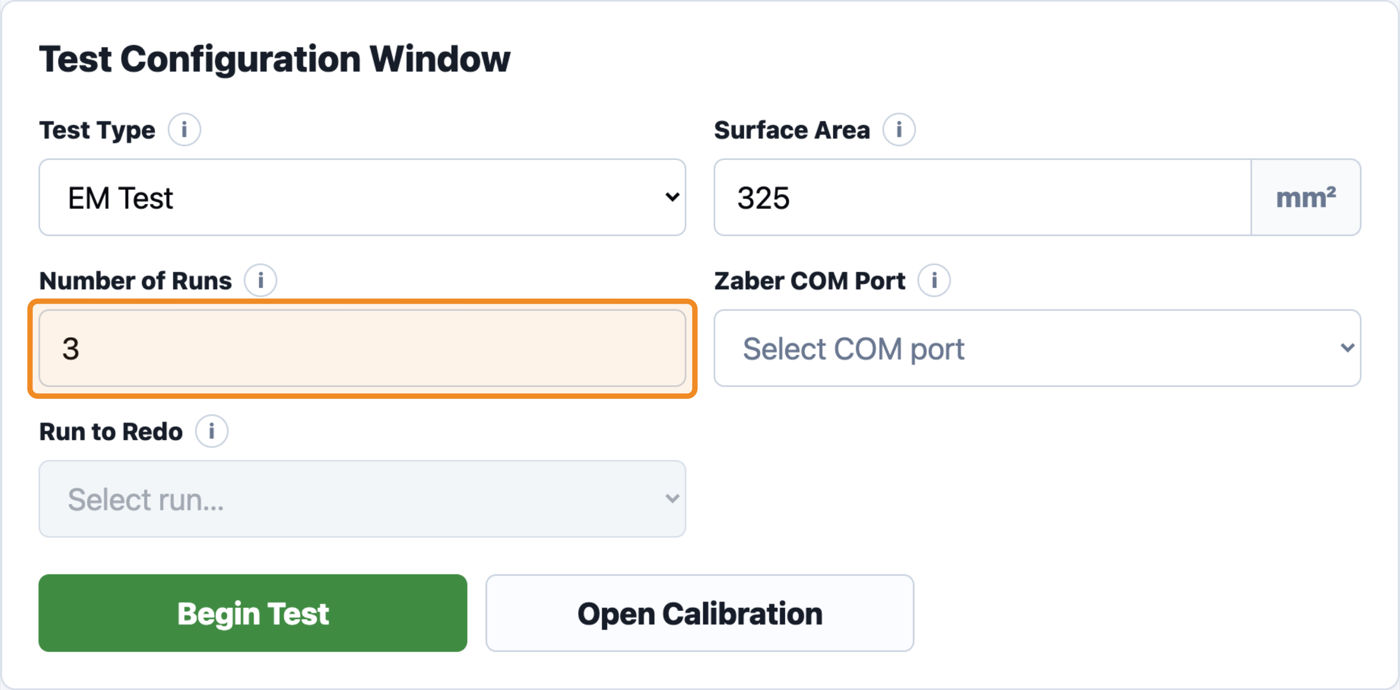
Number of Runs Integer Field

USER STORY

As a test fixture user, I want to choose the number of runs so that repeated testing can be automated.

ACCEPTANCE CRITERIA

- The user can enter the number of runs
- Default number of runs = 3
- The field only accepts positive integers
- The field only enables for the EM Test



FEATURE 1.3 · TEST CONFIGURATION

MUST

STORY 1.3.4

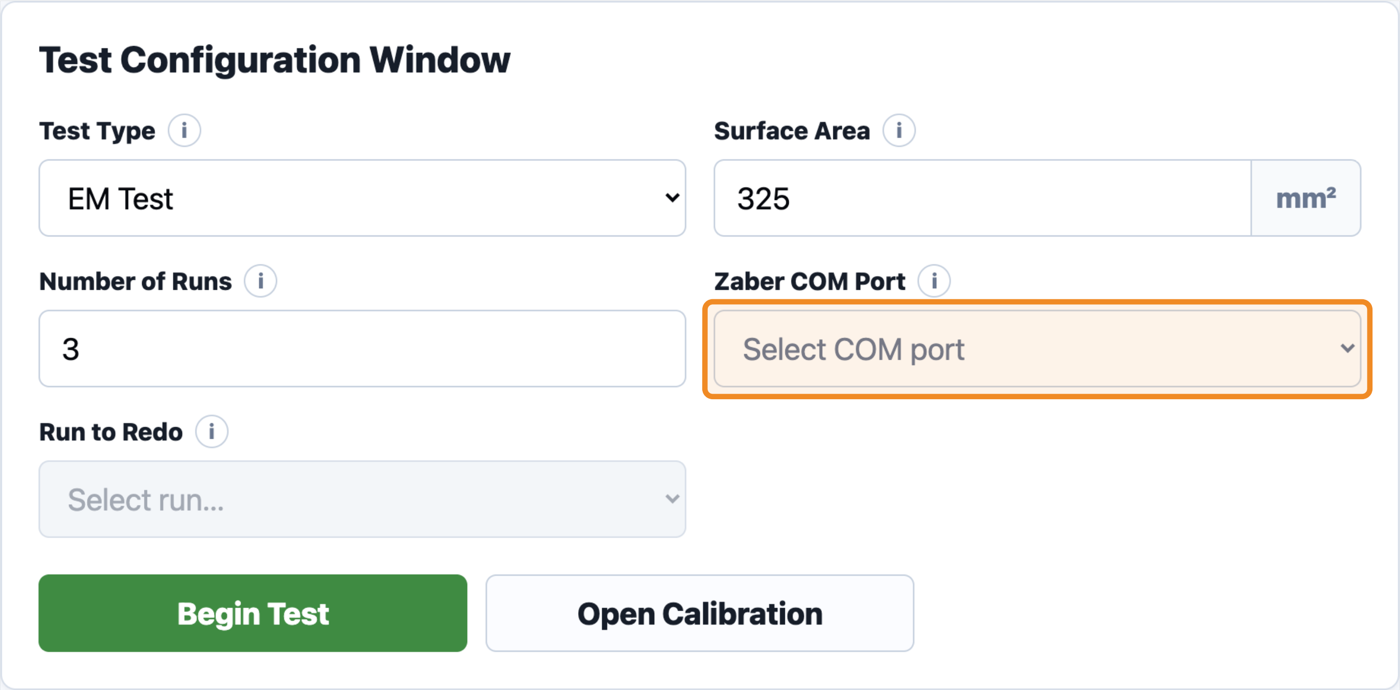
Zaber COM Port Dropdown

USER STORY

As a test fixture user, I want to define the Zaber COM Port so that the software can connect to the correct actuator.

ACCEPTANCE CRITERIA

- The user can enter or select a Zaber COM Port
- Default value = COM3
- The selected COM Port is used during hardware initialization
- The field only enables for the EM Test, Manual Test, and Cyclical Test



FEATURE 1.3 · TEST CONFIGURATION

MUST

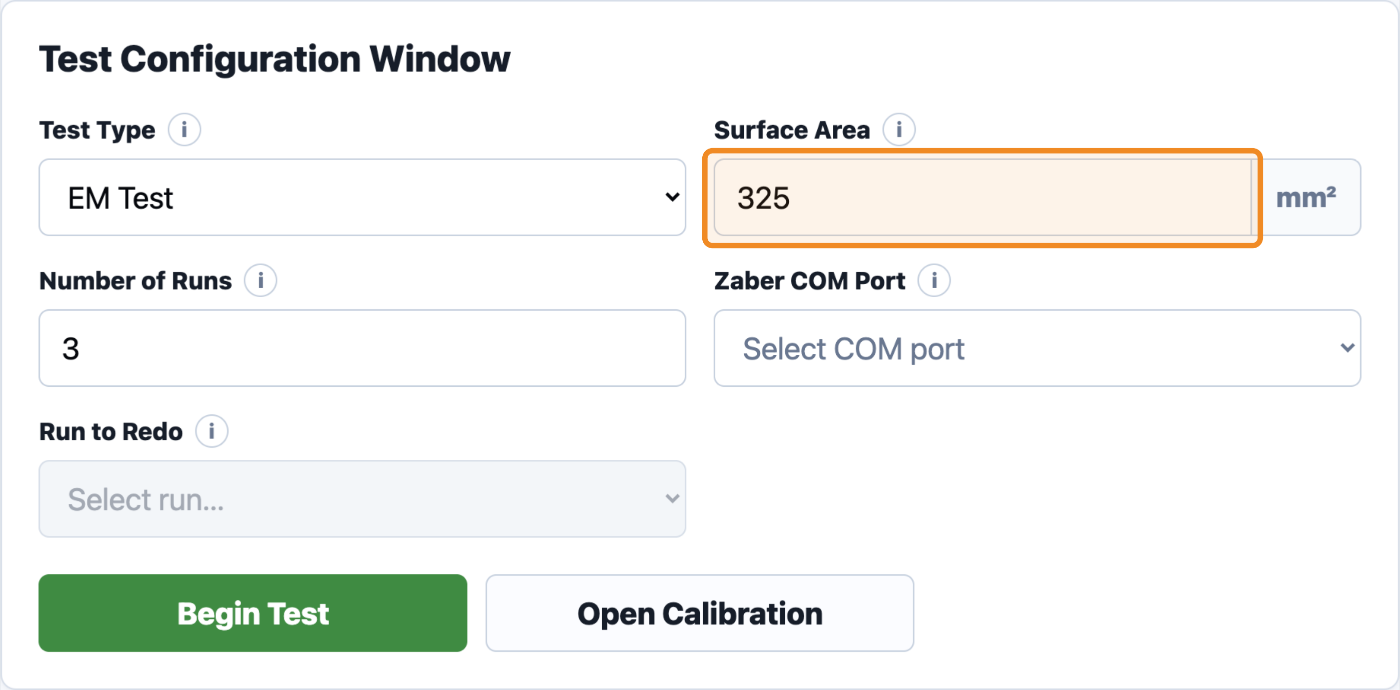
STORY 1.3.5

Surface Area Integer Field

> **USER STORY**
>
> *As a test fixture user, I want to enter the Ecoflex block surface area so that pressure calculations are accurate.*

ACCEPTANCE CRITERIA

- The user can enter a Surface Area value
- Default units are displayed as mm²
- The field only accepts positive numeric values
- The Surface Area value is passed into the analysis workflow



FEATURE 1.3 · TEST CONFIGURATION

STORY 1.3.6

MUST

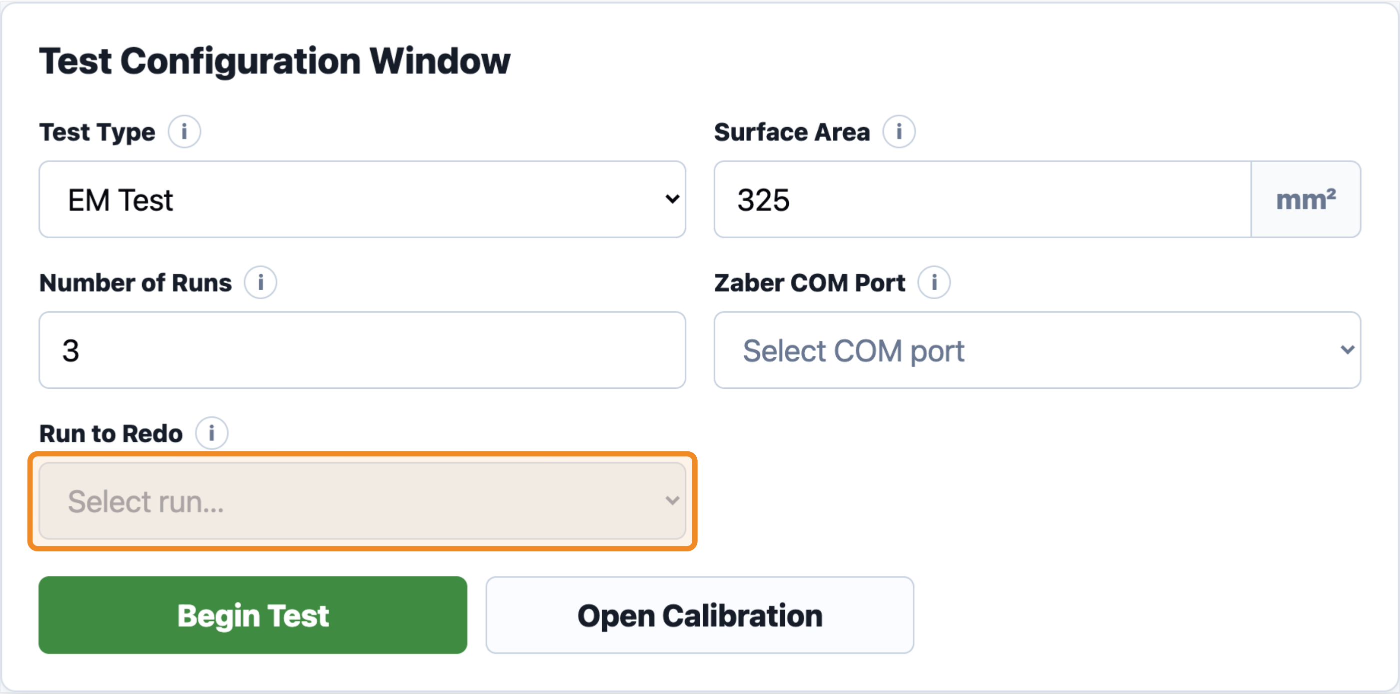
Run to Redo Integer Field

USER STORY

As a test fixture user, I want to enter the specific run number I want to redo so that the system repeats only that selected run. Available only when I choose to redo a run from an existing test.

ACCEPTANCE CRITERIA

- The Run to Redo field is disabled by default
- It becomes enabled only when the user selects "Redo a Specific Run" from the Existing Test Found popup
- In Redo Mode, Test Type is automatically set to EM Test and cannot be modified
- In Redo Mode, Number of Runs is automatically set to 1 and cannot be modified
- The user can enter a positive integer for Run to Redo
- When the test begins, the system repeats only the selected run



FEATURE 1.3 · TEST CONFIGURATION

MUST

STORY 1.3.7

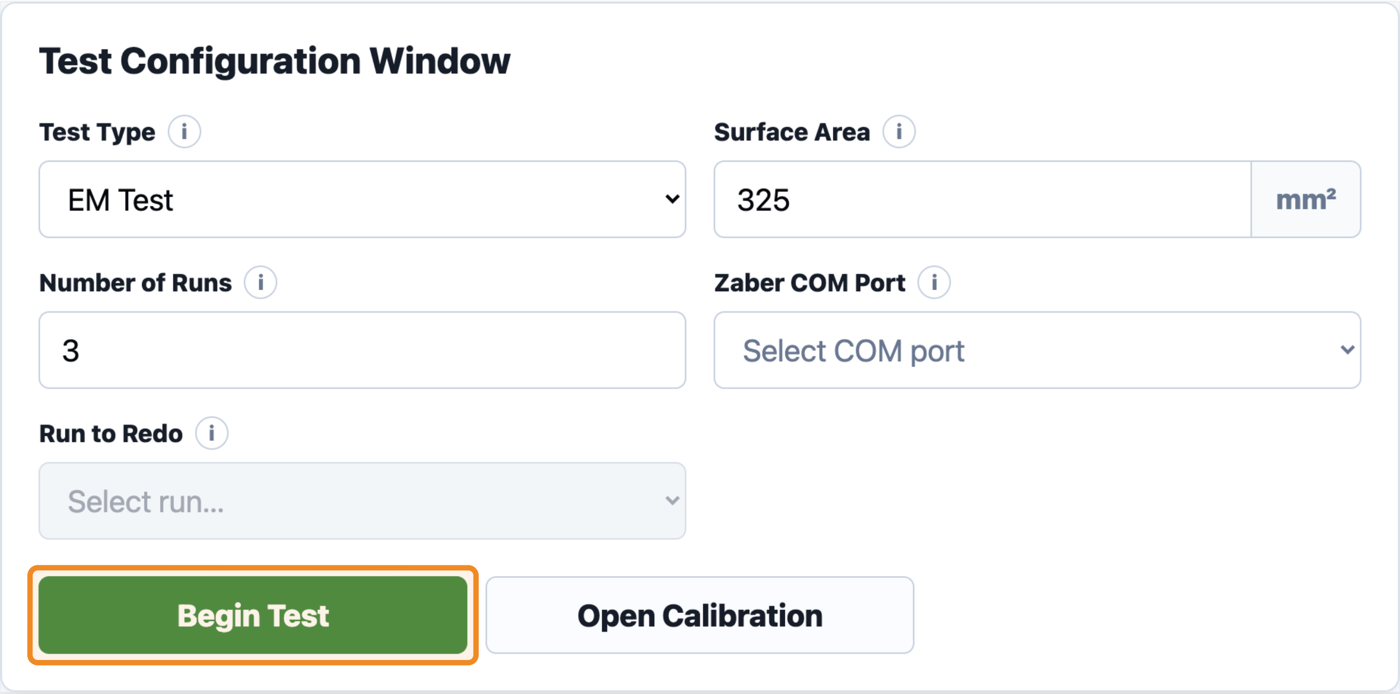
Begin Test Button

USER STORY

As a test fixture user, I want to click Begin Test so that the specified test can begin using the selected configuration settings.

ACCEPTANCE CRITERIA

- Clicking Begin Test starts the testing workflow and opens the appropriate Testing Window (EM/Shear/Manual/Cyclical)
- The system attempts hardware initialization before actuator movement begins
- If Test Type is blank: block, warn "Fill in Test Type before continuing"
- EM + Number of Runs blank or invalid: block with matching validation message
- EM / Manual / Cyclical + Zaber COM Port blank: block, warn "Fill in Zaber COM Port before continuing"
- Surface Area blank: block, warn "Fill in Surface Area before continuing"
- Surface Area invalid: block, warn "Surface Area must be a positive number"
- Redo Mode + Run to Redo blank/invalid: block with matching validation message
- If Redo Mode is active, the system begins only the selected run instead of the full test sequence



FEATURE 1.3 · TEST CONFIGURATION

MUST

STORY 1.3.8

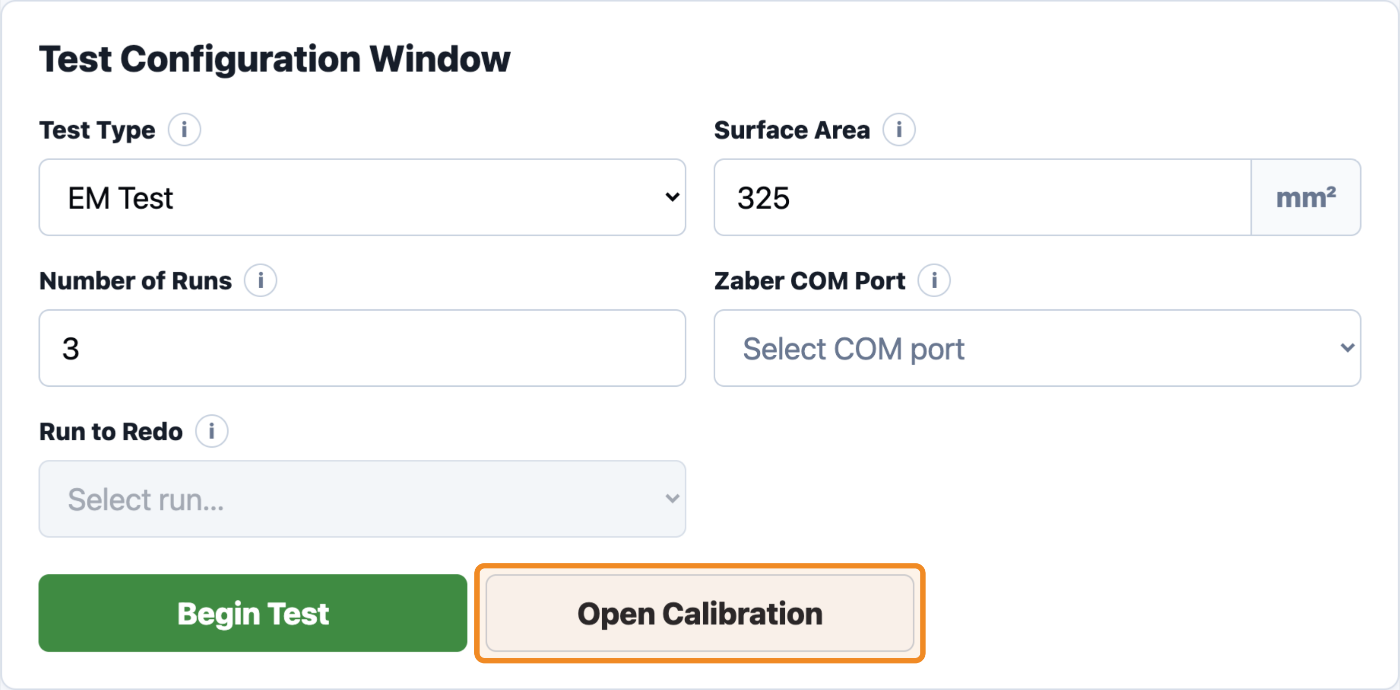
Open Calibration Button

USER STORY

As a test fixture user, I want to open the Calibration Window so that I can perform the Fuji Film calibration before testing.

ACCEPTANCE CRITERIA

- The user can click Open Calibration to open the Calibration Window popup



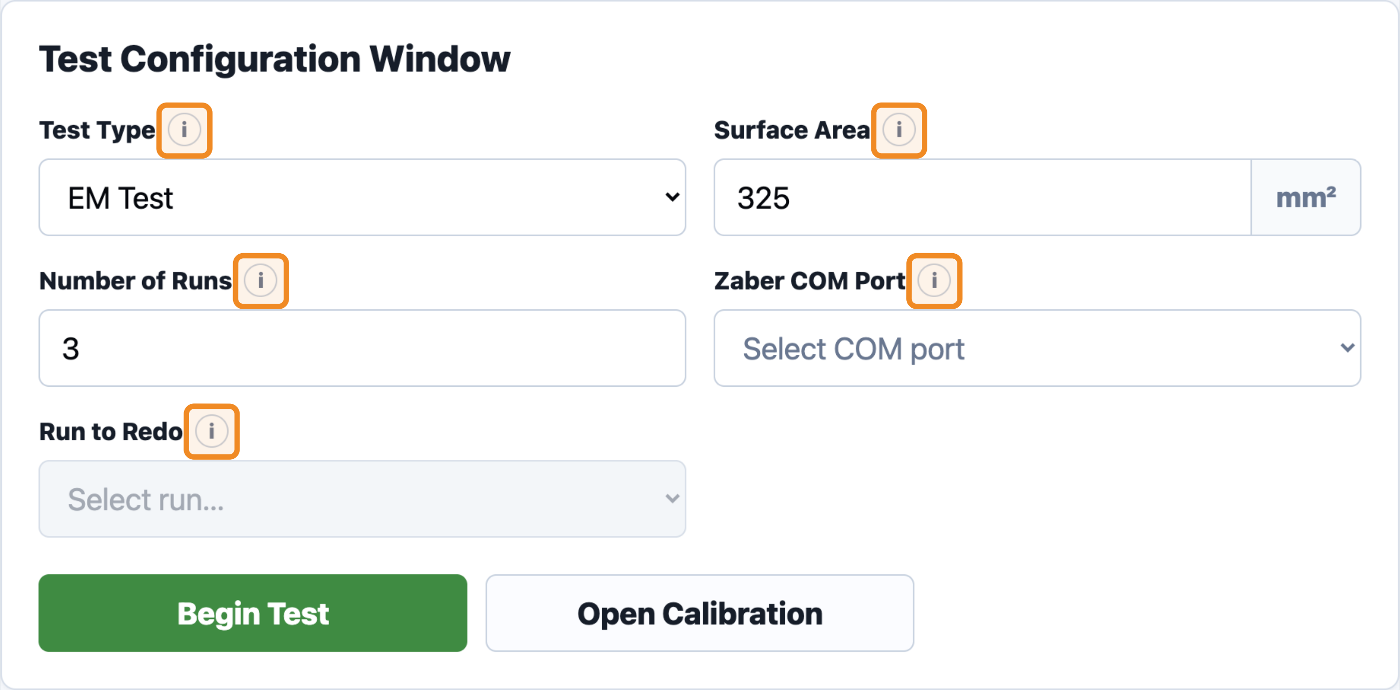
STORY 1.3.9

MUST

Info Icons

USER STORY

As a test fixture user, I want informational help icons throughout the window so that I can quickly understand fields, workflows, and testing terminology without needing external documentation.



INFO ICONS ON THIS PAGE

Test Type: EM Test performs force-controlled compression testing. Shear Test performs live shear-response testing. Manual Test allows direct actuator control. Fatigue Test repeatedly compresses the sensor over many cycles.

Surface Area: Surface Area of the Ecoflex block used to calculate pressure from force measurements.

Number of Runs: Number of EM test runs to complete.

Zaber COM Port: Select the port connected to the Zaber actuator.

Run to Redo: Redo only a specific EM run from an existing test folder. This is only available when ‘Redo a Specific Run’ is selected from the Existing Test Found popup.

FEATURE 1.4

Calibration Page

FEATURE 1.4 · CALIBRATION

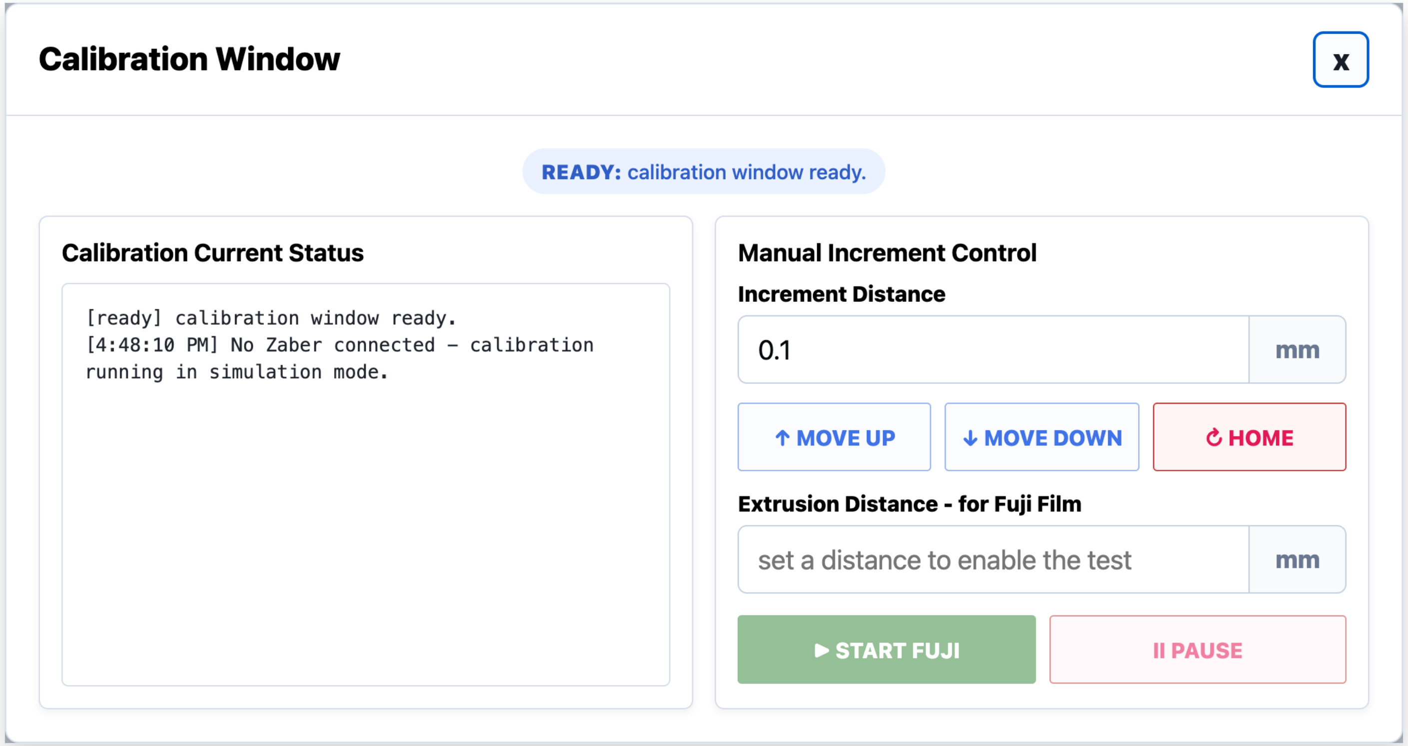
STORY 1.4.1

MUST

Calibration Configuration Window

USER STORY

As a test fixture user, I want a separate Calibration window so that I can safely perform the Fuji Film Test before beginning the specified test.



ACCEPTANCE CRITERIA

- Clicking the "Open Calibration" button opens a popup window titled: "Calibration for Fuji Film Test"

FEATURE 1.4 · CALIBRATION

STORY 1.4.2

MUST

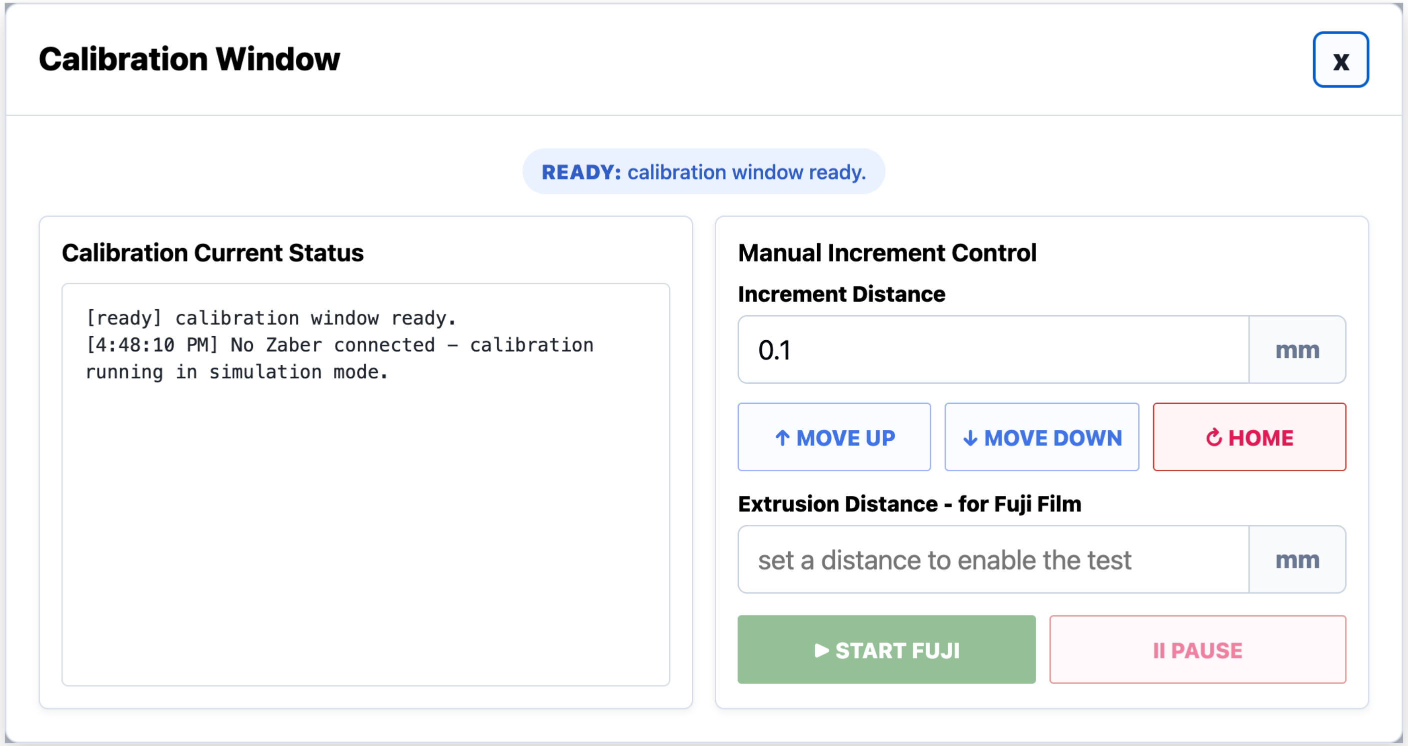
Current Status Display

USER STORY

As a test fixture user, I want to see the current actuator position during calibration so that I can safely adjust the actuator location.

ACCEPTANCE CRITERIA

- Current Status displays Position only, in mm
- Position updates when the user clicks Move Up, Move Down, or Home



FEATURE 1.4 · CALIBRATION

STORY 1.4.3

MUST

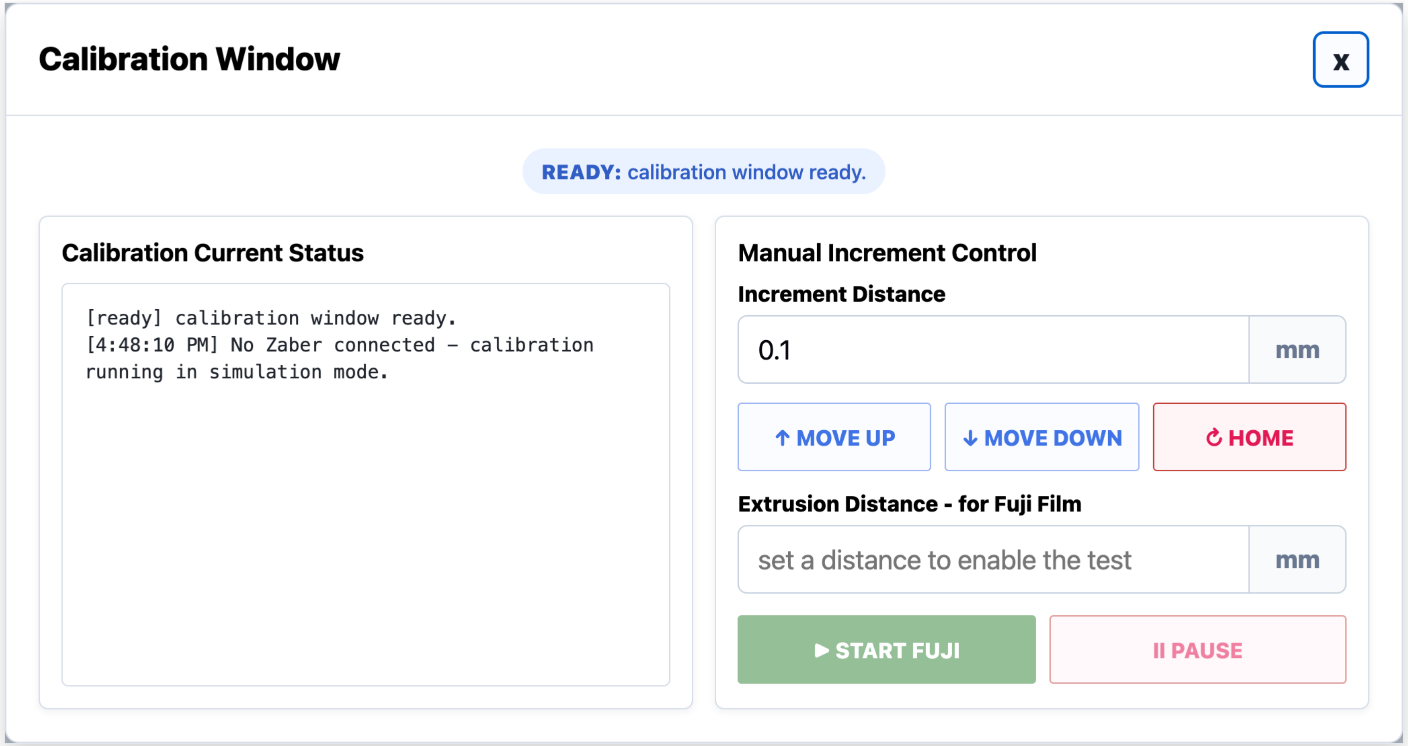
Manual Increment Control

USER STORY

As a test fixture user, I want to move the actuator in small increments so that I can fine-tune the actuator position safely.

ACCEPTANCE CRITERIA

- The user can enter an Increment Distance (mm)
- The user can click Move Up or Move Down; the actuator moves by the entered increment
- Move Up/Down update position step-by-step, with no elapsed-time output
- Increment Distance must remain within a specified maximum allowed range
- If blank: block, warn "Fill in Increment Distance before continuing"
- If exceeds range: block, warn "Increment Distance exceeds the maximum allowed range"
- If non-numeric: block, warn "Increment Distance must be a valid number"



FEATURE 1.4 · CALIBRATION

STORY 1.4.4

MUST

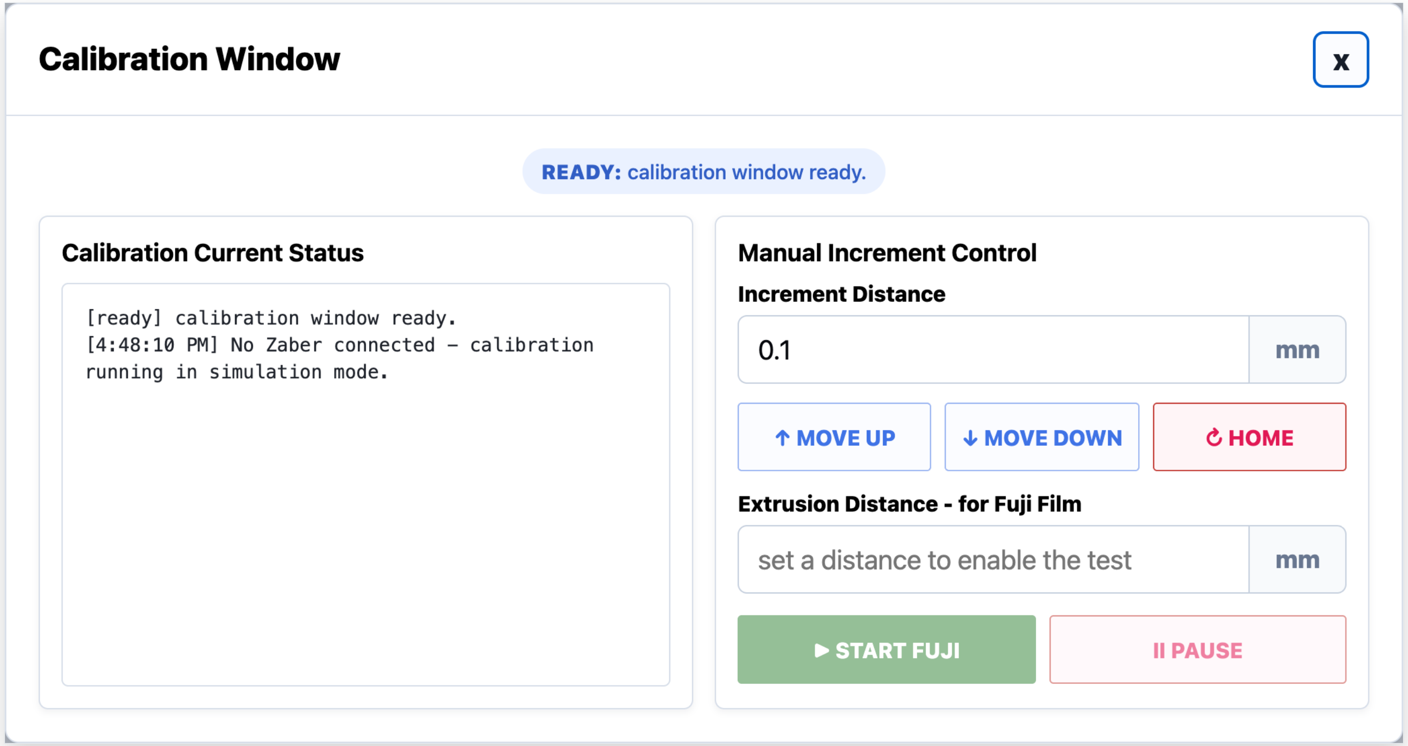
Reset to Home Button

USER STORY

As a test fixture user, I want a Home button so that I can safely return the actuator to its original position.

ACCEPTANCE CRITERIA

- Clicking Home returns the actuator to the home/original position
- The GUI displays that the actuator has reset successfully



FEATURE 1.4 · CALIBRATION

STORY 1.4.5

MUST

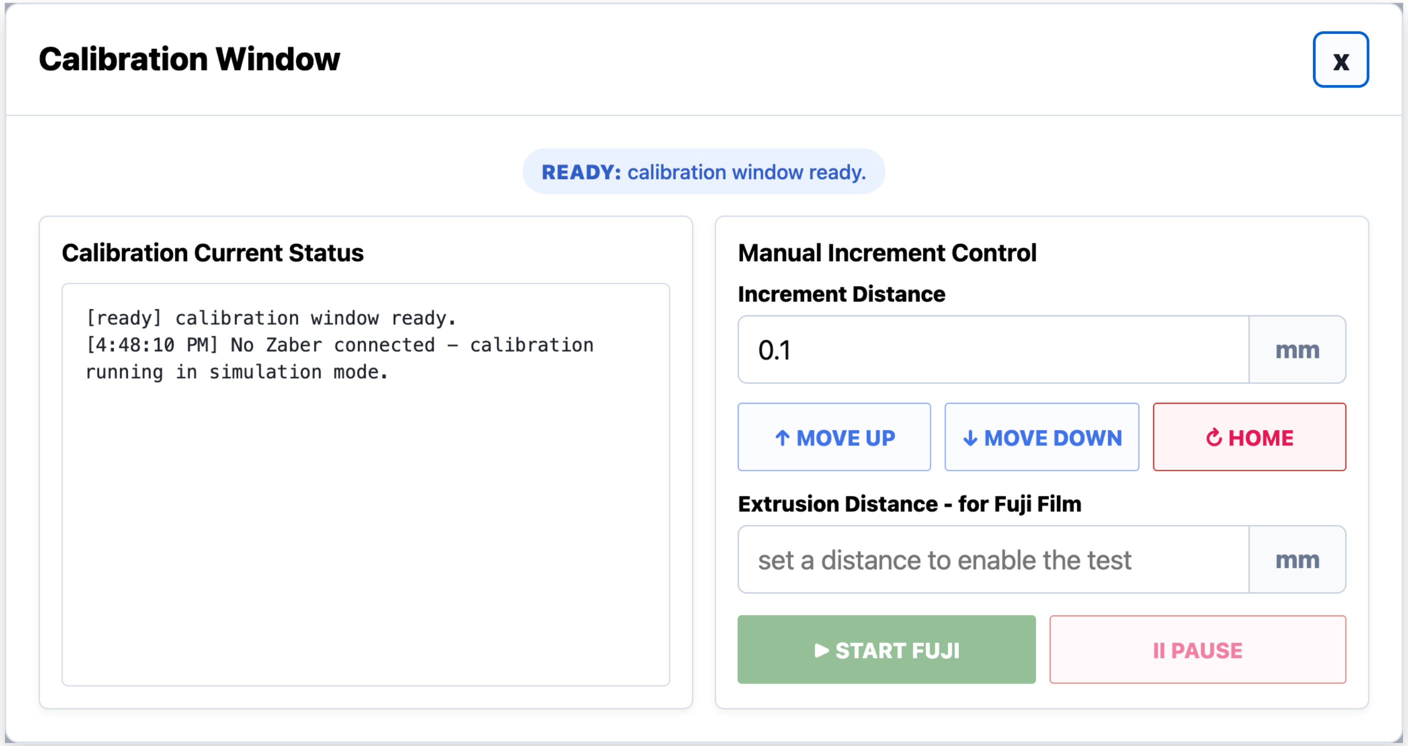
Calibration Test Button

USER STORY

As a test fixture user, I want to run a Calibration Test so that I can calibrate the load cell and actuator setup.

ACCEPTANCE CRITERIA

- The user can click "Begin Calibration Test"
- The actuator moves downward until the target force is reached
- Target Force = 20 N
- Motion stops when the target force is reached



FEATURE 1.4 · CALIBRATION

MUST

STORY 1.4.6

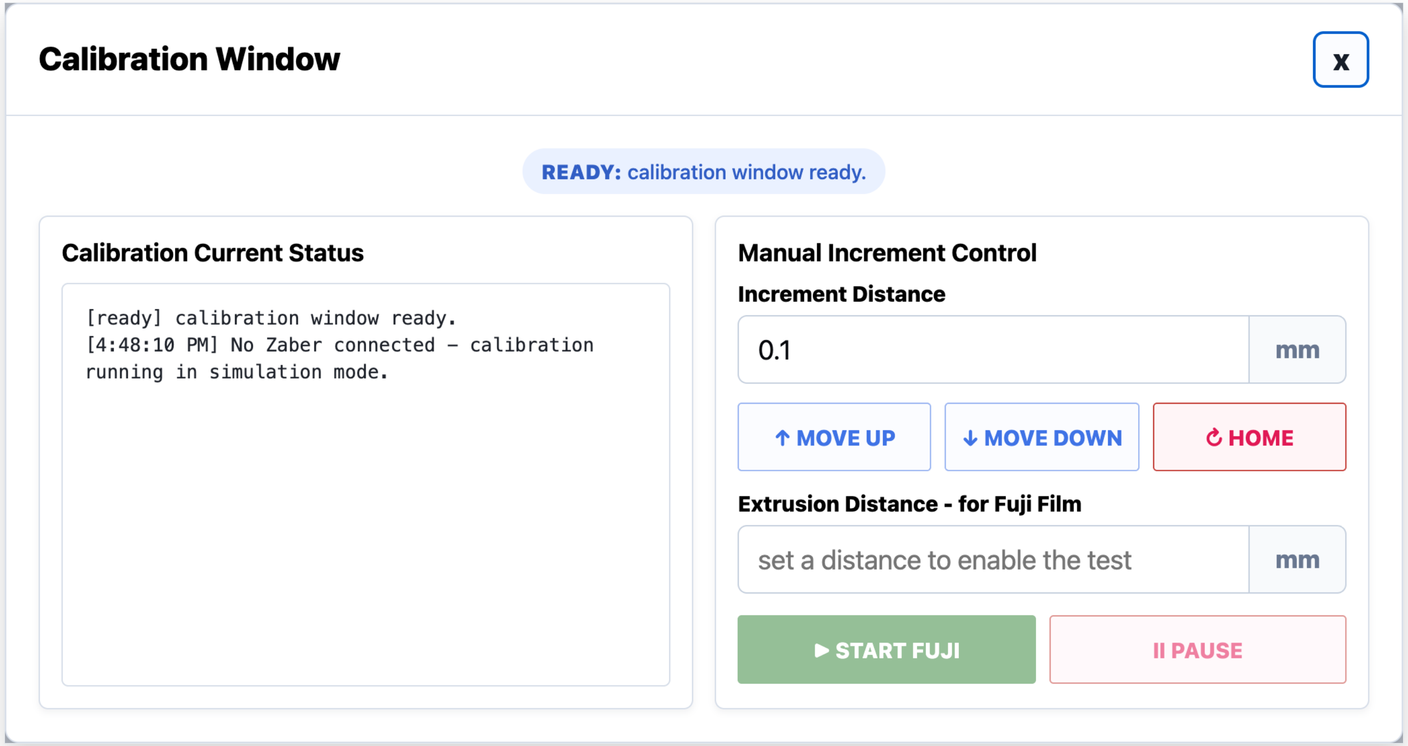
Calibration Test Output Box

USER STORY

As a test fixture user, I want Calibration Test force readings displayed in a table-like format so that I can clearly track force over time during calibration.

ACCEPTANCE CRITERIA

- When the test starts, the output box displays: "Calibration Test started"
- Output box includes column headers: Time (s) | Force (N)
- While running, force values are displayed in the output box with time values
- On reaching target force, the GUI displays under the button: "Calibration Test Completed Successfully"
- If the test stops due to a force spike or error, an error status is displayed instead of a success message



FEATURE 1.4 · CALIBRATION

STORY 1.4.7

MUST

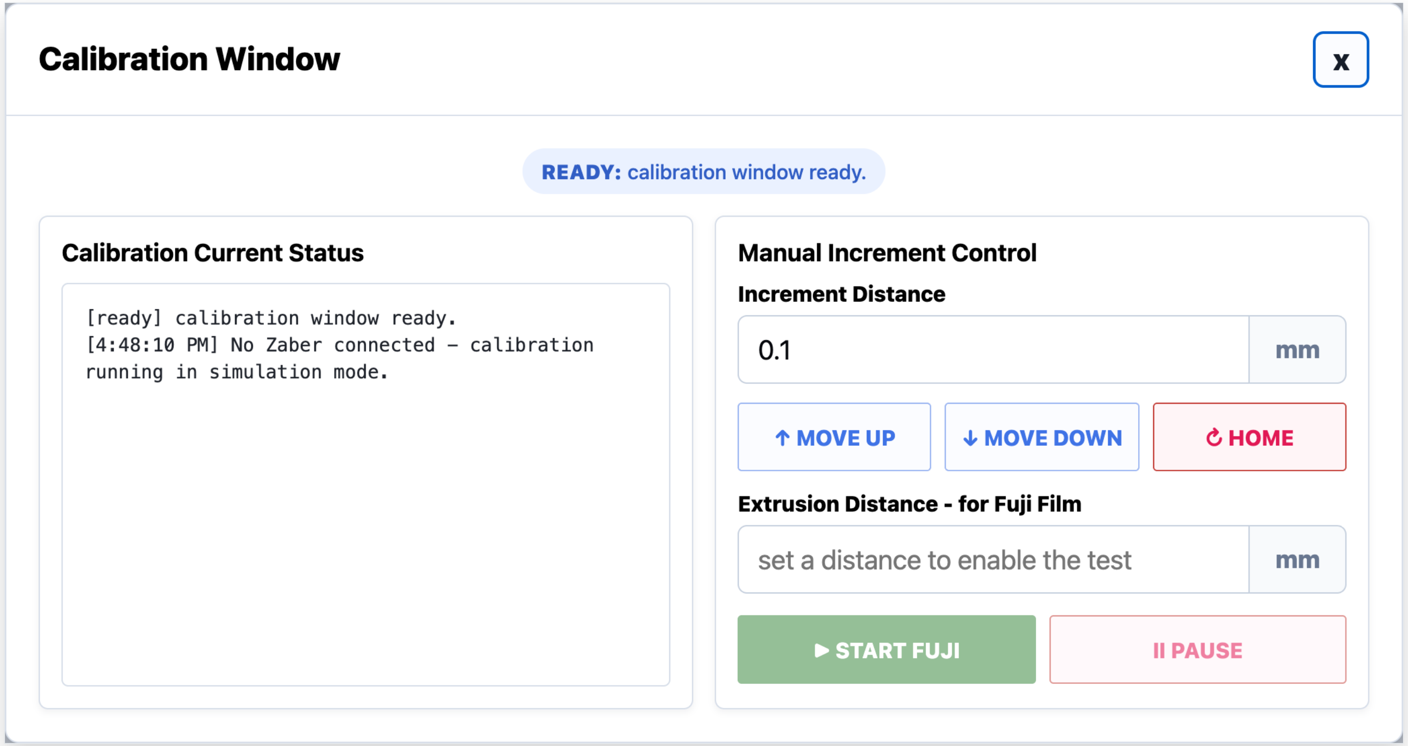
Exit Button

USER STORY

As a test fixture user, I want an Exit button on the Calibration popup window so that I can close the Calibration Window.

ACCEPTANCE CRITERIA

- The Calibration popup window includes an Exit button
- Clicking the Exit button closes only the Calibration popup window
- The Exit button is disabled while the Calibration Test is actively running
- Once the Calibration Test is complete or stopped, the Exit button becomes enabled again



FEATURE 1.5

EM Test Page

FEATURE 1.5 · EM TEST

STORY 1.5.1

MUST

EM Testing Window

USER STORY

As a test fixture user, I want the EM test to open in a dedicated EM Testing Window so that I can monitor and control the active EM test.

ACCEPTANCE CRITERIA

- Clicking Begin Test with Test Type = EM Test opens a new window titled: "EM Testing Window"

FEATURE 1.5 · EM TEST

STORY 1.5.2

MUST

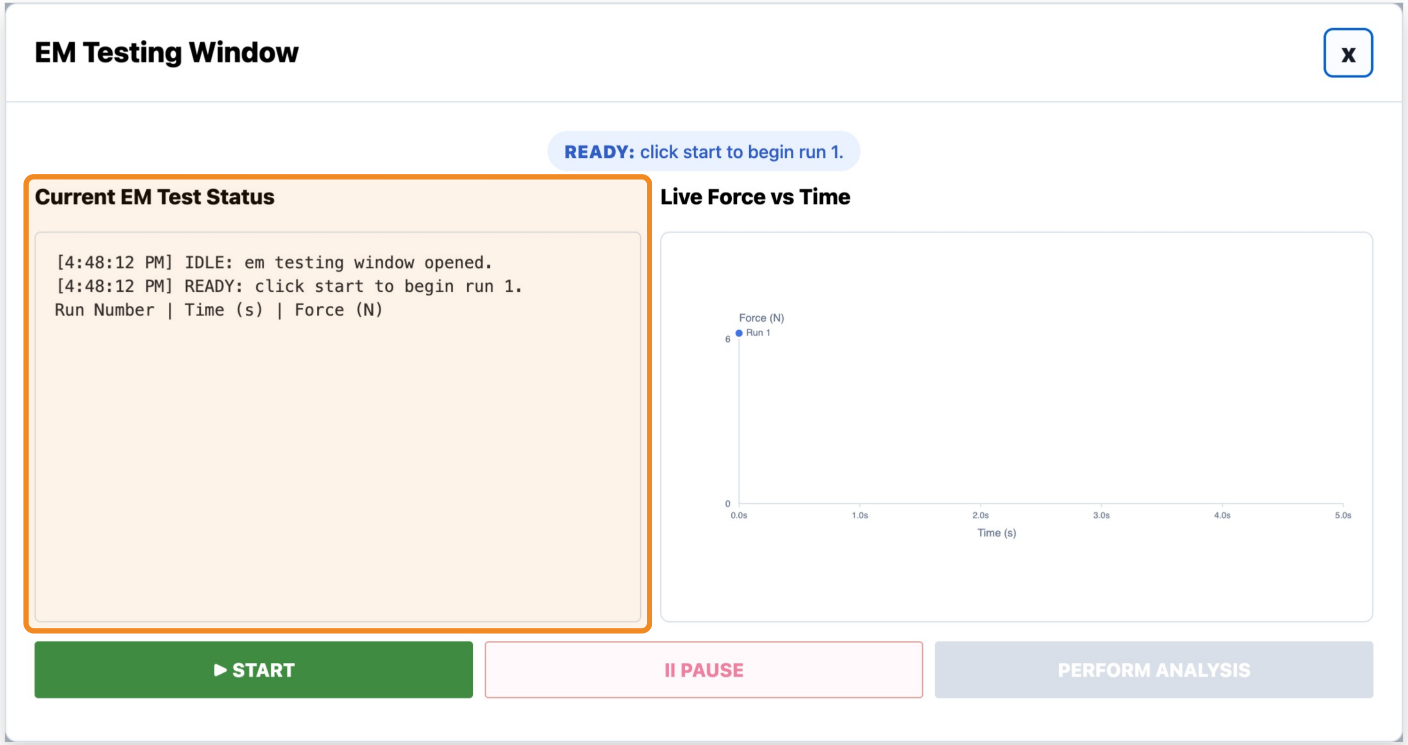
Current EM Test Output Box

USER STORY

As a test fixture user, I want to view the current EM test status so that I can monitor the actuator and testing workflow in real time.

ACCEPTANCE CRITERIA

- The EM Testing Window includes a Current EM Test Status display area
- The output box includes column headers: Run Number | Time (s) | Force (N)
- Force readings are displayed in chronological order (e.g., Run 1 | 4.802 s | 4.941 N)
- Status display updates with key states: IDLE, READY, CONNECTING, RUNNING, PAUSED_RESET_REQUIRED, BETWEEN_RUNS_PAUSED, COMPLETED, ERROR



FEATURE 1.5 · EM TEST

STORY 1.5.3

MUST

Start Button

USER STORY

As a test fixture user, I want to click START so that the EM test begins using the selected configuration settings.

ACCEPTANCE CRITERIA

- Clicking the Start Button begins the EM testing workflow
- The system attempts hardware initialization before movement begins
- The actuator does not move unless initialization succeeds
- Once testing begins, START becomes disabled

FEATURE 1.5 · EM TEST

MUST

STORY 1.5.4

Pause Button

USER STORY

As a test fixture user, I want to pause the current EM test safely so that actuator motion stops and the current run can be repeated.

ACCEPTANCE CRITERIA

- Clicking the Pause Button immediately stops actuator motion
- The actuator returns to the home/original position
- The current run is marked incomplete
- The current run number does not increment
- The current run must be repeated before continuing

FEATURE 1.5 · EM TEST

STORY 1.5.5

MUST

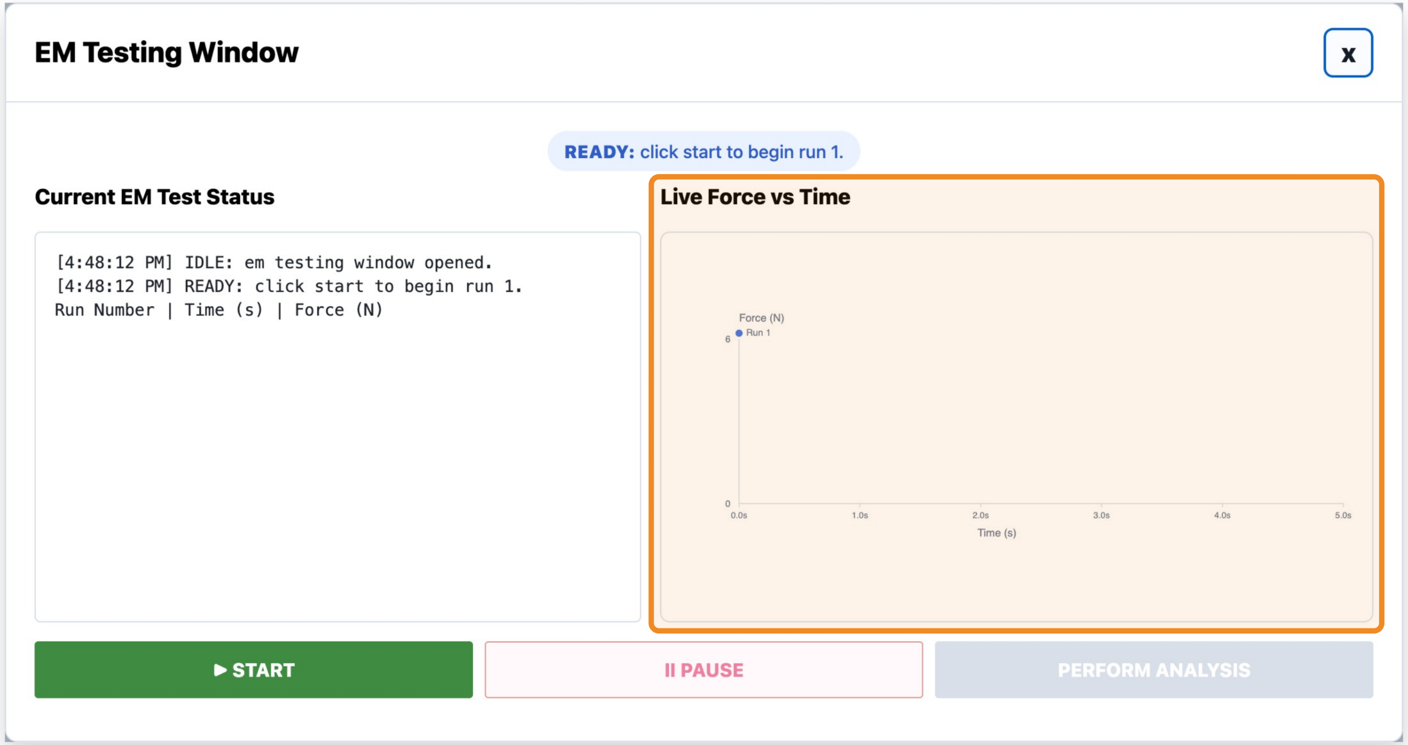
Live Force vs Time Graph

USER STORY

As a test fixture user, I want to view a live Force vs Time graph so that I can visually monitor force trends during the EM test.

ACCEPTANCE CRITERIA

- The EM Testing Window includes a Live Force vs Time graph
- The graph updates in real time while the EM test is running
- The graph displays force in N over time in seconds
- The graph resets appropriately between runs



FEATURE 1.5 · EM TEST

MUST

STORY 1.5.6

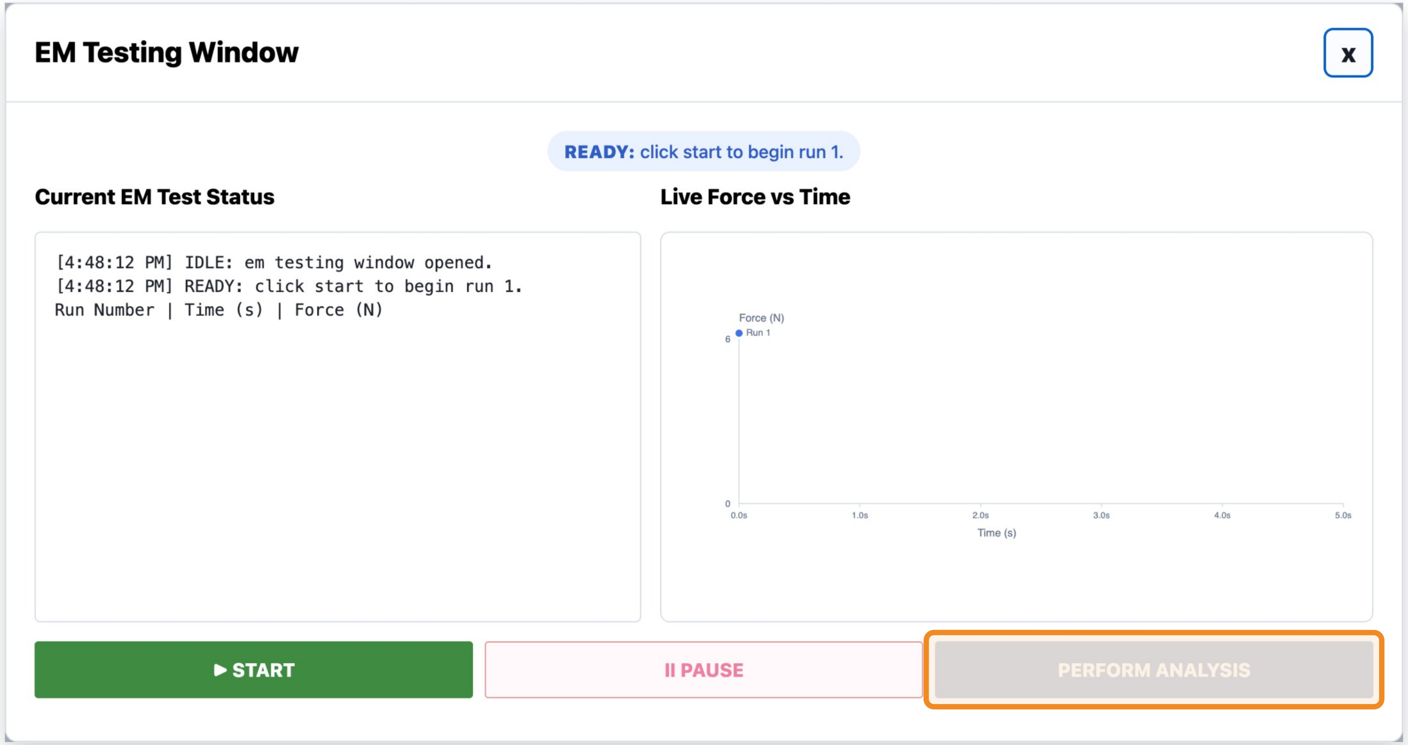
Perform Analysis Button

USER STORY

As a test fixture user, I want to click a Perform Analysis button so that the software can process the completed test data and generate summary statistics and plots.

ACCEPTANCE CRITERIA

- The testing window includes a Perform Analysis button
- Perform Analysis is disabled until the EM test has completed successfully
- Clicking Perform Analysis should show a loading/progress state before opening the EM Analysis Page



FEATURE 1.6

EM Analysis Page

FEATURE 1.6 · EM ANALYSIS

MUST

STORY 1.6.1

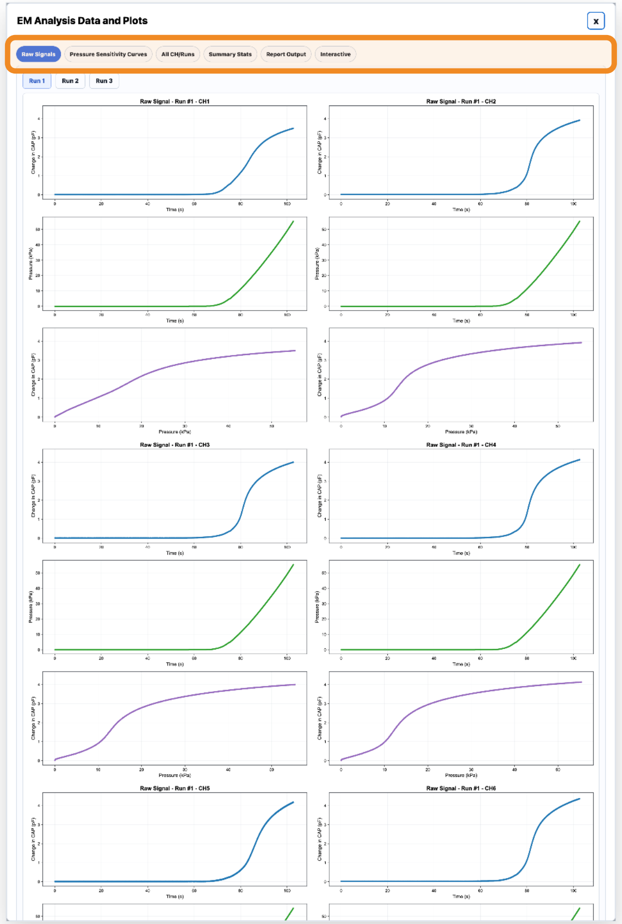
EM Analysis Window

USER STORY

As a test fixture user, I want analysis results to open in a dedicated results window so that I can review processed EM test results after testing completes.

ACCEPTANCE CRITERIA

- After analysis completes, a new window opens titled: "EM Analysis Data and Plots"
- The window displays summary statistics and plots generated from the selected test data
- Users can click between tabs to quickly navigate different EM analysis outputs



FEATURE 1.6 · EM ANALYSIS

STORY 1.6.2

MUST

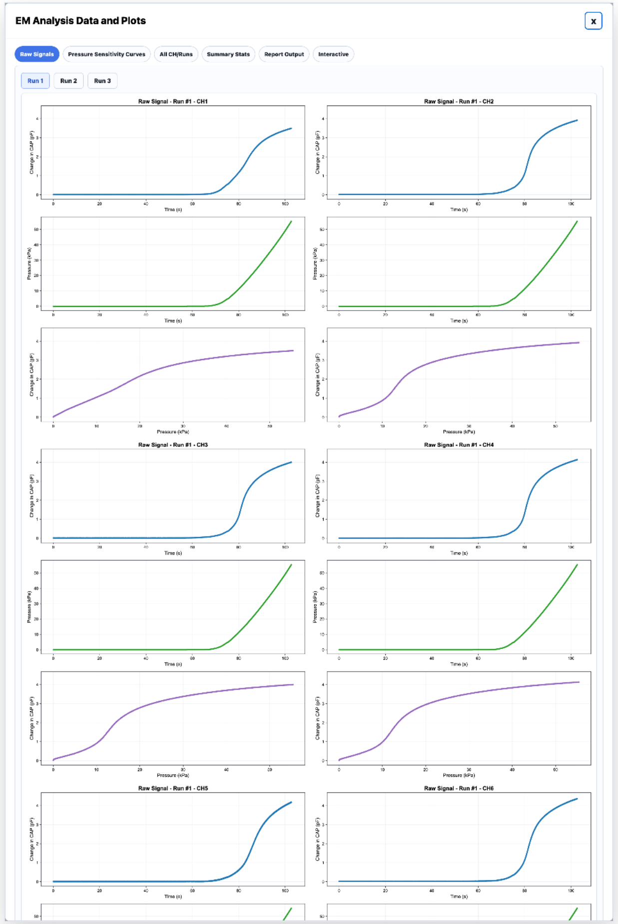
Raw Signals Tab

USER STORY

As a test fixture user, I want to view raw EM test signals so that I can inspect channel behavior before processed analysis is applied.

ACCEPTANCE CRITERIA

- The Raw Signals tab displays raw channel plots for the selected run
- The tab displays all EM channels simultaneously
- Users can switch between runs



FEATURE 1.6 · EM ANALYSIS

MUST

STORY 1.6.3

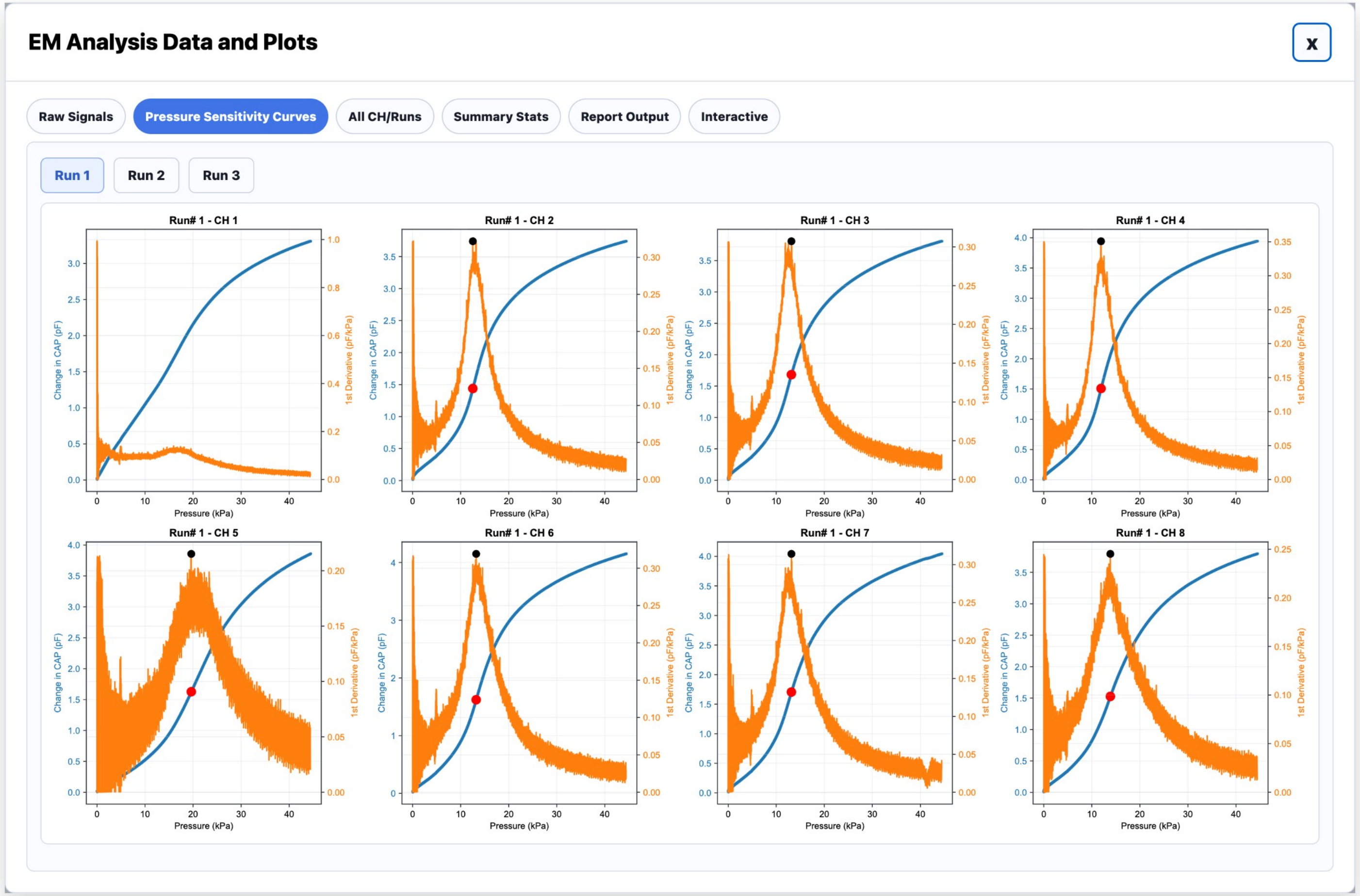
Pressure Sensitivity Curves Tab

USER STORY

As a test fixture user, I want to view pressure sensitivity curves so that I can evaluate how capacitance changes relative to pressure.

ACCEPTANCE CRITERIA

- The Pressure Sensitivity Curves tab displays processed pressure sensitivity plots for all channels
- Each channel plot includes the calculated pressure sensitivity value
- Inflection points are visually highlighted on the plots



FEATURE 1.6 · EM ANALYSIS

STORY 1.6.4

MUST

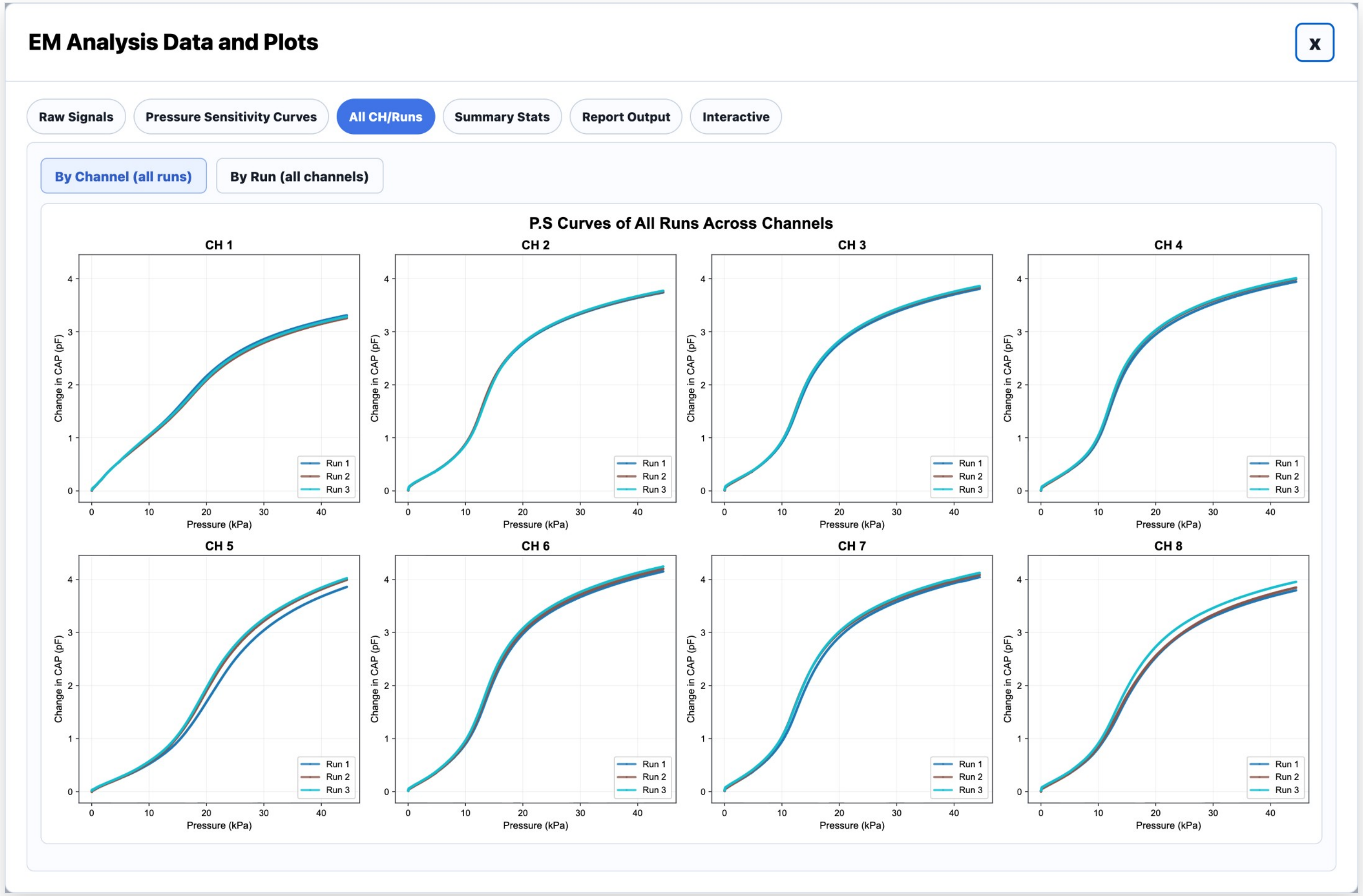
All Channels / Runs Comparison Tab

USER STORY

As a test fixture user, I want to compare all runs and channels simultaneously so that I can evaluate consistency and repeatability across the sensor.

ACCEPTANCE CRITERIA

- The All CH/Runs tab includes comparison views for: By Channel (all runs), By Run (all channels)
- Each plot overlays multiple runs for direct comparison



FEATURE 1.6 · EM ANALYSIS

STORY 1.6.5

MUST

Summary Statistics Tab

USER STORY

As a test fixture user, I want to view summary statistics for all EM channels so that I can evaluate sensor quality quantitatively.

ACCEPTANCE CRITERIA

- The Summary Statistics tab displays a table of calculated metrics for all channels
- Metrics include Mean, Std, COV, Min, and Max for: PS at Inflection, Max kPa, Mean Max CAP
- Every displayed average metric also includes standard deviation
- Every displayed maximum metric also includes minimum values

EM Analysis Data and Plots x

Raw Signals | Pressure Sensitivity Curves | All CH/Runs | Summary Stats | Report Output | Interactive

Summary Stat	CH 1	CH 2	CH 3	CH 4	CH 5	CH 6	CH 7	CH 8
Mean PS at Inflection	0.157	0.324	0.315	0.358	0.230	0.323	0.311	0.246
Std PS at Inflection	0.000	0.006	0.007	0.011	0.013	0.003	0.003	0.009
CoV PS at Inflection	0.00%	1.70%	2.07%	2.98%	5.64%	1.04%	1.08%	3.83%
Min PS at Inflection	0.157	0.320	0.307	0.351	0.215	0.319	0.308	0.237
Max PS at Inflection	0.157	0.330	0.320	0.370	0.239	0.326	0.314	0.256
Mean Max kPa	2.048 kPa	12.695 kPa	12.875 kPa	11.656 kPa	19.215 kPa	13.351 kPa	12.676 kPa	13.357 kPa
Std Max kPa	0.000 kPa	0.155 kPa	0.246 kPa	0.229 kPa	0.351 kPa	0.112 kPa	0.495 kPa	0.508 kPa
CoV Max kPa	0.00%	1.22%	1.91%	1.97%	1.82%	0.84%	3.90%	3.81%
Min Max kPa	2.048 kPa	12.536 kPa	12.697 kPa	11.483 kPa	19.003 kPa	13.231 kPa	12.167 kPa	12.839 kPa
Max Max kPa	2.048 kPa	12.846 kPa	13.156 kPa	11.916 kPa	19.620 kPa	13.451 kPa	13.156 kPa	13.855 kPa
Mean Max CAP	3.566 pF	4.033 pF	4.129 pF	4.264 pF	4.424 pF	4.524 pF	4.405 pF	4.190 pF
Std Max CAP	0.024 pF	0.018 pF	0.027 pF	0.031 pF	0.079 pF	0.041 pF	0.038 pF	0.077 pF
CoV Max CAP	0.68%	0.44%	0.66%	0.73%	1.79%	0.92%	0.85%	1.85%
Min Max CAP	3.543 pF	4.016 pF	4.100 pF	4.229 pF	4.334 pF	4.481 pF	4.368 pF	4.120 pF
Max Max CAP	3.591 pF	4.051 pF	4.154 pF	4.288 pF	4.483 pF	4.564 pF	4.443 pF	4.274 pF

FEATURE 1.6 · EM ANALYSIS

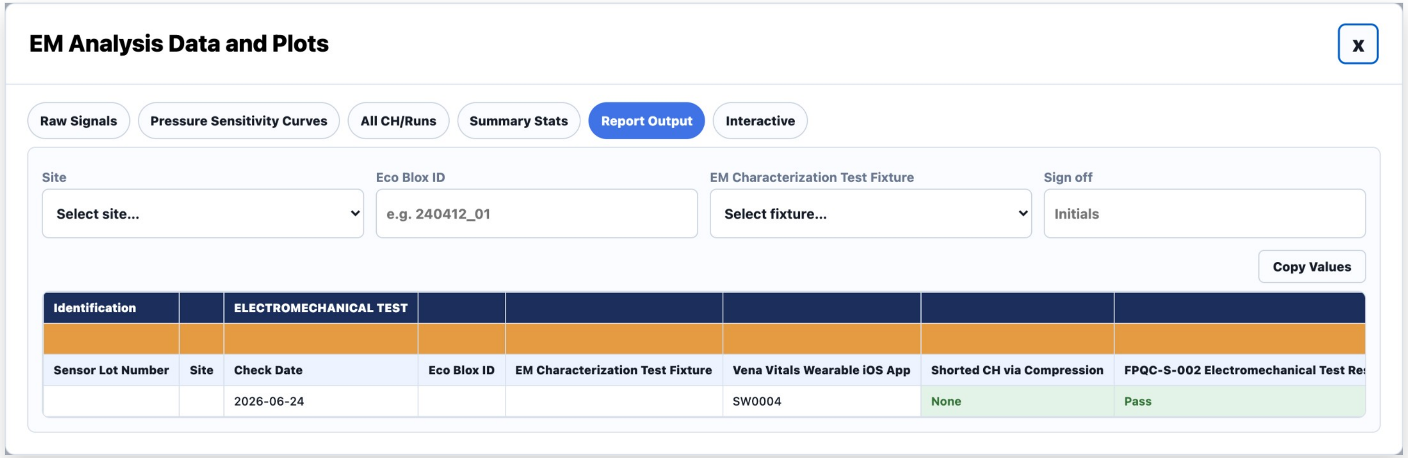
MUST

STORY 1.6.6

Report Output Tab

USER STORY

As a test fixture user, I want a report-style output tab so that I can easily copy, export, and review finalized EM testing results.



ACCEPTANCE CRITERIA

- The Report Output tab displays a formatted summary report table
- Includes: Lot Number / Sensor ID, Electromechanical Standards Status, Software Version, Date Tested, Fabrication Type, Eco Block ID, Run Number Tested, Run Number Counted
- Shorted Channel Status: the analysis pipeline automatically checks for potential channel shorting behavior
- The tab includes a "Copy Values" button
- Clicking "Copy Values" copies only the data row (not the headers) to quickly transfer results into reports

FEATURE 1.7

Shear Test Page

FEATURE 1.7 · SHEAR TEST

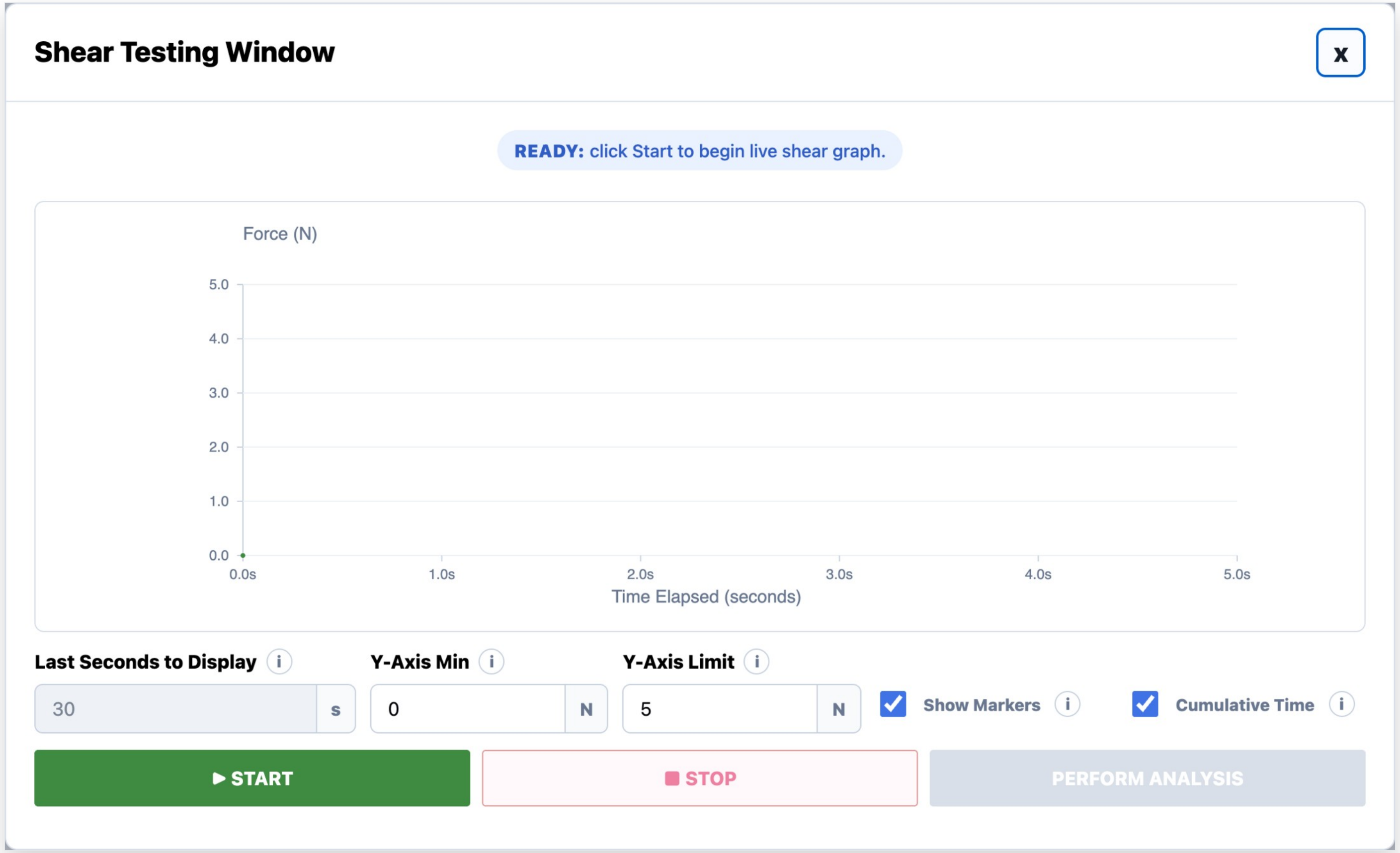
MUST

STORY 1.7.1

Shear Testing Window

USER STORY

As a test fixture user, I want the Shear Test to open in a dedicated Shear Testing Window so that I can monitor and control the active Shear test.



ACCEPTANCE CRITERIA

- Clicking Begin Test with Test Type = Shear Test opens a new window titled: "Shear Testing Window"

FEATURE 1.7 · SHEAR TEST

MUST

STORY 1.7.2

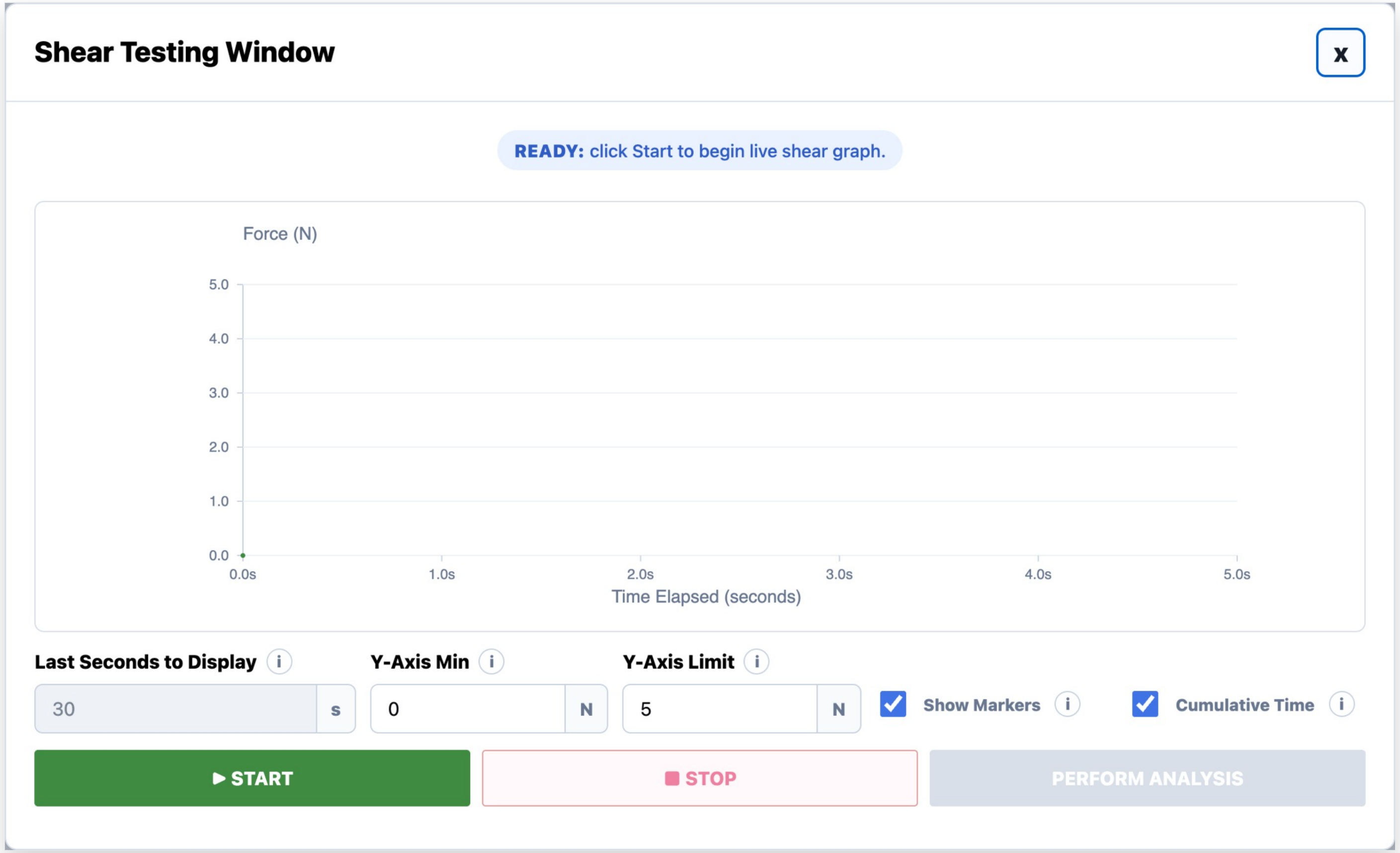
Live Shear Graph Display

USER STORY

As a test fixture user, I want to view the live Shear graph so that I can monitor force behavior in real time during testing.

ACCEPTANCE CRITERIA

- Graph X-axis is labeled: "Time Elapsed (seconds)"
- Graph Y-axis is labeled: "Force (N)"
- The graph updates in real time during the Shear test
- The graph displays live force data from the load cell
- Hovering over the graph displays the current time and force values (e.g., Time: 0.000 s, Force: 0.000 N)



FEATURE 1.7 · SHEAR TEST

STORY 1.7.3

MUST

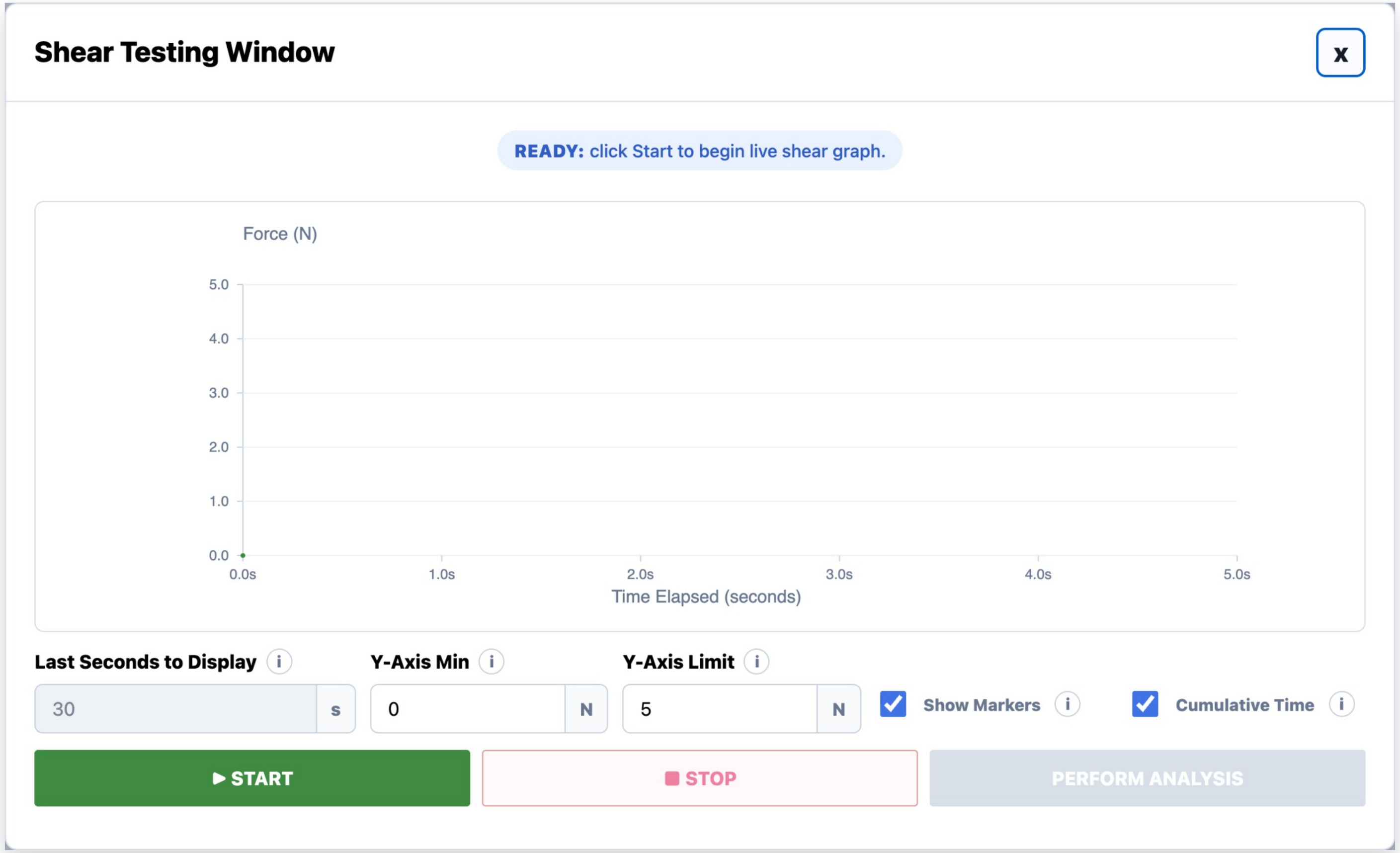
Graph Display Controls

USER STORY

As a test fixture user, I want to control the graph display settings so that I can customize how the Shear test data is visualized.

ACCEPTANCE CRITERIA

- The user can edit the "Last Seconds to Display" field
- The user can edit the "Y-Axis Min" field
- The user can edit the "Y-Axis Limit" field
- The user can toggle the "Show Markers" checkbox to display visible graph markers
- The user can toggle the "Cumulative Time" checkbox
- Changing these settings updates the graph display



FEATURE 1.7 · SHEAR TEST

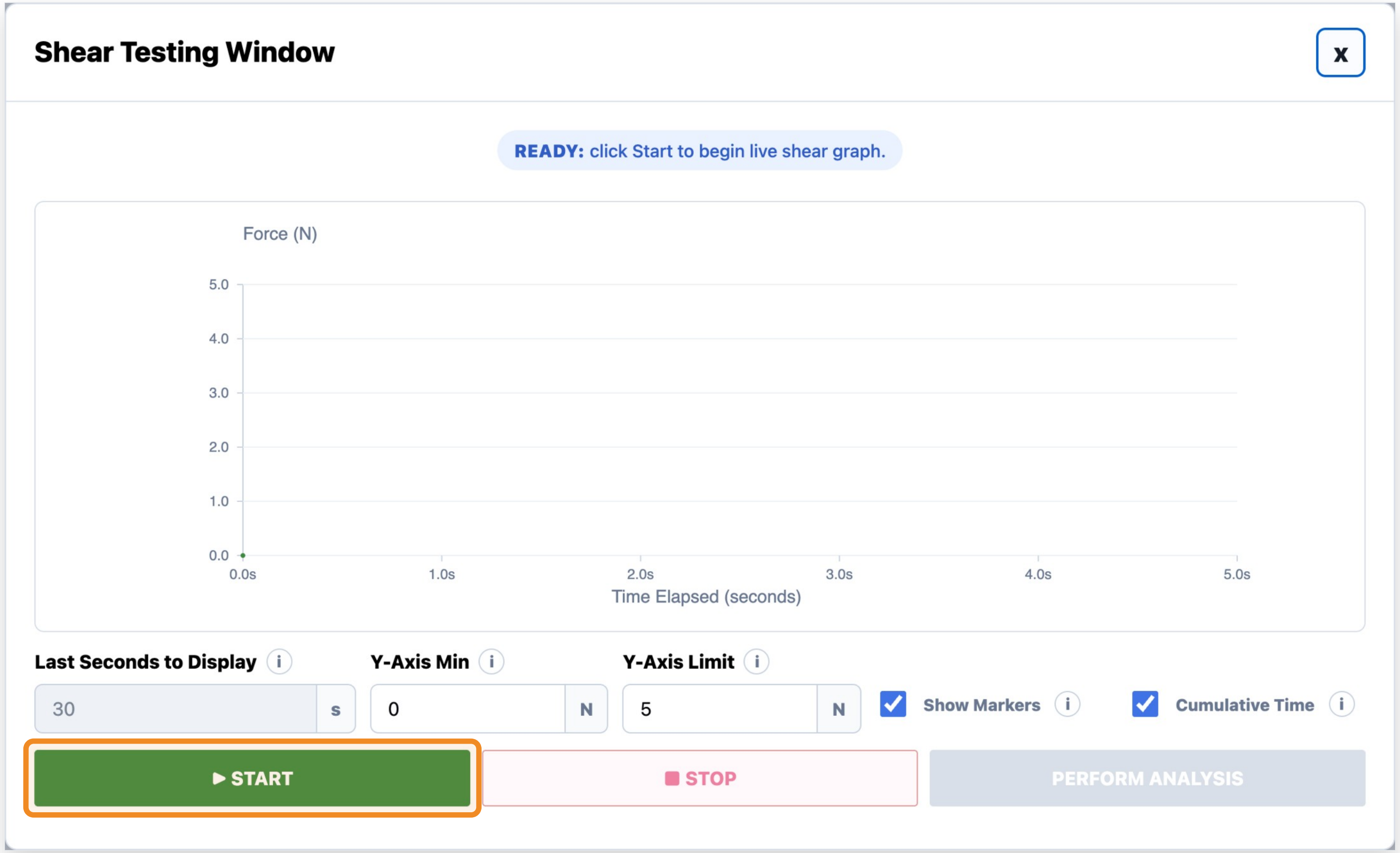
STORY 1.7.4

MUST

Start Button

USER STORY

As a test fixture user, I want to click START so that the live Shear test begins.



ACCEPTANCE CRITERIA

- Clicking START begins the Shear testing workflow
- The graph begins updating in real time
- START becomes disabled while the test is running, and after PAUSE when the actuator is moving
- Clicking Start after a paused/incomplete run begins a new active run dataset

FEATURE 1.7 · SHEAR TEST

MUST

STORY 1.7.5

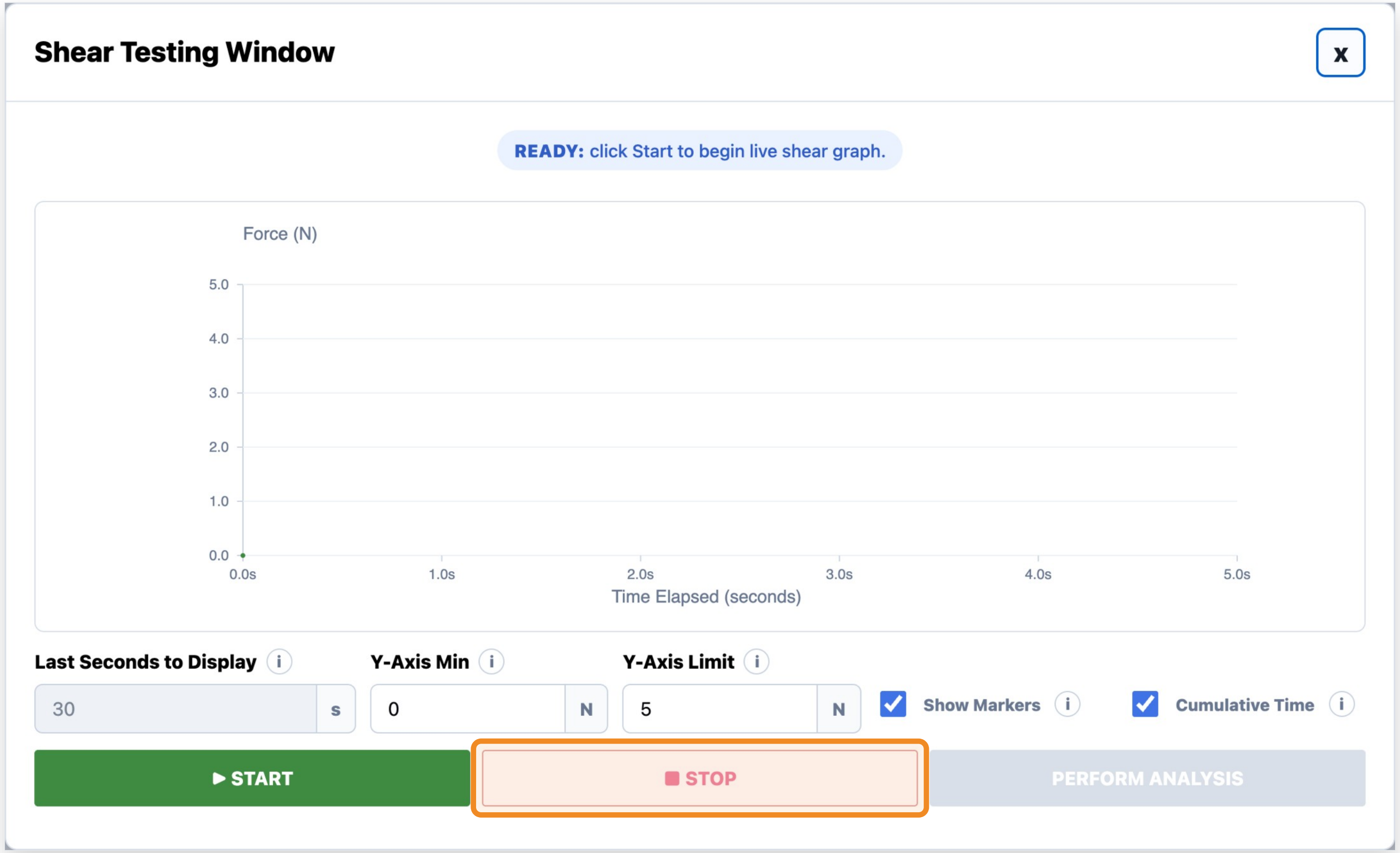
Stop Button

USER STORY

As a test fixture user, I want to stop the Shear test so that I can stop graph updates and actuator movement safely.

ACCEPTANCE CRITERIA

- Clicking STOP stops the active Shear workflow
- START becomes enabled after Stop is clicked.
- PERFORM ANALYSIS becomes enabled only after START has been clicked, valid data has been collected, and STOP has been clicked.
- If the user clicks Start again after stopping, the previous stopped/incomplete graph data is replaced by the newest run dataset.
- Perform Analysis uses the newest valid stopped run dataset.



FEATURE 1.7 · SHEAR TEST

STORY 1.7.6

MUST

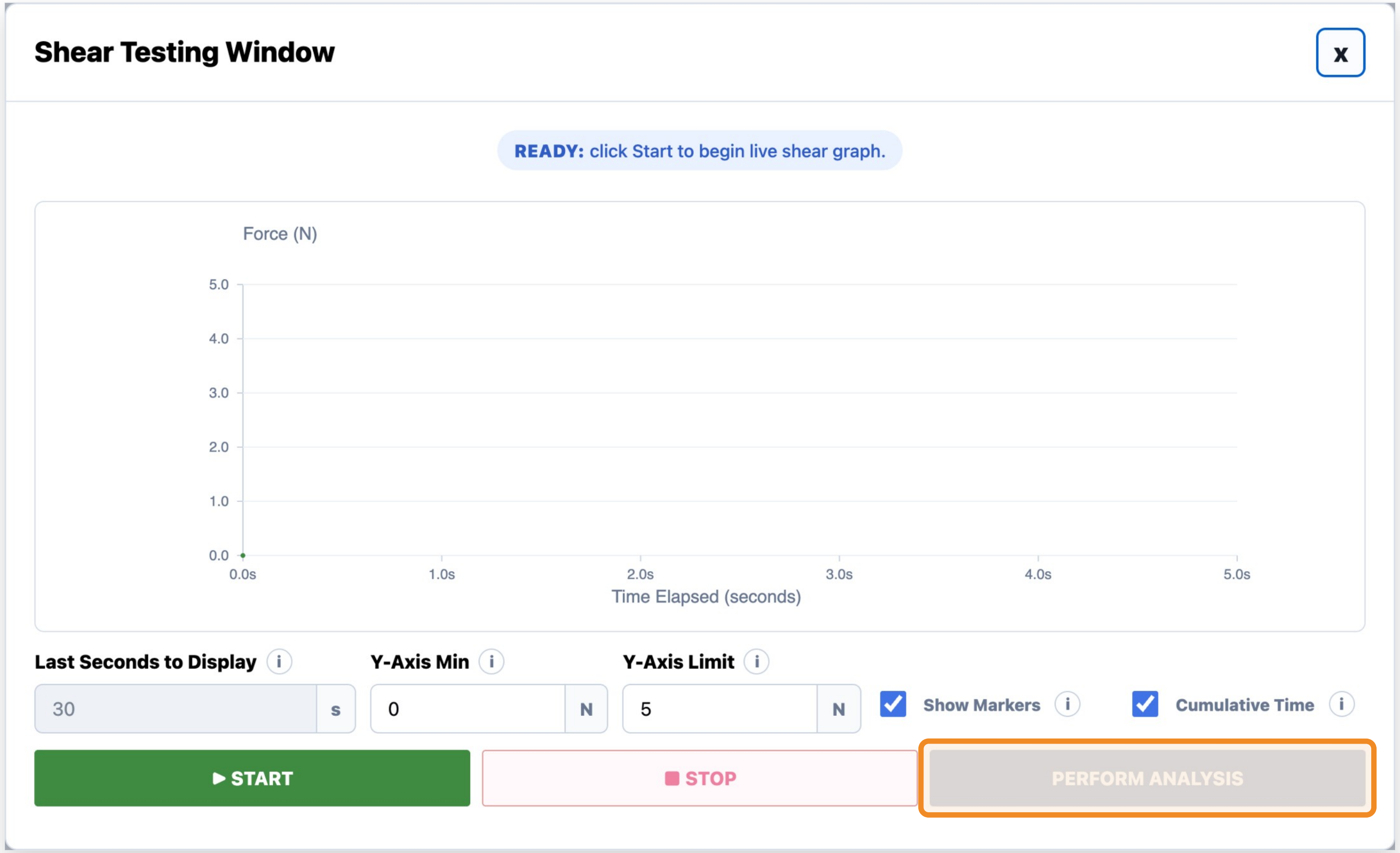
Perform Analysis Button

USER STORY

As a test fixture user, I want to click Perform Analysis after test data has been collected so that the software can process the latest available test data and generate summary statistics and plots.

ACCEPTANCE CRITERIA

- The Testing Window includes a Perform Analysis button
- Perform Analysis is disabled before testing starts, then becomes enabled after Start has been clicked, valid test data has been collected, and Stop has been clicked.
- If user pauses the test and restarts the run, the previously paused/incomplete graph data is not saved as the active dataset
- When clicked, analysis is generated using the newest/latest valid run data
- Clicking Perform Analysis shows a loading state before opening the Shear Analysis Page.



FEATURE 1.7 · SHEAR TEST

STORY 1.7.7

MUST

Info Icons

USER STORY

As a test fixture user, I want informational help icons throughout the window so that I can quickly understand fields, workflows, and testing terminology without needing external documentation.

INFO ICONS ON THIS PAGE

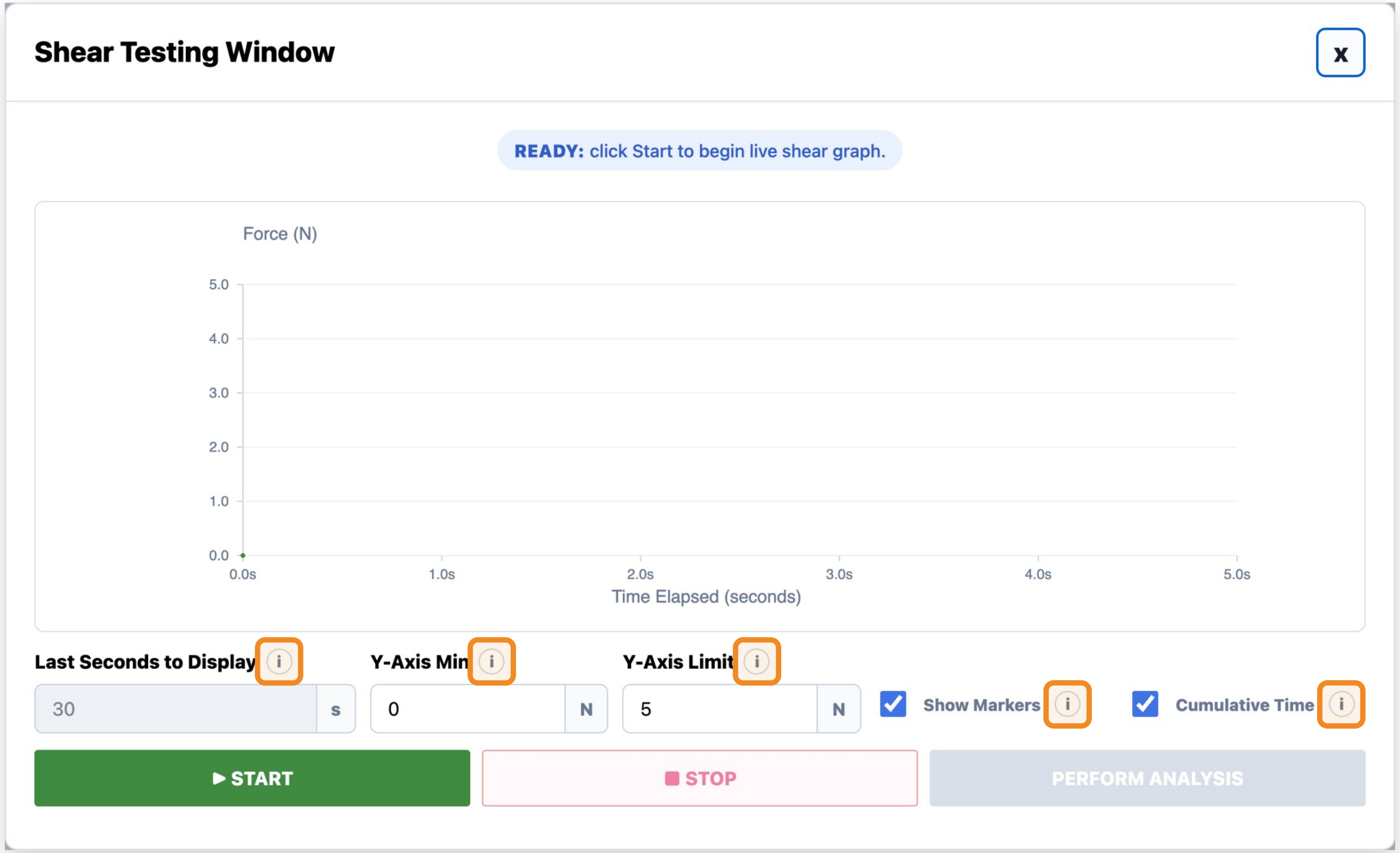
Last Seconds to Display: When Cumulative Time is off, the shear graph shows only the last selected number of seconds.

Y-Axis Min: Lowest force value shown on the shear graph y-axis.

Y-Axis Limit: Highest force value shown on the shear graph y-axis.

Show Markers: Shows individual live force readings on top of the graph line.

Cumulative Time: When on, the shear graph shows the entire run from start to finish and ignores Last Seconds to Display.



FEATURE 1.8

Shear Analysis Page

FEATURE 1.8 · SHEAR ANALYSIS

STORY 1.8.1

MUST

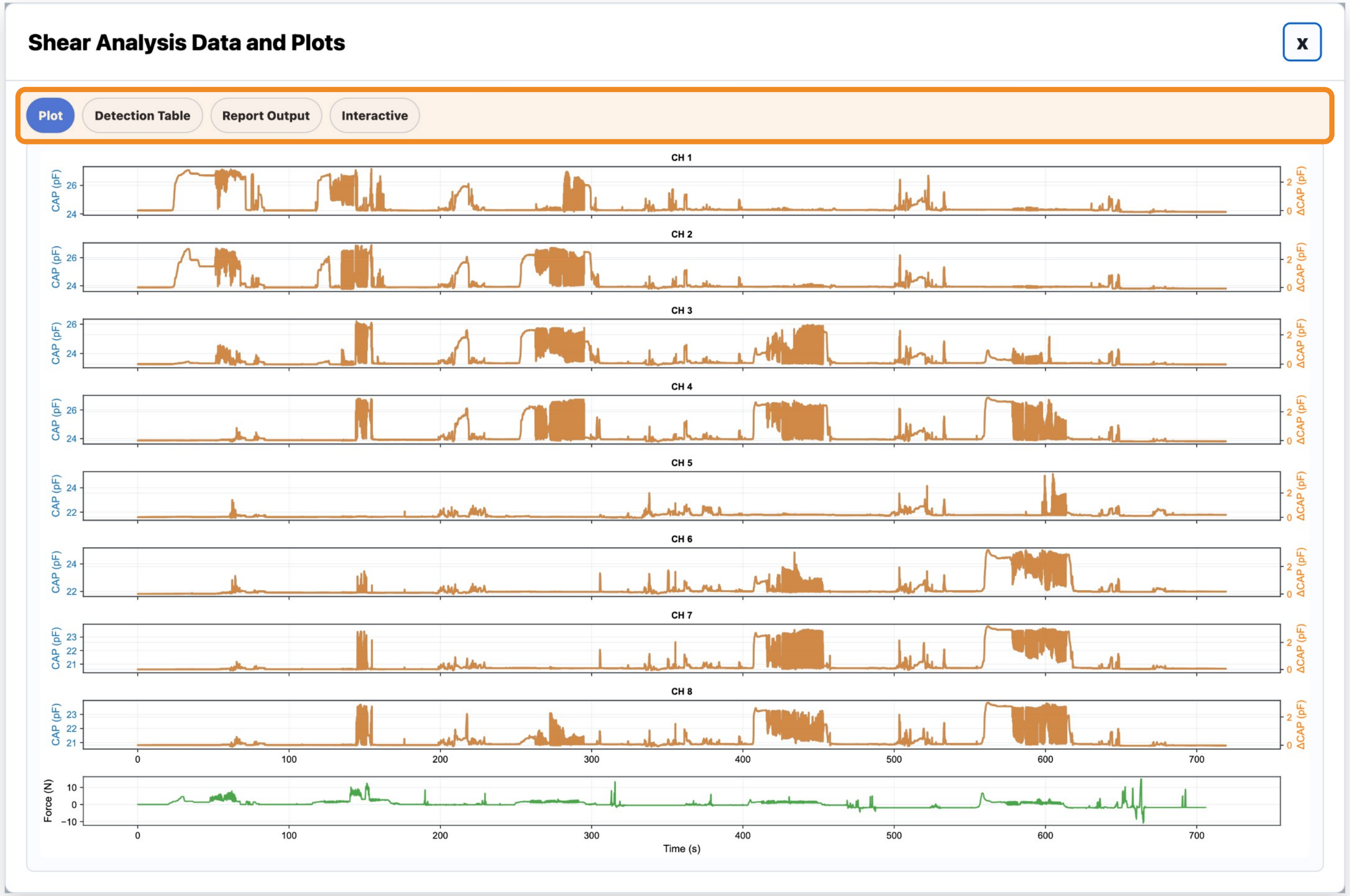
Shear Analysis Window

USER STORY

As a test fixture user, I want analysis results to open in a dedicated results window with tabs so that I can review processed Shear Test results in an organized way.

ACCEPTANCE CRITERIA

- After analysis completes, a new window opens titled: "Shear Analysis Data and Plots"
- Shear Analysis tabs may include: Plots, Failure Checks, Report Output
- Users can click between tabs to view different plots and metrics



FEATURE 1.8 · SHEAR ANALYSIS

STORY 1.8.2

MUST

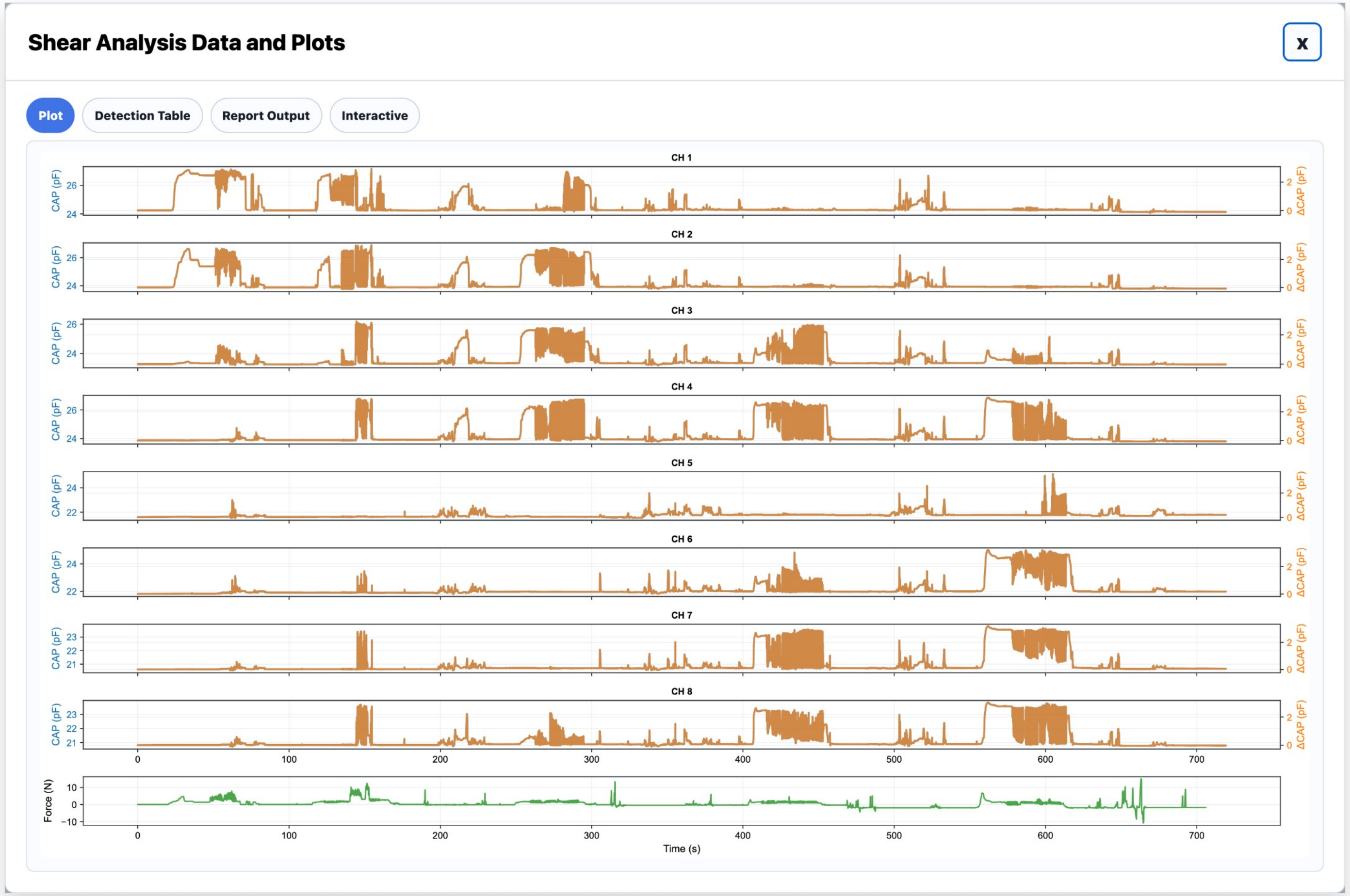
Plot Tab

USER STORY

As a test fixture user, after clicking Perform Analysis on a shear test I want to view a plot summary of the test.

ACCEPTANCE CRITERIA

- The tab displays a graph for each capacitance channel and one final graph for outputted force over time
- The X axis for each graph is Time (seconds)
- The Y axis for each capacitance channel includes both Capacitance (pF) and delta Capacitance (pF)
- The final graph Y axis is Force (Newtons)



FEATURE 1.8 · SHEAR ANALYSIS

STORY 1.8.3

MUST

Detection Table Tab

USER STORY

As a test fixture user, I want to know which capacitance channels had negative values and which had a large delta after performing a shear test.

ACCEPTANCE CRITERIA

- The tab displays a grid where each column is a capacitance channel
- One row shows negative values (pF) if detected, or None
- One row shows delta values (pF) if a high delta was detected, or None
- A high delta is defined as greater than 10 pF

FEATURE 1.8 · SHEAR ANALYSIS

MUST

STORY 1.8.4

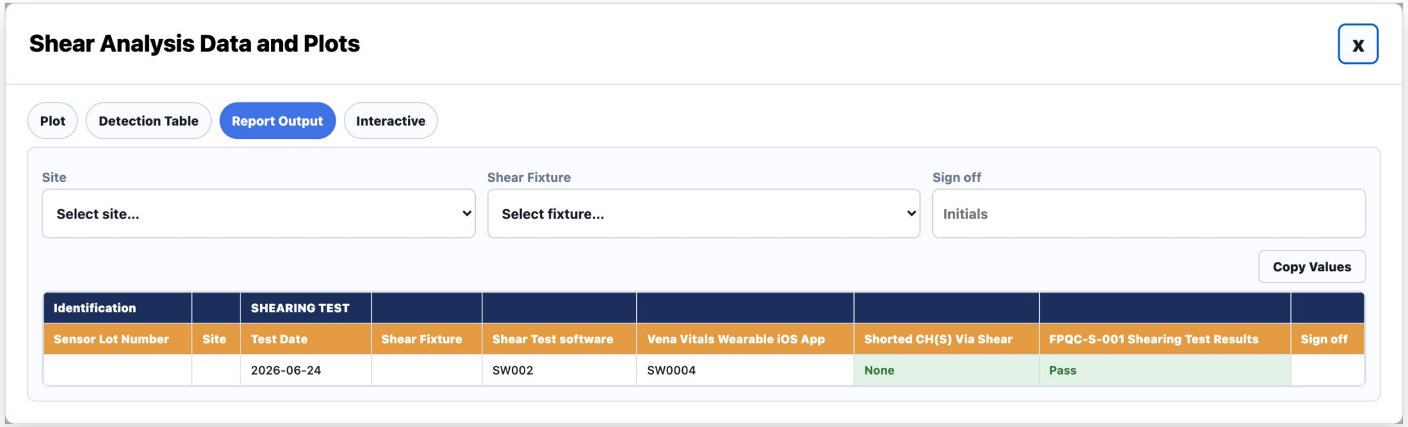
Report Outputs Tab

USER STORY

As a test fixture user, I want to easily copy information from the Shear analysis summary so I can document it later.

ACCEPTANCE CRITERIA

- The tab includes: Sensor ID, Test Result (Pass/Fail), Shorted Channels Ch 1-N, Test Date DD/MM/YEAR, Test Taker Initials, Test Version
- User can copy values via the "Copy Values" button
- Copied values should be pastable into Excel format



FEATURE 1.9

Manual Test Page

FEATURE 1.9 · MANUAL TEST

STORY 1.9.1

MUST

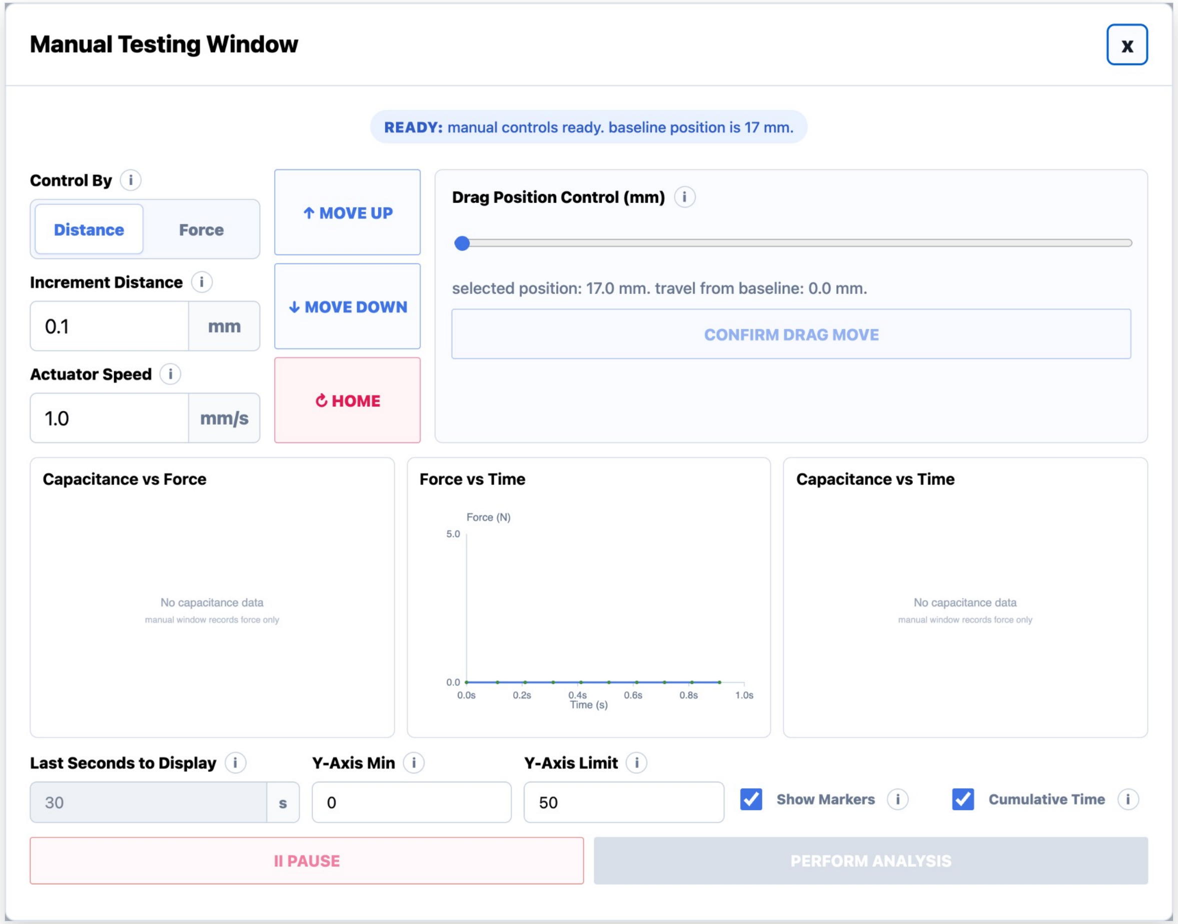
Manual Testing Window

USER STORY

As a test fixture user, I want the Manual Test to open in a dedicated Manual Testing Window so that I can manually control the Zaber stage and perform custom tests.

ACCEPTANCE CRITERIA

- Clicking Begin Test with Test Type = Manual Test opens a new window titled: "Manual Testing Window"
- The window includes manual control settings, actuator movement controls, live graphs, graph display controls, Pause, and Perform Analysis



FEATURE 1.9 · MANUAL TEST

STORY 1.9.2

MUST

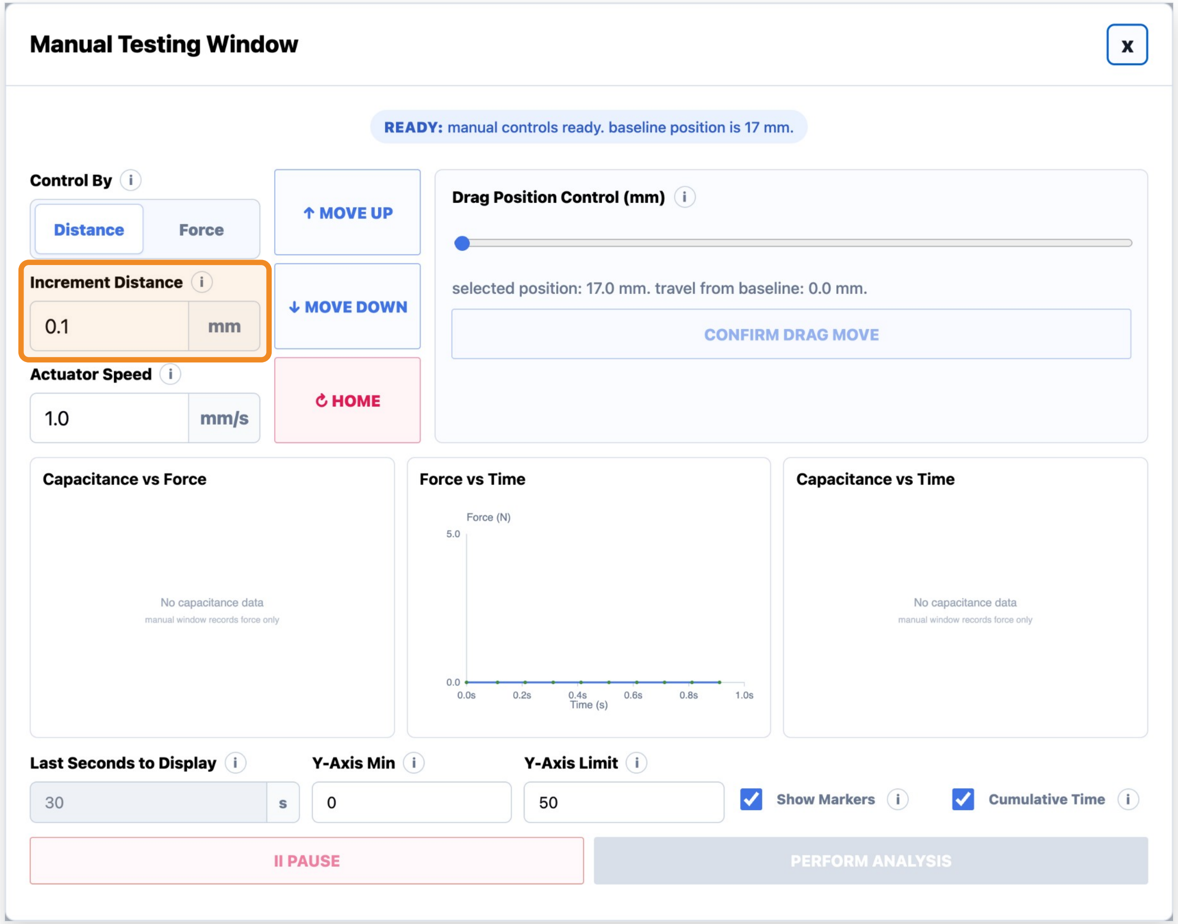
Control By Distance

USER STORY

As a test fixture user, I want to manually control actuator movement by distance so that I can perform precise step-by-step actuator positioning and exploratory testing.

ACCEPTANCE CRITERIA

- When "Control By = Distance" is selected, the window displays: Increment Distance, Actuator Speed, Move Up, Move Down, Home
- User enters Increment Distance (mm) and Actuator Speed (mm/s)
- Move Up moves the actuator upward using the specified Increment Distance and Actuator Speed
- Move Down moves the actuator downward using the specified Increment Distance and Actuator Speed
- Home returns the actuator to the home/original position
- Current position and live graphs update after movement
- Movement controls are temporarily disabled while the actuator is moving; re-enabled when motion completes



FEATURE 1.9 · MANUAL TEST

STORY 1.9.3

MUST

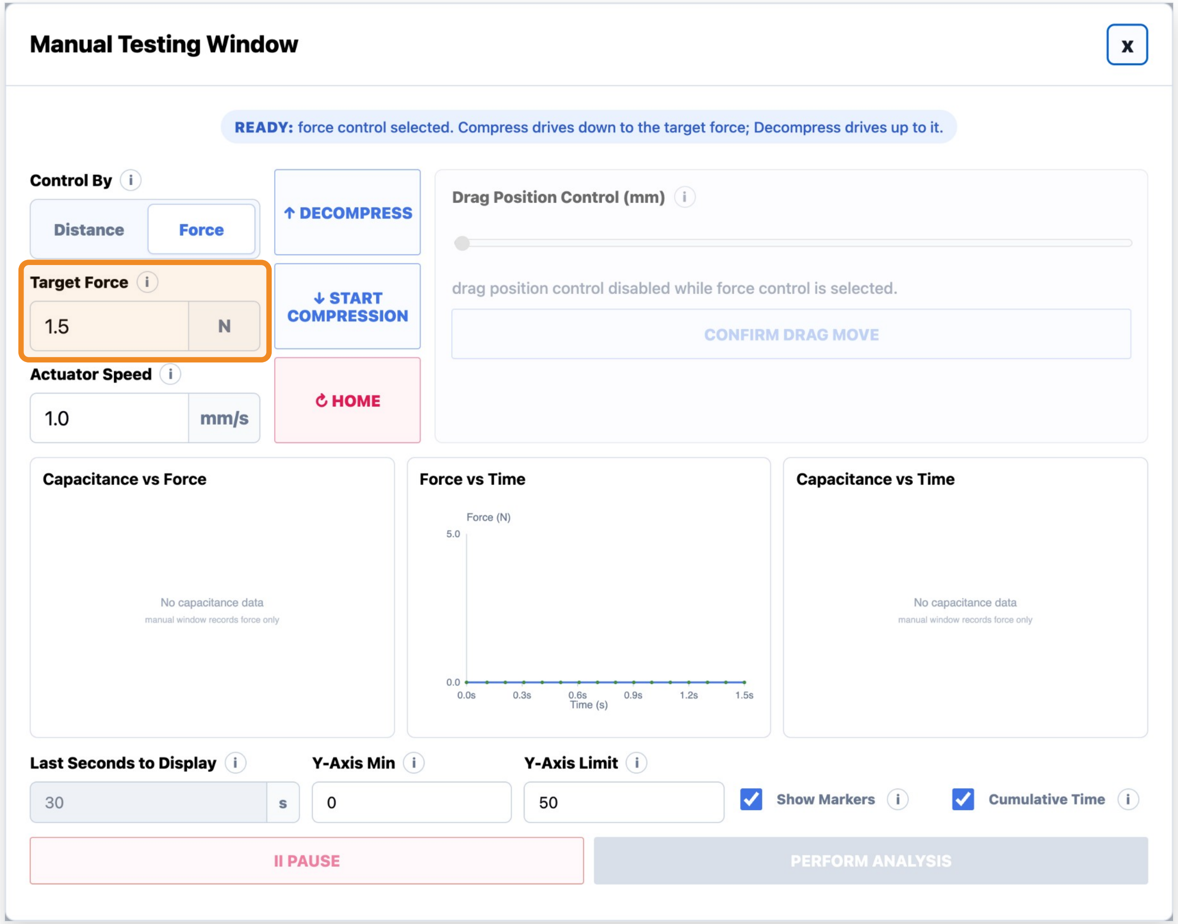
Control By Force

USER STORY

As a test fixture user, I want to manually control actuator movement by force so that I can perform force-based compression testing and exploratory sensor characterization.

ACCEPTANCE CRITERIA

- When "Control By = Force" is selected, the window displays: Target Force, Actuator Speed, Release, Start Compression, Home
- User enters Target Force (N) and Actuator Speed (mm/s)
- Start Compression begins actuator compression toward the selected Target Force
- Release releases the applied compression or moves the actuator upward
- Home returns the actuator to the home/original position
- Live graphs update during compression and release behavior
- Movement controls are temporarily disabled while the actuator is moving or compressing



FEATURE 1.9 · MANUAL TEST

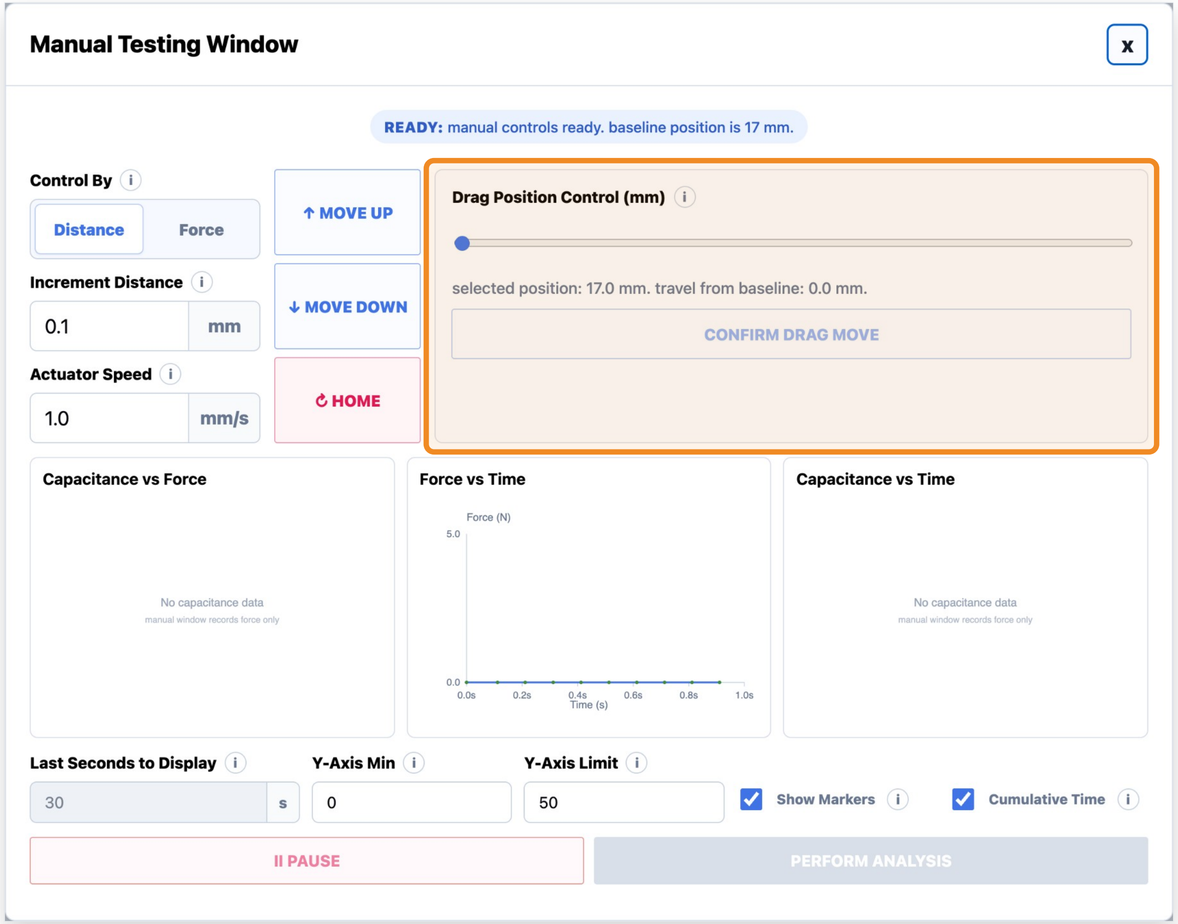
MUST

STORY 1.9.4

Drag Position Control

USER STORY

As a test fixture user, I want to stage actuator movement using a slider so that I can preview the desired position before physically moving the actuator.



ACCEPTANCE CRITERIA

- The window includes a Drag Position Control slider
- Dragging the slider only stages a move and does not move the actuator
- The readout displays the selected position and travel from baseline
- Confirm Drag Move remains disabled until the slider changes
- The actuator only moves after the user clicks Confirm Drag Move
- Movement controls and Confirm Drag Move are disabled during active movement
- Current position and live graphs update after movement completes

FEATURE 1.9 · MANUAL TEST

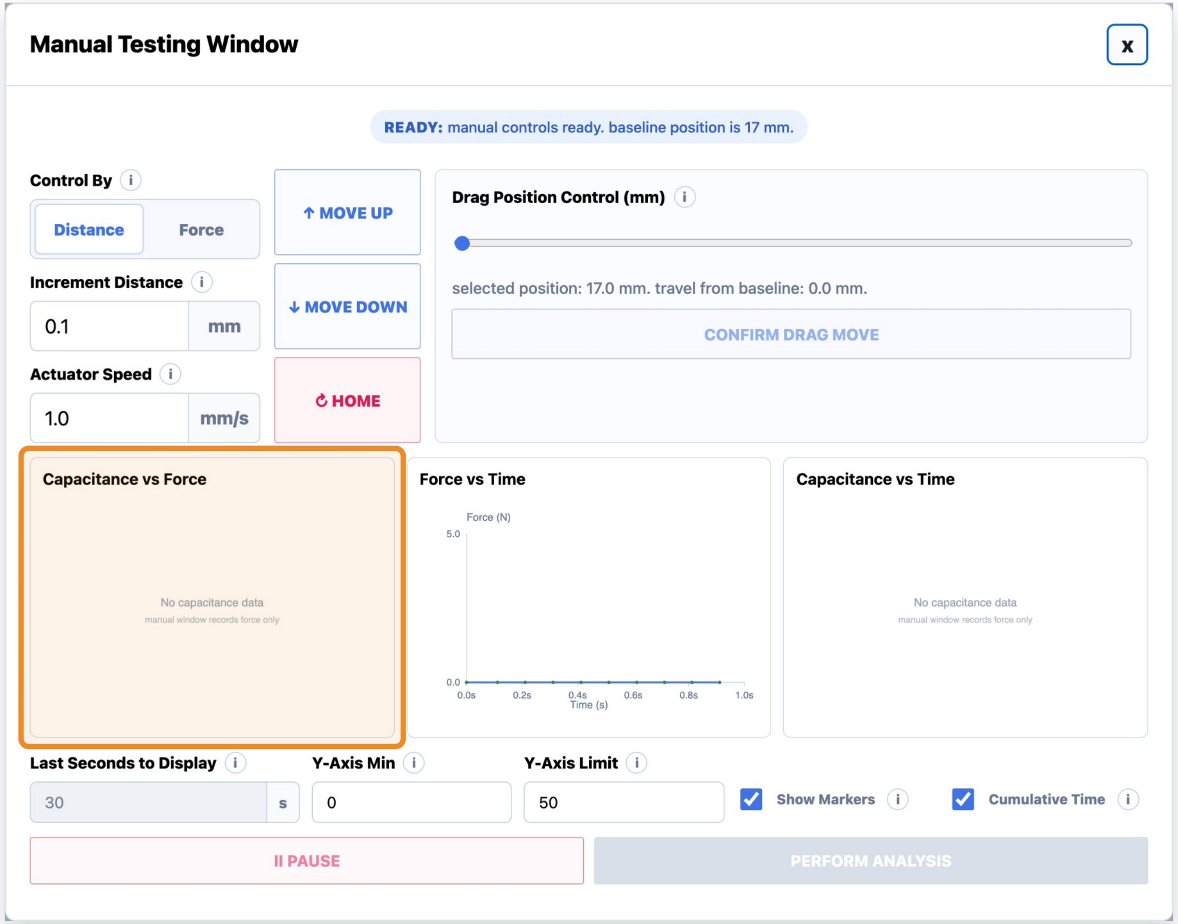
MUST

STORY 1.9.5

Manual Test Live Graphs

USER STORY

As a test fixture user, I want live graphs during manual testing so that I can monitor sensor behavior while manually moving the actuator.



ACCEPTANCE CRITERIA

- The window displays three live graphs: Capacitance vs Force, Force vs Time, Capacitance vs Time
- Graphs update when movement or compression occurs

FEATURE 1.9 · MANUAL TEST

MUST

STORY 1.9.6

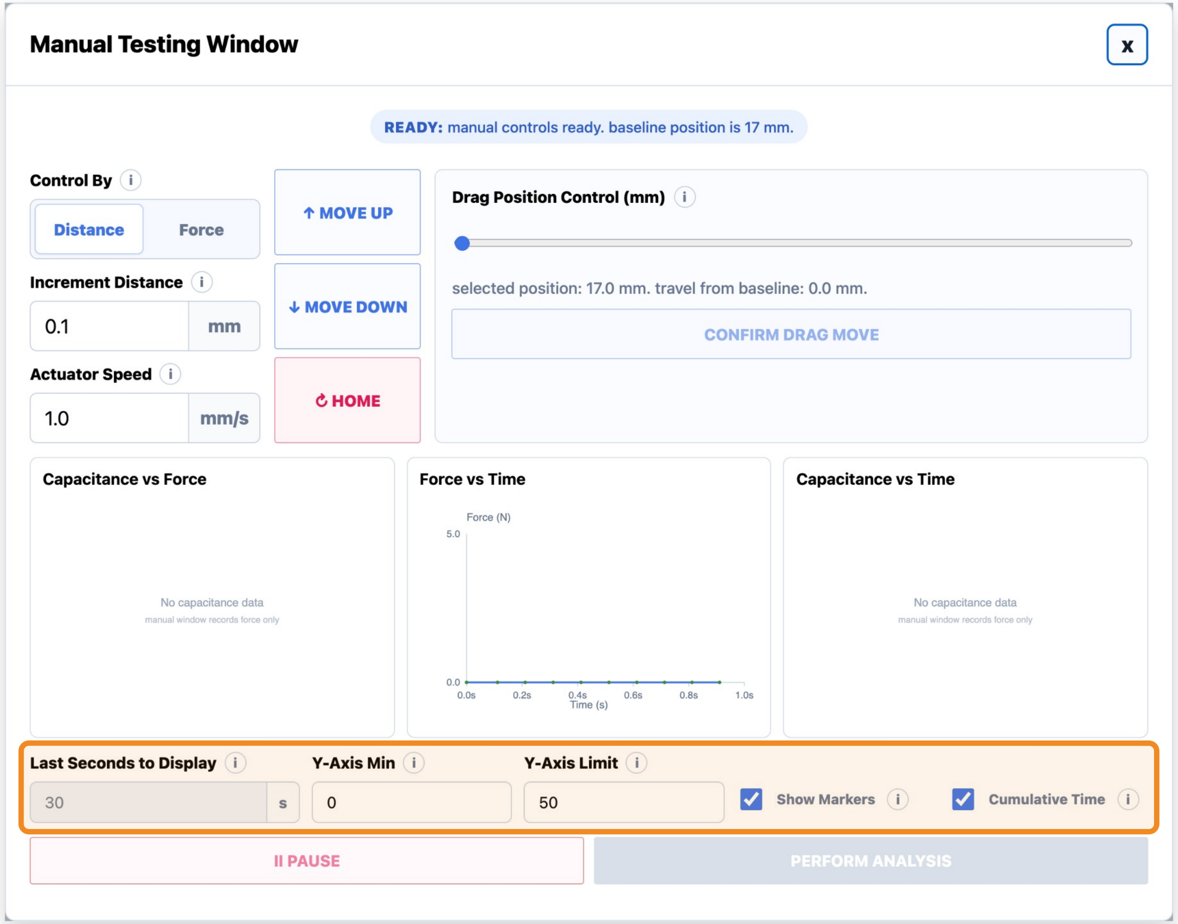
Graph Display Controls

USER STORY

As a test fixture user, I want graph display controls so that I can customize how live data is visualized.

ACCEPTANCE CRITERIA

- The user can edit Last Seconds to Display
- The user can edit Y-Axis Min
- The user can edit Y-Axis Limit
- The user can toggle Show Markers
- The user can toggle Cumulative Time



FEATURE 1.9 · MANUAL TEST

STORY 1.9.7

MUST

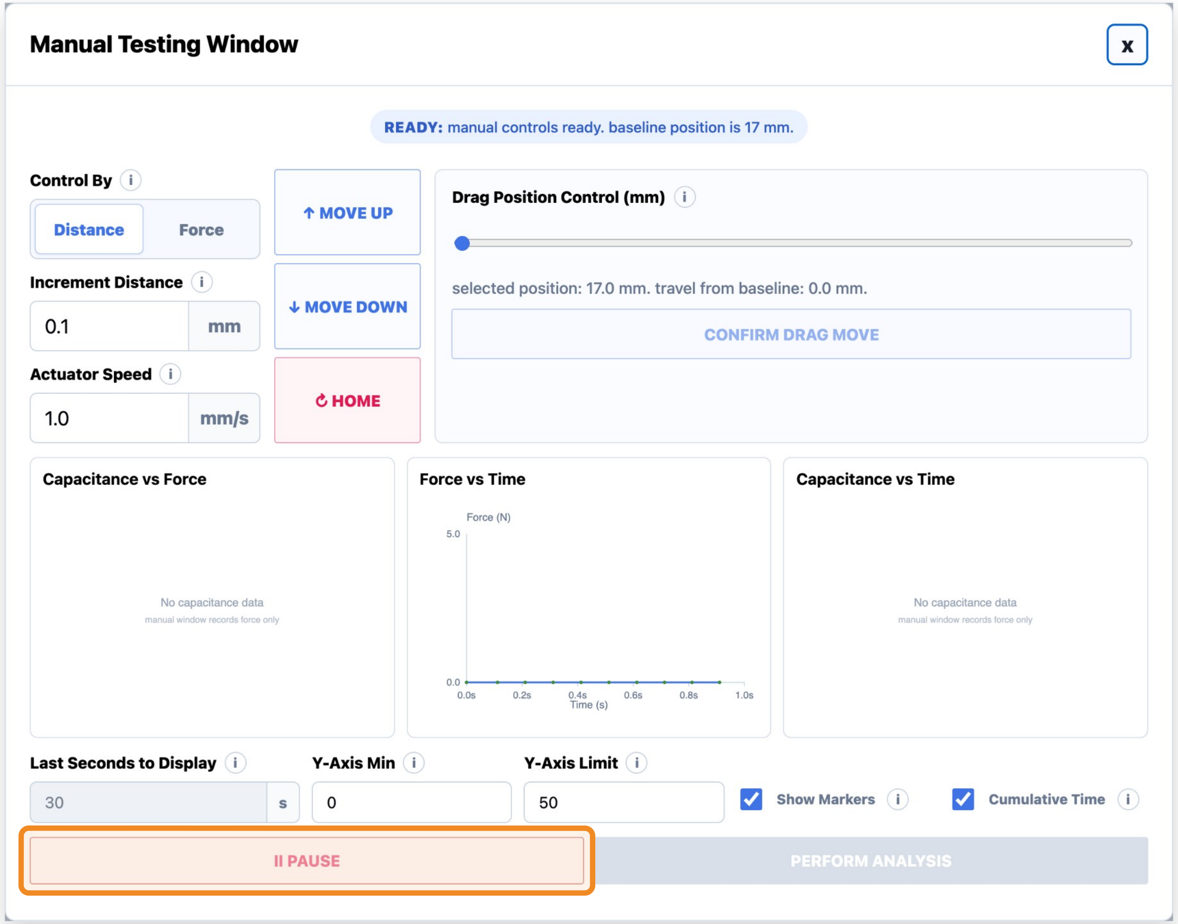
Pause Button

USER STORY

As a test fixture user, I want to pause Manual Testing so that I can stop active movement or graph updates safely.

ACCEPTANCE CRITERIA

- Clicking Pause stops active actuator movement
- Graph updates pause while the workflow is paused
- Movement controls remain locked until the system is safe/ready again



FEATURE 1.9 · MANUAL TEST

STORY 1.9.8

MUST

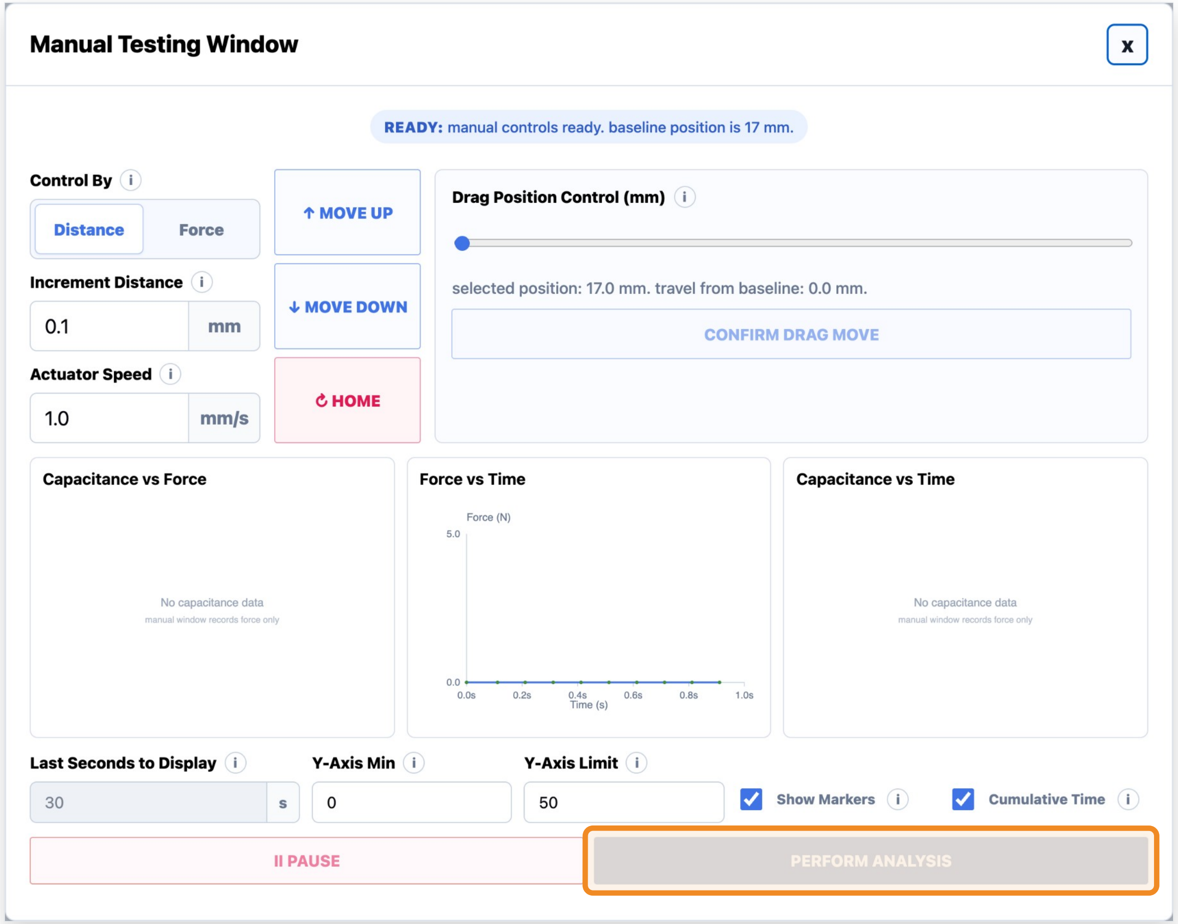
Perform Analysis Button

USER STORY

As a test fixture user, I want to perform analysis on manual test data so that I can review results from the current preview/manual dataset.

ACCEPTANCE CRITERIA

- Perform Analysis becomes available after manual test data exists
- Clicking Perform Analysis generates manual analysis from the current preview data
- Shows a loading state before opening the Manual Analysis Page



FEATURE 1.9 · MANUAL TEST

STORY 1.9.9

MUST

Info Icons

USER STORY

As a test fixture user, I want informational help icons throughout the window so that I can quickly understand fields, workflows, and testing terminology without needing external documentation.

INFO ICONS ON THIS PAGE

Control By: Distance mode moves the actuator by a chosen millimeter increment. Force mode simulates compression until the selected target force is reached.

Increment Distance: Distance used for each Move Up or Move Down click. Move Down adds distance from the actuator baseline; Move Up subtracts it.

Actuator Speed: Assumed actuator travel speed used to estimate elapsed time on the manual preview graphs. Default is 2.0 mm/s.

Drag Position Control (mm): Drag to choose a target position from 17 mm baseline to 50.8 mm max (full actuator travel). The actuator only moves after Confirm Drag Move is pressed.

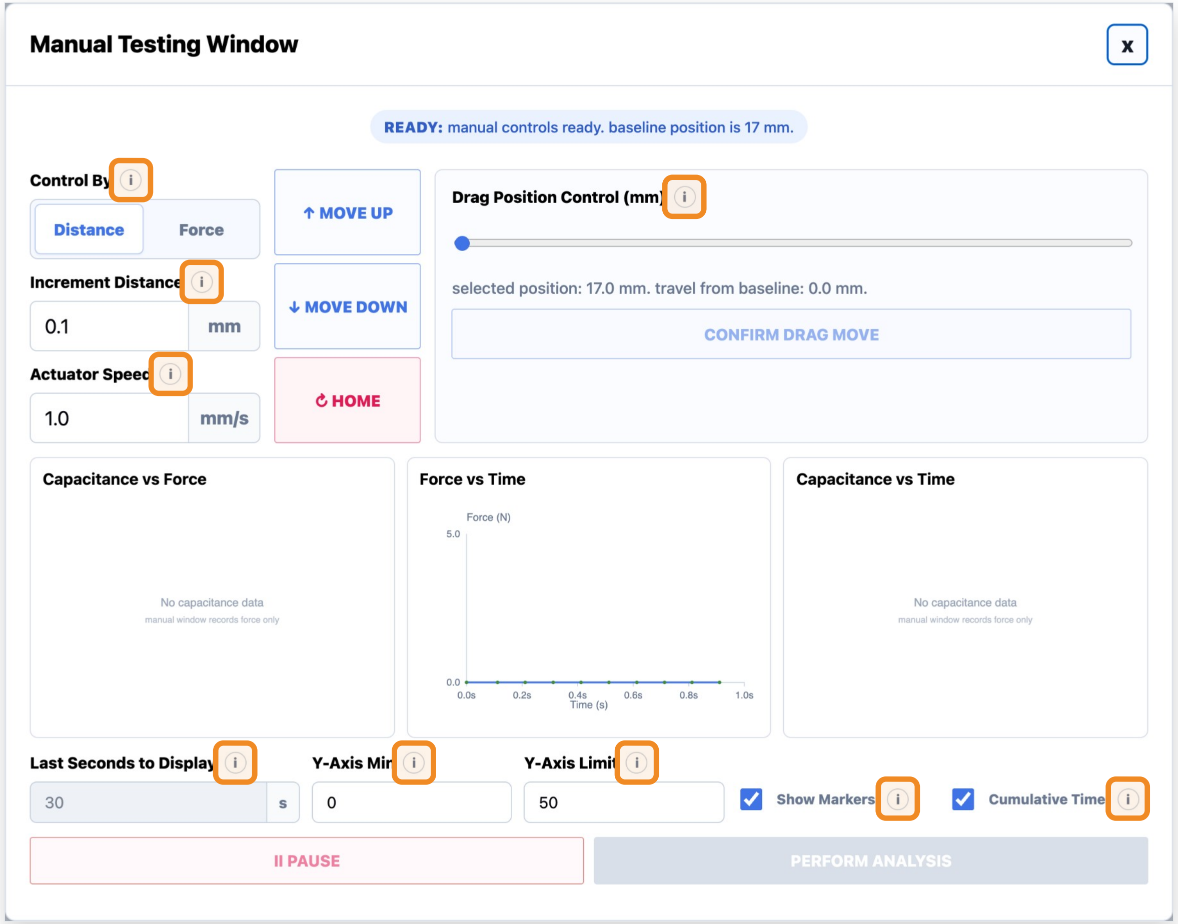
Last Seconds to Display: When Cumulative Time is off, time-based graphs show only the last selected number of seconds.

Y-Axis Min: Lowest value shown on the manual test graph y-axis.

Y-Axis Limit: Highest value shown on the manual test graph y-axis.

Show Markers: Shows individual data points on top of the graph line.

Cumulative Time: When on, the graph shows the entire run from start to finish and ignores Seconds to Display.



FEATURE 1.10

Manual Analysis Page

FEATURE 1.10 · MANUAL ANALYSIS

STORY 1.10.1

MUST

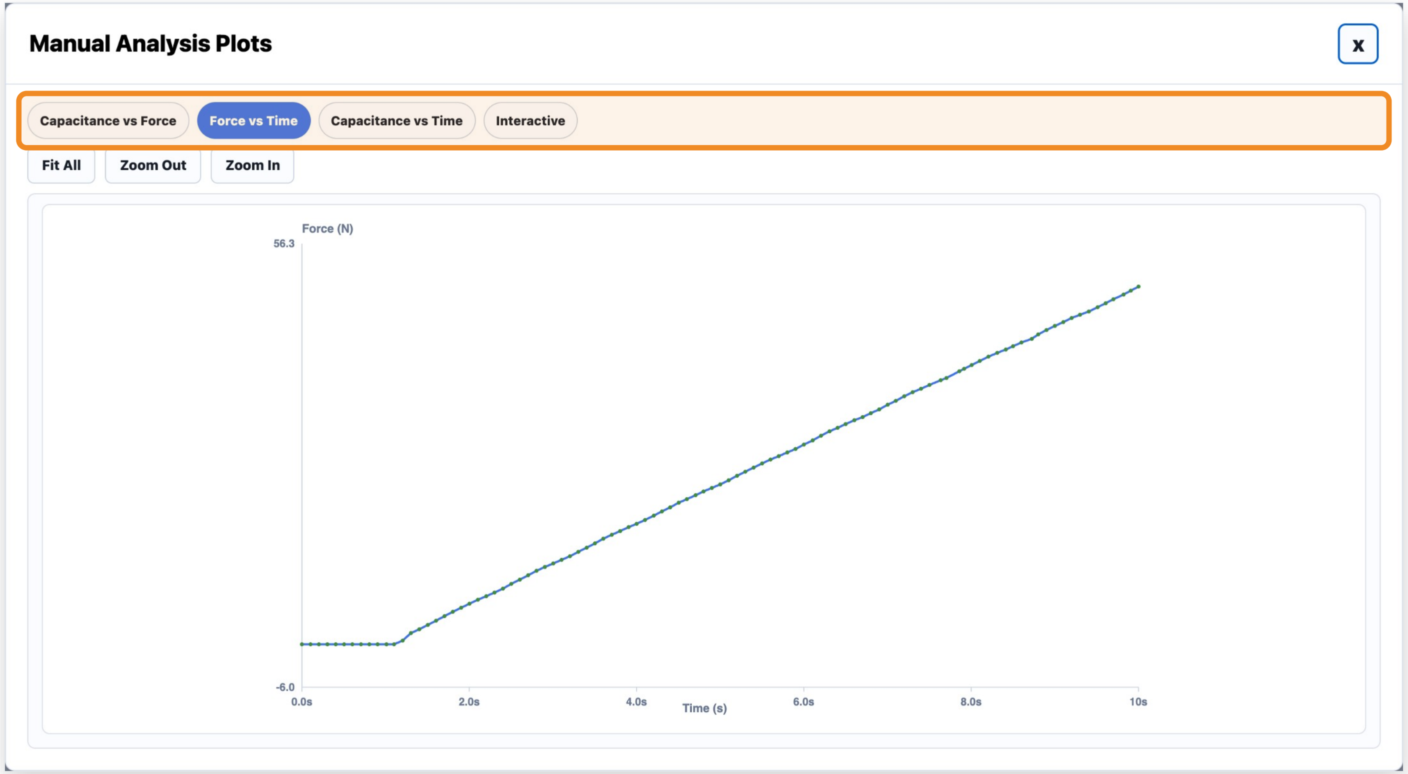
Manual Analysis Window

USER STORY

As a test fixture user, I want Manual Test results to open in a dedicated analysis window so that I can review the data collected during manual actuator movement.

ACCEPTANCE CRITERIA

- Clicking Perform Analysis from the Manual Testing Window opens a new window titled: "Manual Analysis Data and Plots"
- The window displays plots generated from the current manual test data
- The window includes tabs for different manual analysis plots



FEATURE 1.10 · MANUAL ANALYSIS

MUST

STORY 1.10.2

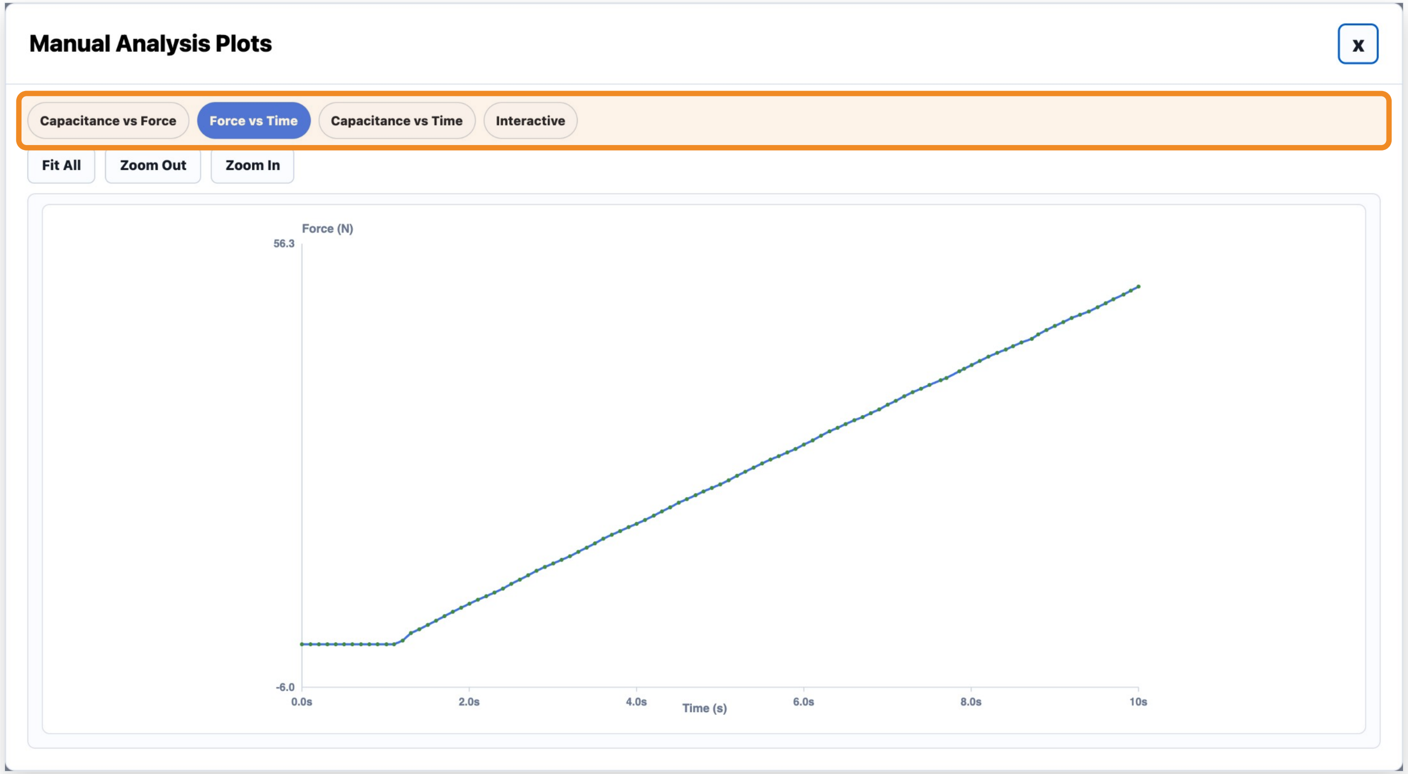
Manual Analysis Plot Tabs

USER STORY

As a test fixture user, I want Manual Analysis plots organized into tabs so that I can switch between different sensor relationships easily.

ACCEPTANCE CRITERIA

- Tabs include: Capacitance vs Force, Force vs Time, Capacitance vs Time
- Clicking each tab updates the displayed plot
- Plots are generated from the current manual test data
- Data points are shown with markers and connected by a line



FEATURE 1.11

Cyclical Test Page

FEATURE 1.11 · CYCLICAL TEST

STORY 1.11.1

MUST

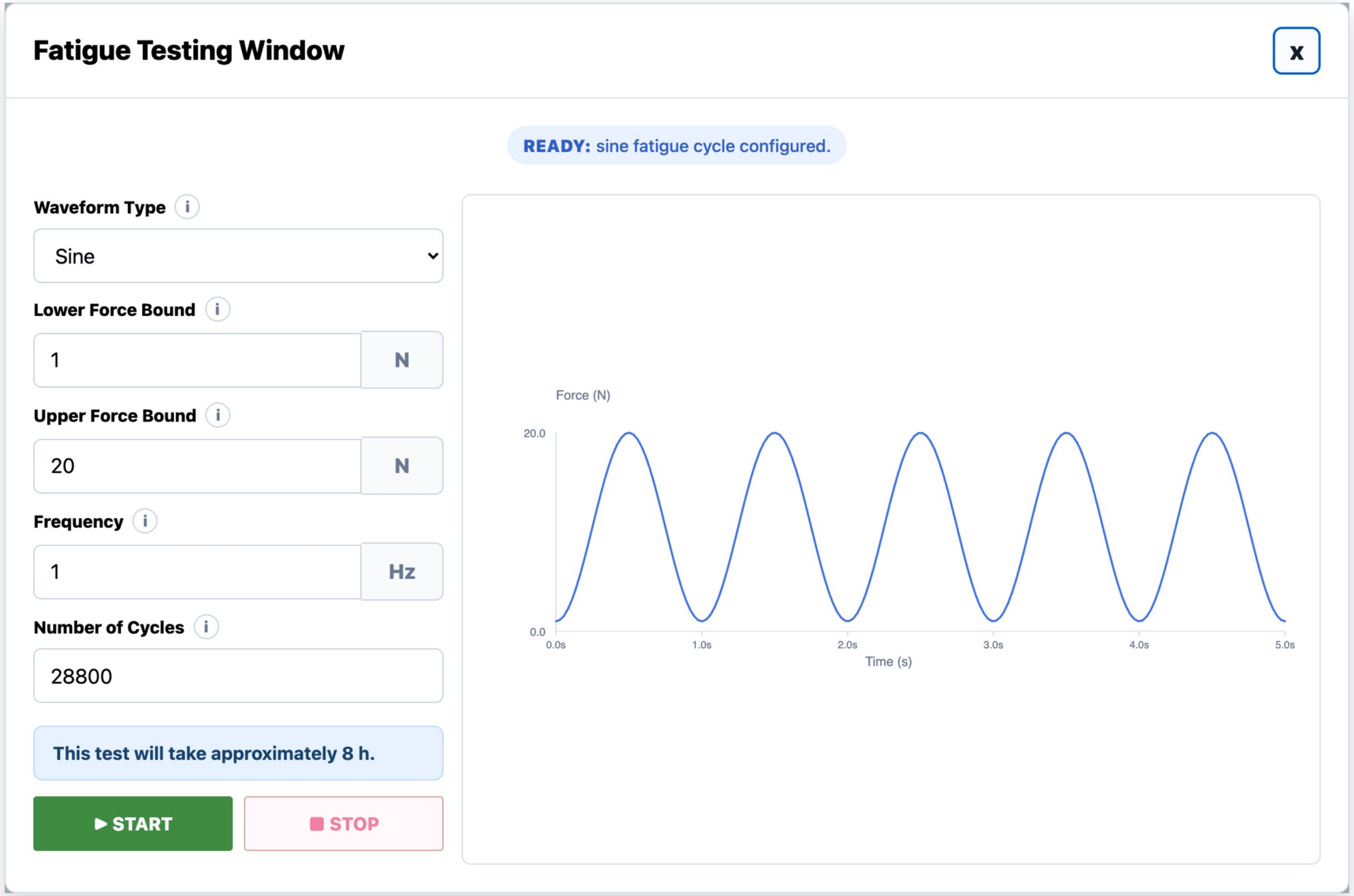
Cyclical Testing Window

USER STORY

As a test fixture user, I want the Cyclical Test to open in a dedicated Cyclical Testing Window so that I can monitor and control a cyclical test.

ACCEPTANCE CRITERIA

- Clicking Begin Test with Test Type = Cyclical Test opens a new window titled: "Cyclical Testing Window"



FEATURE 1.11 · CYCLICAL TEST

MUST

STORY 1.11.2

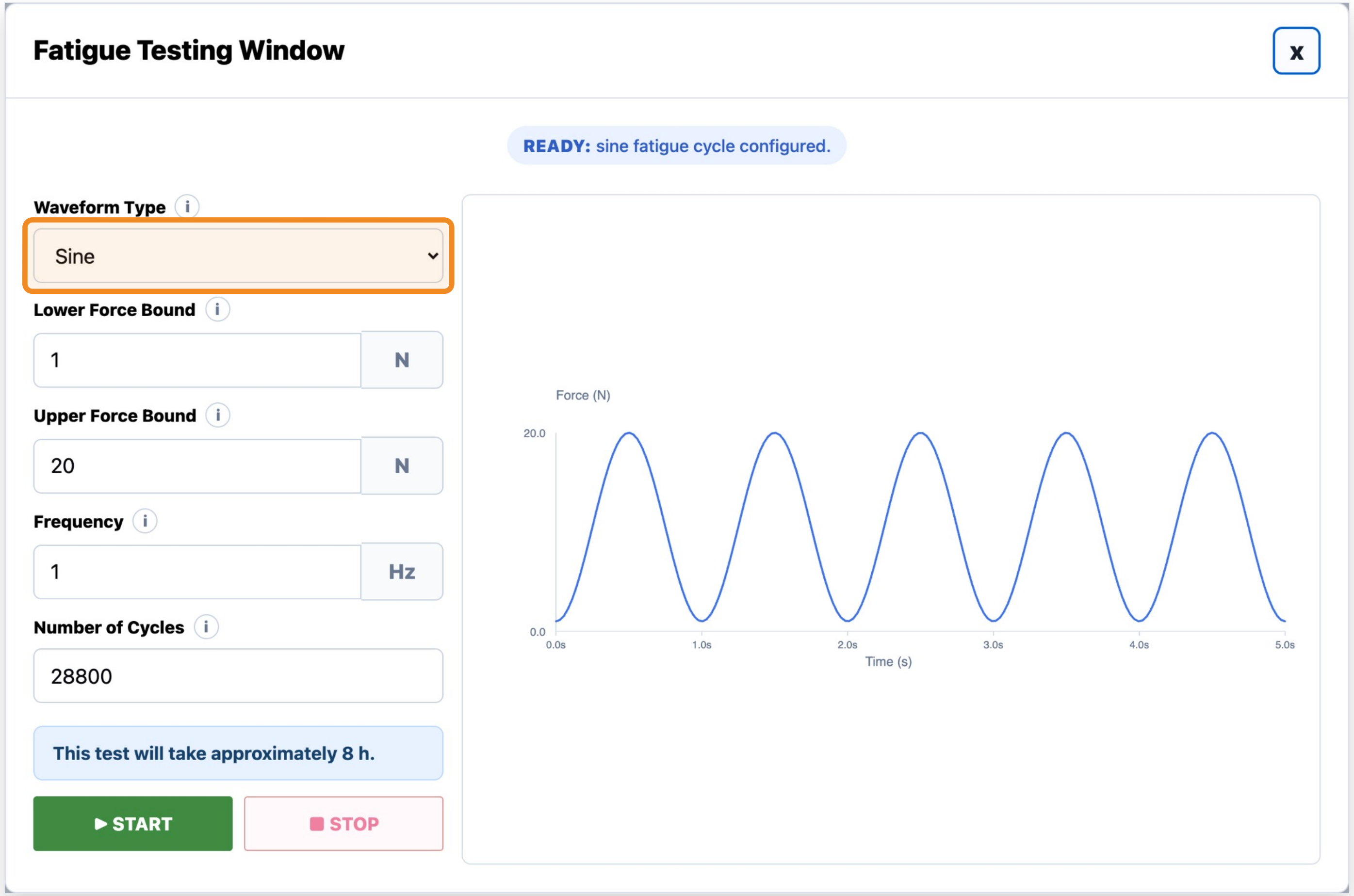
Waveform Type Dropdown

USER STORY

As a test fixture user, I want to perform cyclical tests with different waveform outputs.

ACCEPTANCE CRITERIA

- User can click a "Waveform Type" dropdown menu
- Options include: Sine, Square



FEATURE 1.11 · CYCLICAL TEST

STORY 1.11.3

MUST

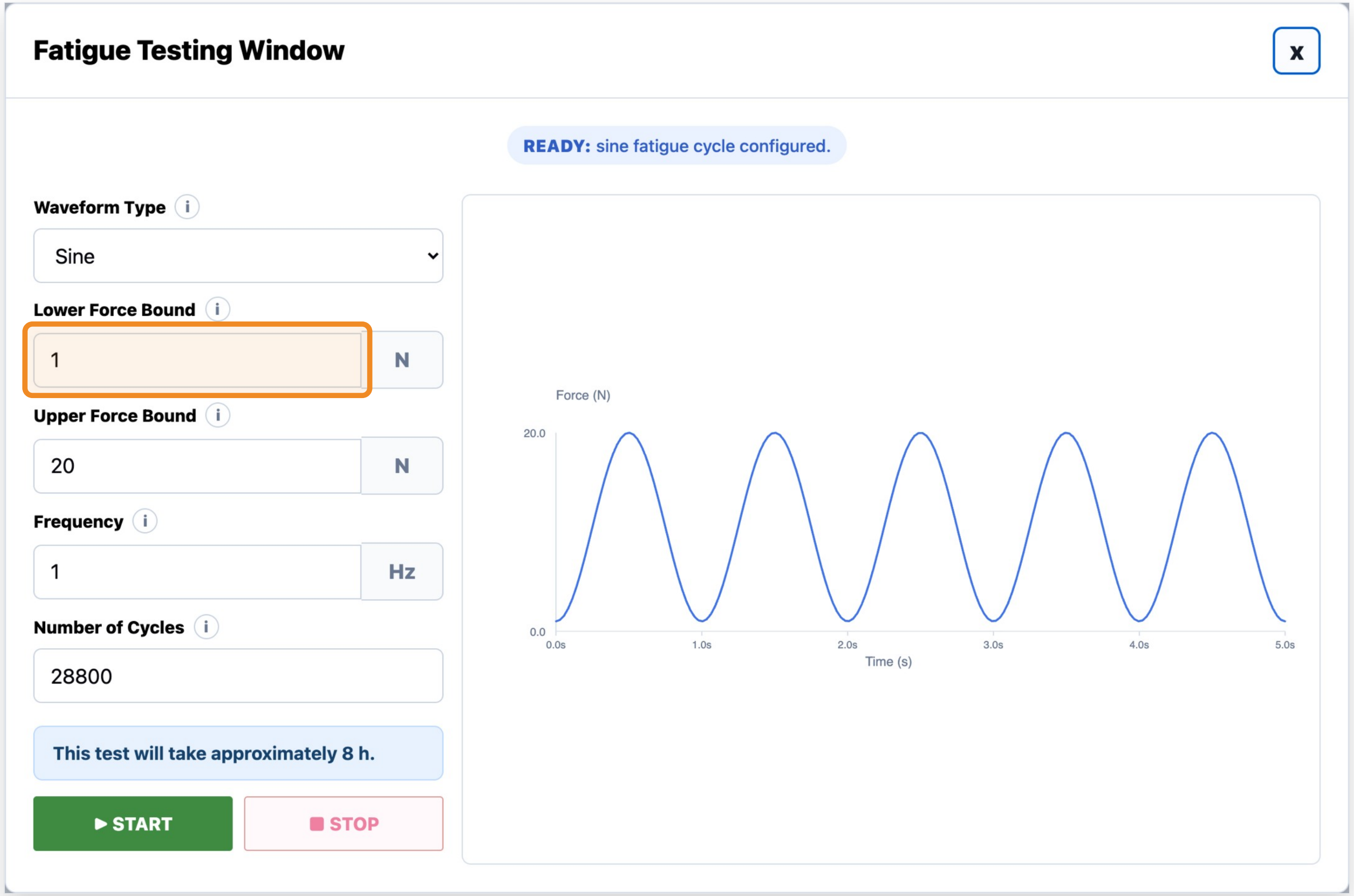
Lower Force Bound Integer Field

USER STORY

As a test fixture user, I want to control the lower bound of the waveform during a cyclical test.

ACCEPTANCE CRITERIA

- User enters a positive integer into the "Lower Force Bound (N)" field
- The integer must be strictly greater than 0
- The integer must be strictly lower than the Upper Force Bound (N) value
- Otherwise, an error message displays informing the user the lower bound is larger than the upper bound
- Default lower bound is 1 Newton



FEATURE 1.11 · CYCLICAL TEST

STORY 1.11.4

MUST

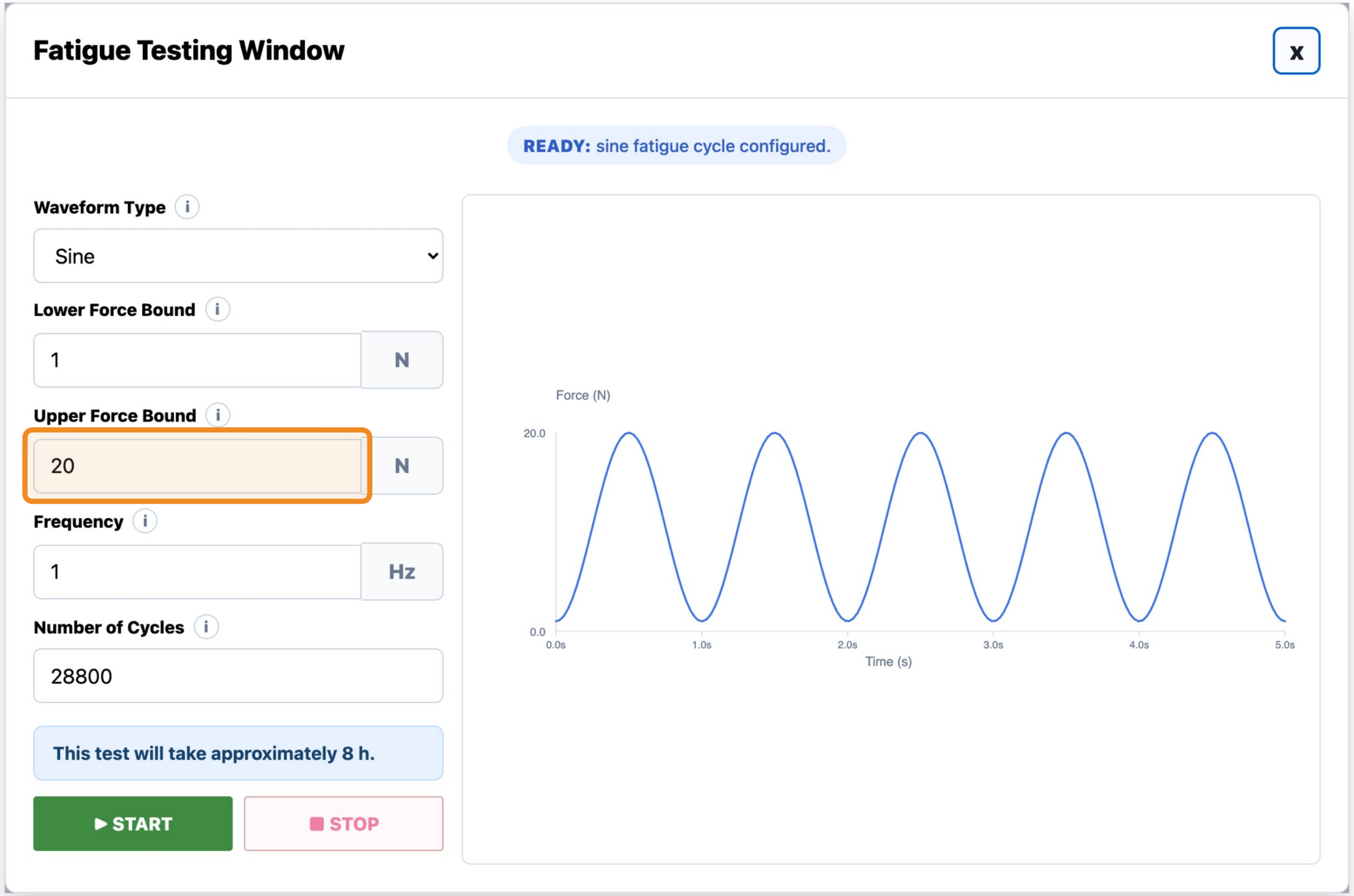
Upper Force Bound Integer Field

USER STORY

As a test fixture user, I want to control the upper bound of the waveform during a cyclical test.

ACCEPTANCE CRITERIA

- User enters a positive integer into the "Upper Force Bound (N)" field
- The integer must be lower than or equal to 32 Newtons
- The integer must be strictly higher than the Lower Force Bound (N) value
- Otherwise, an error message displays informing the user the upper bound is lower than the lower bound
- Default upper bound is 20 Newtons



FEATURE 1.11 · CYCLICAL TEST

MUST

STORY 1.11.5

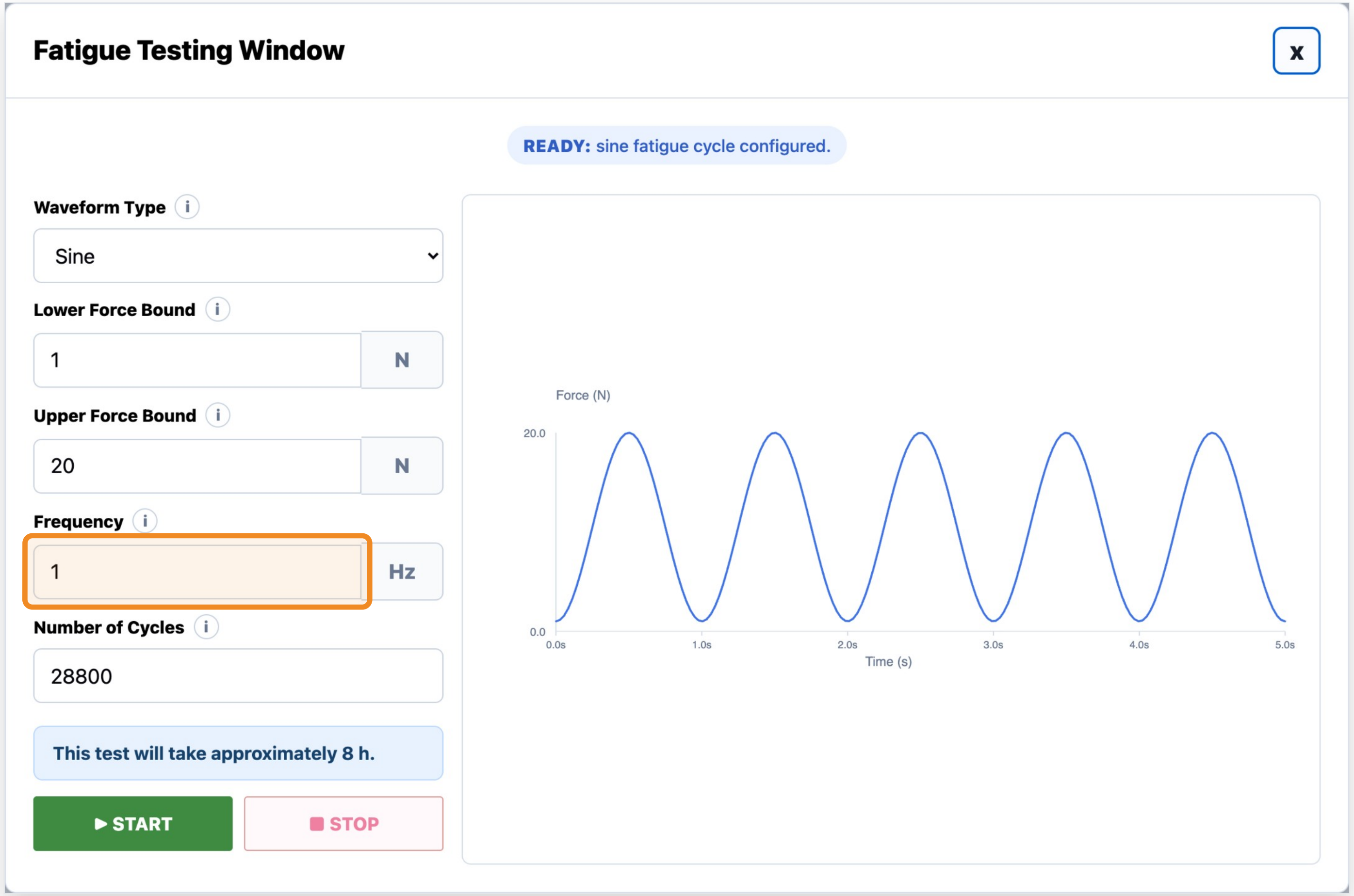
Frequency Integer Field

USER STORY

As a test fixture user, I want to control how many cycles occur per second by manipulating the frequency of the cyclical waveform.

ACCEPTANCE CRITERIA

- User enters a positive integer into the "Frequency (Hz)" field
- The integer represents the number of wave cycles that occur per second
- Default frequency is 1 Hertz



FEATURE 1.11 · CYCLICAL TEST

MUST

STORY 1.11.6

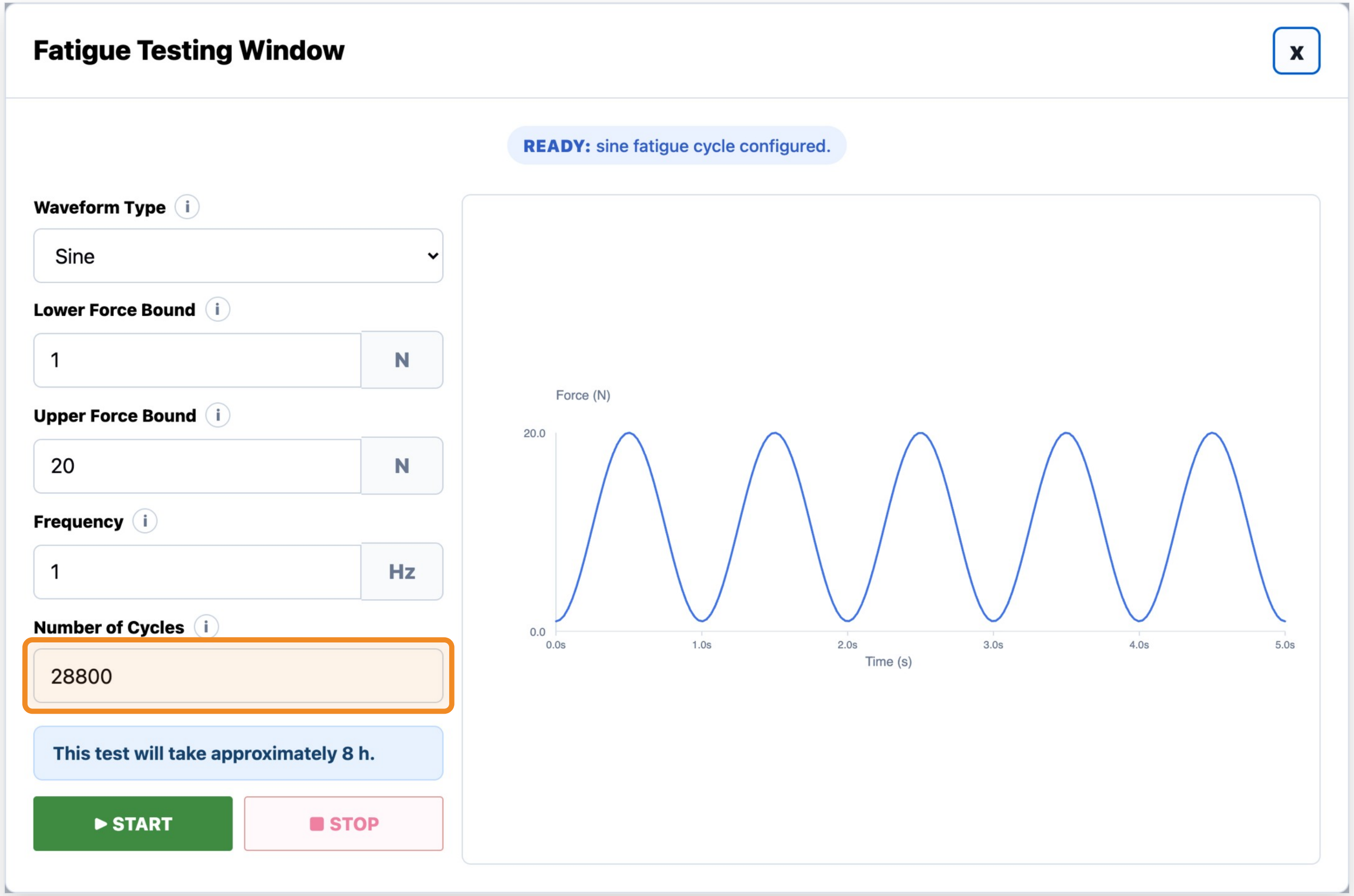
Number of Cycles Integer Field

USER STORY

As a test fixture user, I want to control the number of cycles that should be completed before the test ends.

ACCEPTANCE CRITERIA

- User enters a positive integer into the "Number of Cycles" field
- A message below the field displays the test duration computed from Frequency and Number of Cycles
- Default is 28,800 cycles (8 hours at 1 Hz)



FEATURE 1.11 · CYCLICAL TEST

MUST

STORY 1.11.7

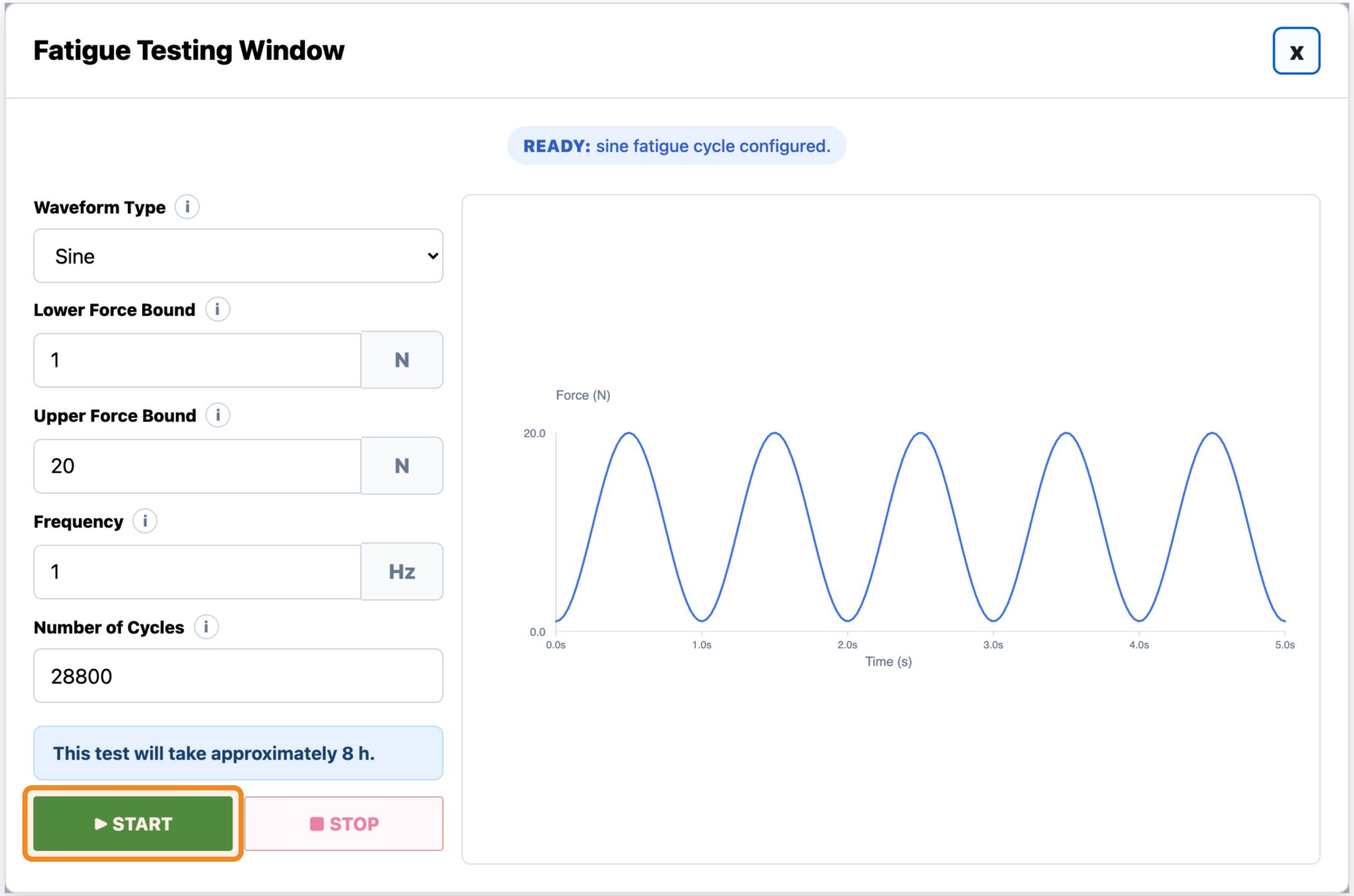
Start Button

USER STORY

As a test fixture user, I want to start a cyclical test.

ACCEPTANCE CRITERIA

- User clicks a "Start" button to begin the Cyclical test
- All input fields must be filled in with valid values for the test to begin
- Once Start is clicked, all input fields are disabled



FEATURE 1.11 · CYCLICAL TEST

STORY 1.11.8

MUST

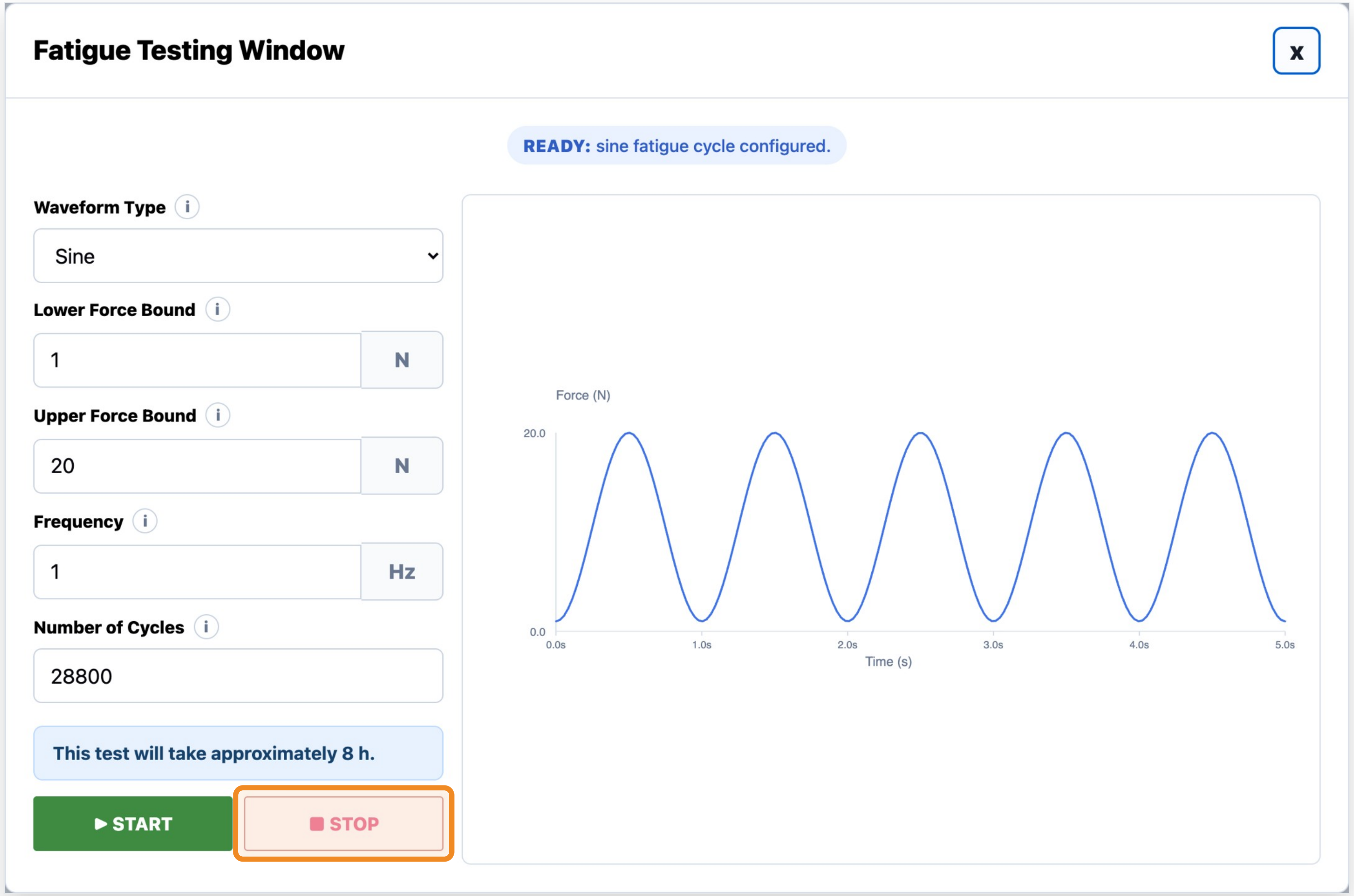
Stop Button

USER STORY

As a test fixture user, I want to pause a cyclical test while it is running.

ACCEPTANCE CRITERIA

- User clicks a "Stop" button to pause the Cyclical test while it is occurring
- Stopping deletes the previous data and restarts the cyclical test



FEATURE 1.11 · CYCLICAL TEST

STORY 1.11.9

MUST

Display Graph

USER STORY

As a test fixture user, I want to look at a live graph of the Cyclical test while it is running.

ACCEPTANCE CRITERIA

- The right side of the page displays a live Force vs Time graph while a cyclical test is running
- The graph is blank when nothing is running
- Completing the cyclical test saves the graph to the previously selected save folder

FEATURE 1.11 · CYCLICAL TEST

STORY 1.11.11

MUST

Info Icons

USER STORY

As a test fixture user, I want informational help icons throughout the window so that I can quickly understand fields, workflows, and testing terminology without needing external documentation.

INFO ICONS ON THIS PAGE

Waveform Type: Choose the force waveform used to cycle the actuator between the lower and upper force bounds. Sine is a smooth press/release. Square holds at each bound.

Lower Force Bound: Lowest force target in each fatigue cycle. Example: cycle between 1 N and 20 N.

Upper Force Bound: Highest force target in each fatigue cycle. The actuator compresses the sensor repeatedly up to this force.

Frequency: Number of fatigue cycles per second.

Number of Cycles: Total fatigue cycles to run. Default is 28800 cycles.